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# PROMOTING STUDENT COMMUNITY WITH INCREASED INFORMATION VISIBILITY IN DISTANCE EDUCATION

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# Abstract

With the advancement of technology, more and more degree programs are available in the form of online education. Despite its top ranking in online bachelor's degree programs, Pennsylvania State University's World Campus presents a variety of challenges and opportunities related to students' experiences, feelings, and operation as a "community" of learners analogous to what is experienced by resident students. Towards the vision of creating a socially integrated online environment for students to work towards their online degree, I hypothesize that increased visibility of students' relevant information for social interaction and shared identity attributes contributes to a more cohesive "felt" community.

To investigate this general hypothesis, I first studied the potential connections and information needs among online students, as well as the practices for developing social relationship from the perspectives of instructors and students. Second, I designed and carried out formative evaluations of several interactive prototypes using visualizations of student data; these were designed to enhance awareness of peers with respect to different aspects of similarity or other grounds for affinity (e.g., shared timezone or geographic location, shared educational goals).

In a related effort, I developed and validated two constructs of community that could be used to assess felt community in the context of online learning context, including attention to both feelings of affinity or personal ties and beliefs about the capacities of the online community. The instruments were created to assess the impact of my design efforts at making such connections visible.

Finally, working from my earlier findings I designed a new set of tools that included both a peer communication and collaboration space built in Slack Workspace *WeAre!* and a custom set of web visualizations that convey different facets of similarity to one's peers, which is aimed to inform and invite social interaction and to promote community feelings. I fielded this last set of designs to World Campus students, collecting a mix of self-report measures (e.g., pre- and post-usage measures; open-ended comments from tool users) and a range of activity data gathered during tool use (e.g., messages exchanged, access to the custom Web tools). Combining across these multiple forms of data, and contrasting the survey data of those who did and did join the Slack space, I identified factors that influence the adoption and use of Slack Workspace, and also investigated the effect of design interventions in promoting felt community. I discuss who was most likely to take advantage of the Slack *WeAre!* space, how they used it, and to what extent the custom Web tools were useful in raising awareness and promoting interaction.

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# Chapter 1 Introduction

## 1.1 Background

With advances in technology and broadened access to educational resources, university degree programs delivered online have drawn considerable attention and efforts, with regular increases in enrollment [Kena et al., 2015] and comparable, or even better learning outcomes [Shachar & Neumann, 2003; Snyder & Dillow, 2015] compared with residential educational programs. Using a meta-analysis approach over 12 years, Shachar & Neumann [2003] reported that in two thirds of the cases analyzed, students taking courses by distance education out-performed traditional counterparts with respect to their final score. However, distance education continues to suffer from the mere fact of distance: separation of instructors and students in terms of time and space leads to feelings of being socially removed from the situation [Guo et al., 2010]. This social isolation is a major contributor to the retention problems common in online education [Carr, 2000; Tinto, 1975]. Ashar & Skenes [1993] suggest that learning goals attract adults to an online program, but it is the presence of a social environment that makes them persist.

At the same time, productive social interaction is an important component of collaborative learning; being part of a learning community leads to fuller engagement with the class and dialogue about class content [Clegg et al., 2013]. Not surprisingly, social presence and social interaction predict students' learning performance in CSCL [Xing et al., 2015]. Xing et al. [2015] found that learners who demonstrate higher social skills engage in greater amounts of social interaction online, which in turn mediates the impacts of technology support for learning. For example, online learners who report stronger feelings social belonging report better learning outcomes than those with weaker feelings [Kizilcec & Halawa, 2015]. These findings suggest that is is important to foster communities for online learners, despite (and in some sense because of) the physical distance between students and teachers, or among learners themselves.

Studies have shown that learners in Massive Open Online Courses (MOOCs) are missing many aspects of social connection to their peer learners [Zheng et al., 2015]. However, even established online educational institutions struggle to offer an integrated and cohesive "student experience" for online learners, who are typically more diverse and geographically distributed than those pursuing resident education. The resulting impacts on motivation and satisfaction have important implications for learning outcomes, both in the short term (e.g., how to formulate and design for learning objectives) and over the longer term (e.g., personal or professional development).

The remainder of this chapter is organized as follows: 1.1.1 summarizes the origin and development of distance education as the overall context for my dissertation project. I follow that with a discussion of supporting concepts and constructs, including the notion of "school" in traditional contexts as well as the online contexts in section 1.1.2 and how communities can play a role in learning in 1.1.4. Finally, in section 1.2 I introduce the distance education program at Pennsylvania State University (World Campus) as the context for my work, closing in section 1.3 with a statement of my overall research objectives.

### 1.1.1 Distance Education

Distance education has proliferated in recent decades and continues to expand the number of learners who are drawn to options for online learning; the information and technology supports for online students using are also quickly expanding. Of 2.06 million students enrolled in 2012, 14 percent took at least one distance education course as part of their education, which in general included a mix of in-person and distance education classes [Snyder & Dillow, 2015].

Based on education statistics in 2013, about 4.6 million undergraduates participated in distance education, 2.0 millions of whom took distance education courses exclusively. 23% of post-baccalaureate students who were enrolled in degree-granting postsecondary institutions participated through distance education exclusively, an additional 31% of whom participated in at least one distance education course. For undergraduate enrollment in the 2013-2014 academic years, there were 11.34% students enrolled exclusively in distance education (6% were enrolled in programs located in the same state, and 4% were in a different state), whereas 26.5% were enrolled in at least some distance education; both of these are increases compared with previous years [Kena et al., 2015].

These current statistics cast doubt on the frequent criticisms of the usefulness of distance education. Instead they can be seen as extending older trends that have often documented increasing enrollments of online education over the years [Allen & Seaman, 2007; Tallent-Runnels et al., 2006].

Despite the consistently increasing number of distance education students , it was not until 1972 that attempts to define distance education started: in a presentation to the World Conference of the International Council for Correspondence Education (ICCE) meeting, Moore defined distance education as the set of teaching-learning relationships that are characterized by separation between learners and teachers, in contrast with "contiguous teaching" which means teaching occurs in the presence of the learners [Moore & Kearsley, 2011, p. 207]. With separation of geographic locations comes the separation of psychological and communication space; Moore [1993] proposed the notion of "transaction distance" to describe the perceived psychological distance between instructors and students. He further attributed the extent of transactional distance to three sets of constructs: dialogue (or interaction), structure (guidance, rules, routines enforced by instructors or environment) and learner autonomy (or self-directed learning).

The benefits of teachers' immediacy and presence that are typically found and acknowledged in traditional classrooms do not apply in the same way for online learning environment, due to the (possibly) larger class size, the limited attention from a single instructor who impart knowledge directly or indirectly, and the separation created by time and space Anderson et al. [2001]; Richmond et al. [2006]. As Moore indicated in his discussion of learner autonomy, distance education that only considers factors that affect teaching would be flawed [Moore & Kearsley, 2011, p. 213]. He further argued that the idiosyncrasies and independence of learners should be accepted as a valuable resource rather than as a distractive nuisance and encouraged cultivating efforts from instructors for students to become active learners with the capability of developing their own learning plans in a community environment.

Since Moore's pioneering work, researchers have introduce various lens and relationships to characterize distance education [Andrews & Tynan, 2010; Dibiase & Rademacher, 2005]. Among these, many have shifted focus from a guided didactic conversation between teachers and students to a community model with distributed responsibility and power [Garrison et al., 2001]. Taking a cost-effect perspective, Anderson developed equivalency theory (Anderson 2003), proposing that one of the three interaction modes (instructor-learner, learner-learner and learner-content) can be provided at a high level to produce deep and meaningful formal learning.

In contemporary education settings, technology and information have become prevalent and widely accessible to student populations, yielding more possibilities for students to obtain needed support on their own, thereby empowering students to a considerable degree. Consequently, online interactions among students (whether for class-specific or more personal purposes) may be starting to exert considerable impact; this possibility needs further investigation. More broadly, I argue that the focus should shift from traditional instructor-centered approach to a supportive learner-community.

### 1.1.2 School as a Place for Learning

School is seen as an agency for cultivating self-efficacy for many different types of cognitive tasks [Bandura, 1993]. College education, as part of post secondary education oriented toward preparing learners for the job market or for advanced academic pursuit, undertakes an essential role not only in cultivating students' cognitive competencies and their academic and professional knowledge, but also in providing a transition into the society at large. Bandura [1993] characterized college as a place where knowledge and thinking skills are continually tested, evaluated and socially compared; and where teachers' interpretations of students' needs and actions affect their social construction of what is happening within the learning community, including how students perceive their own performance, sense of belonging and expected behaviors. Schools are complex social environments where students share beliefs, fears, values and norms [Hofman et al., 2001]. However, when university education "goes virtual", the social fabric of learning may be weakened. If online students do not feel they are part of a community, they may feel isolated, anxious, defensive and unwilling to take risks in their learning [Wegerif, 1998].

My vision is that all learners - including students learning at a distance[1] - should feel like members of a community, trusting that they will be heard and treated sympathetically by their fellows, so that they will confidently reach out for collaboration and social support as needed. Because distance learners are interacting and working together with separations of distance and potentially time as well, the creation of community will likely depend on digital tools that evoke and support feelings of shared identity and connectedness.

### 1.1.3 Challenges for Connecting Online Learners

Computer-Supported Collaborative Learning (CSCL) environments have emerged as virtual "places" for online interaction and learning to take place. However, one pitfall recognized for CSCL environments is that the social interactions and experiences of online students are often taken for granted [Kreijns et al., 2003]. For example, MOOC instructors report that they are most concerned with students' interactions with content and academic contributions as evidence of course performance [Stephens-Martinez et al., 2014]; social data are of little interest to them [Zinn & Scheuer, 2006]. A qualitative study of novice online instructors suggests that they simply overlook the presence or level of community in their classes [Conrad, 2004]. Underscoring the inclination to overlook the social psychological aspects of CSCL, Kreijns et al. [2003] argue that possible benefits of social but non-task interaction in online learning have been ignored. Thus I see a gap between the expected benefits of social interchange

[1] Although the term "distance learning" encompasses much broader contexts, my dissertation here focuses on the instance of online degree-earning program provided by universities.

in online learning and the tendency for instructors to ignore or fail to promote such behaviors.

In response to such concerns, researchers studying online learning have been exploring social factors that may impact online learning. For example, learners' sense of belonging within an online program appears to be positively associated with their learning success [Kizilcec & Halawa, 2015]. However, cultivating the interpersonal relationships that could promote feelings of belonging takes time [Walther, 2002], and time is a limited resource that distance learners struggle to manage [Liu et al., 2007; Rabourn et al., 2015]. Researchers and designers must consider these cost-benefit tradeoffs and build social mechanisms that respect distance learners' real world values and interests [Overbaugh & Nickel, 2011]. Different from MOOCs, online degree-earning programs by nature span longer period of time for the process of taking multiple courses and obtaining credits to graduate with a degree or a certificate. Therefore, pursuing and developing community in an online context of degree program is likely to be more attractive than in other contexts; likewise, empirical research situated in such a case for online learning is also appropriate providing that time is an anticipated resource and investment for not only academic degrees, but also in social relationships [Walther, 2002].

### 1.1.4 Communities for Learning

Community is by nature a context- and case-specific construct, and is often defined across a range of dimensions, such as location, social norms, types of social interaction, and vision. In this section, I briefly introduce a few useful definitions from the literature, and draw relevant concepts primarily from the domains of interest in virtual world and distance learning environment.

Using the general context of urban society, Lyon and Driskell reviewed many different definitions of community [Lyon & Driskell, 2011]. They extracted three common elements from this vast set: shared places, distinctive social interaction, and common ties. In an early synthesis of network communities, Mynatt and her colleagues suggested a loose consensus around community as "a multi-dimensional, cohesive social computing that includes, in varying degrees: shared spatial relations, social conventions, a sense of membership and boundaries, and an ongoing rhythm of social interaction" [Mynatt et al., 1997]. They also delineated network community as a unique constellation of characteristics that included technology mediation, persistence, multiple interaction styles, capability for real-time interaction, and multi-user. Further, they offered three dimensions that might guide the design of network communities in the future: sense of (virtual) space in analogy to the concept of location, dynamics across being "real" and "virtual" space, and evolution over time. [Rheingold, 1993, 5] defines virtual communities as a "social aggregation that emerges from the Internet when enough people carry on public discussions long enough with sufficient human feeling, to form personal relationships in cyberspace". These accounts of community share the depiction of three important elements: *regular visits of a (virtual) place*, *members' social interaction* and *time* (e.g. "ongoing", "long", "persistence", "evolution").

As a sub-type of community, learning community also has been given multiple meanings in different online contexts. In its broadest sense, it captures all the stakeholders involved in distance education [Conrad, 2005; Seufert et al., 2002]. Other researchers have scoped it instead to focus on the social ties and interactions across instructors and learners [Garrison et al., 1999; Garrison & Arbaugh, 2007; Swan, 2002]. Organizing teachers within their own professional community as a learning community of peers has also been found to benefit teachers in regard to mutual support and reciprocal interaction [Barab et al., 2003]. Finally, learning community is sometimes used to refer to the student learners, as they experience activities and learn together [Liu et al., 2007; Mitchell & Sackney, 2011]. This last definition is particularly apropos to my dissertation, as I am focused primarily on mechanisms for building community within an online student population.

A related concept found in the literature describing and defining learning community is *virtual campus*; this is used to refer to a group of online students driven by the common pursuit of earning an academic degree from a recognized university. In this dissertation, I define the target of interest, i.e. the learning community as comprising students who are, or have been, enrolled in a virtual campus and who are taking classes at a distance. Although the universe of distance education is still small in scope compared with traditional residential classrooms or schools, its emergence and growth is very likely to have significant impacts on approaches and outcomes of education activities [Snyder & Dillow, 2015]. Importantly, my focus on university-provided distance education differs from Massive Open Online Courses (MOOCs), in that the students receive credits, and often degrees or diplomas as ending products. However it shares similarities with MOOCs in the mechanisms for knowledge transfer from instructors and peer learners, as well as its characteristic separation of time and space.

## 1.2 Research Context: The Case of World Campus

In the context of distance education, specifically as offered by Pennsylvania State University (aka, PSU), online education is packaged and marketed as a university "campus", analogous to other physical campus locations (e.g., in Altoona or Debois, other towns in Pennsylvania). Therefore, the context I am exploring here is both a virtual site within the university's rich landscape of offerings (there are 24 physical campuses across the state in addition to the primary University Park campus), and a typical online degree program delivered by a major university. Note that any given degree obtained from online or from a physical campus is equivalent in terms of the course specifications, program requirements, final diploma and acknowledgement. However, as previous research has indicated, the feelings of community in an online school context are significantly lower than those felt in a residential counterpart [Rovai et al., 2004]. Therefore, providing a holistic community experience for an online university student that is comparable to that experienced by residential students remains a significant challenge for researchers and practitioners.

My dissertation project We Are![2] has been conducted in a distance education program called World Campus (WC) at PSU. As a high-quality, paid degree program, WC is typical of modern distance education programs, in that all communication

[2]We Are! is a research project aimed at building identity, participation and connectivity among World Campus students. It was initially supported by The Center for Online Innovation in Learning (COIL) `https://coil.psu.edu/blog/we-are-building-identity-participation-and-connectivity-within-world-campus-cohorts/.Thenamecomesfromtheuniversitybondingchantthatissofamiliartostudents,staffandalumni.`

and coursework activities are computer-mediated, students are separated by time and space, and students may take just one or a few classes each term as they work toward degree requirements.

The WC tuition cost is often covered by a student's employer, personal savings or a mix of both. There are no mandatory in-person meetings or face-to-face events. When I started the WeAre! project in Spring 2016, WC had an enrollment of 17912 students; a large majority (87.9%) are part-time. Most of these students (85.0%) are "adult learners" ($\geqslant$ 25 years of age), who are often working full-time and/or are responsible for a range of family concerns in addition to their online coursework. In addition, 21% have military ties. With respect to other demographic characteristics of WC students: 48.4% are female; 70.1% White, 7.0% Black, 4.5% Asian; 36.0% reside in the state where university is located; 57.5% are undergraduate students.

### 1.2.1 University Infrastructure for World Campus

Pennsylvania State University's World Campus hosts a large number of degree and certificate programs [3], supporting both undergraduate and graduate education (see Figure D.1). WC courses typically enroll 30-60 students per class section with one lead instructor. Students enrolled in Penn State World Campus undergraduate courses have reported spending approximately 8 to 12 hours per week on readings and assignments for a 3-credit courses (longer time is required for graduate degree programs).

Students need to meet some basic technical requirement to succeed in their online learning experience as listed in the program website (e.g. Updated Operation Systems, Browser, Office Software, Internet access). As a Penn State student, they log into Course Management System, Canvas[4], for access to the class materials (e.g., syllabus, assignments, exams). Through the Canvas Dashboard, online learners can also interact with their peers and instructors via Canvas [5], Groups, and Discussion boards. Most WC courses use Big Blue Button, Zoom or Adobe Connect for video-

[3] There is a total of 201 programs at the time of this dissertation writing in 2019.

[4] Canvas is a popular learning management system (LMS) used by 3,000 universities, school districts and institutions around the world `https://www.canvaslms.com`; institutions can integrate other tools as listed in `https://www.eduappcenter.com`.

[5] Students also have their Microsoft Outlook web portal to manage their PSU Email

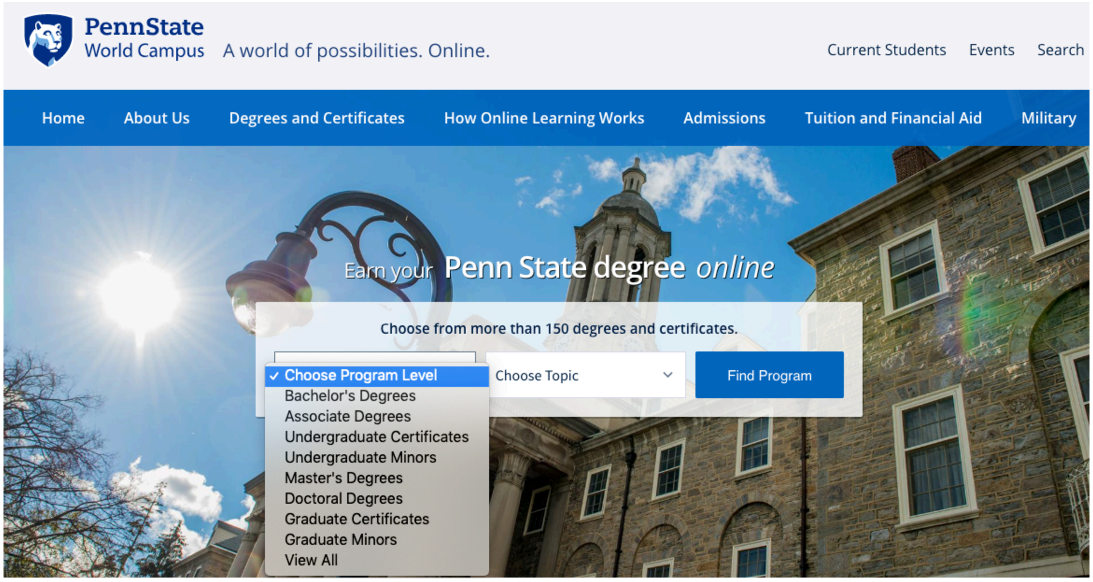



**Figure 1.1** Penn State `World Campus` program website page

conferencing. External collaboration tools (e.g., Google Drive) and communication tools (e.g. Yammer, Messenger, Google Hangout and Slack) are common, and are selected according to the preferences of individual students or groups, see Figure 1.2.

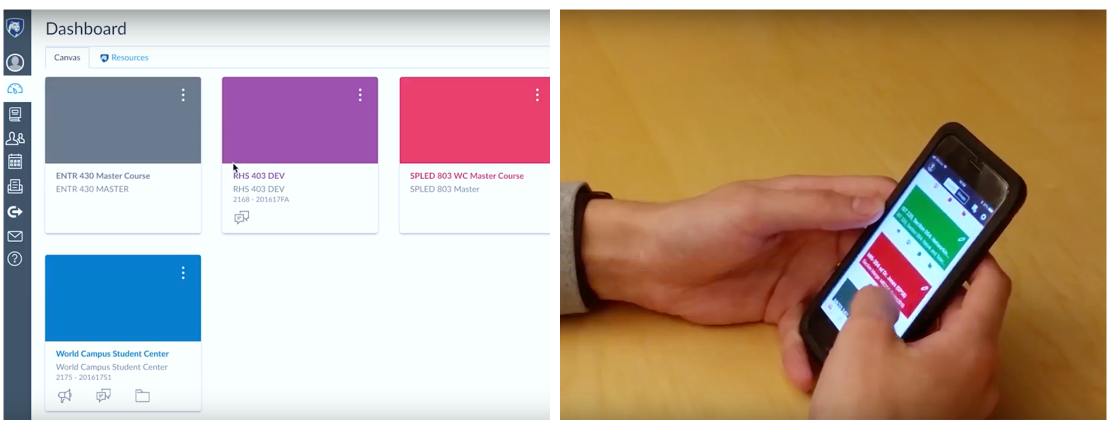



**Figure 1.2** The Learning Management System Canvas offered by Pennsylvania State University, used both in residential and online courses

There are also social media accounts administrated by staff or student clubs in Twitter, blog websites, and Facebook group page. During the completion of my dissertation, I also found that students have more options to connect beyond the classroom, such as student learning center (a Discussion board module accessible to all World Campus students [6]), and LionLink, a Alumni Network tool for all Penn State students and alumni [7].

## 1.3 Research Objectives

To facilitate the WC student community in growing and evolving their space, I have carried out a design-based research project that has explored requirements for community building, followed by the design, prototyping, and evaluation of interactive systems to facilitate the online students' social connectivity. My overarching design goal has been to engage WC students with information about themselves and their peers in ways that create and enrich feelings of community. Guided by a growing body of design rationale and implications from existing literature related to online learning environments, as well as design frameworks for creating and sustaining community, I have explored the design space on enhancing visibility through interactive tools that I hypothesize will foster distance education communities. Historically, the theoretical facets of community building have been applied largely to geographic communities (e.g., towns or neighborhoods) or to general interest-based or practice-based online communities; so one conceptual innovation has been my adaptation of existing community informatics frameworks drawn from these community settings to the online community that may be possible and useful in the context of distance education.

My inquiry around design and technological solutions towards the goal of promoting social integration among distance learners has fueled my dissertation research. Further, with this work, I have sought to contribute a substantial body of empirical data and findings to inform the design of community building for distance education, and to address the psychological and cognitive issues elaborated in my dissertation, such as lack of sense of community in virtual learning environments, disengagement

[6] Students need to subscribe to the discussion in Canvas on their own, not automatically enrolled as a member

[7] https://psulionlink.peoplegrove.com/hub/lionlink/home

due to separation in time and space as a distance learner.

# Chapter 2 Literature Review

This chapter covers the theoretical foundations and empirical studies that have informed my dissertation project. These bodies of work contribute in different ways to the problem I have chosen to study, namely the building of community in the context of distance education that is delivered online. In the following, I first review concepts of community that are useful in studying distance education, and computer-mediated communication (Section 2.1). Next, I discuss empirical studies that have probed the relationships that may be involved in community building for distance education (Section 2.2). Next, I review techniques and design mechanisms that researchers have investigated in efforts to support distance education (Section 2.3). Finally, I synthesize the previous sections to offer my own conceptual framework for investigating the presence and role of community in distance education (Section 2.4).

## 2.1 Community in Distance Education

This section presents theoretical frameworks related to distance education and community informatics in general, followed by issues tied more specifically to my goal of community building among online learners in the context of distance education. I focus primarily on the students, while acknowledging that many other important factors also play a crucial role, such as teaching strategies, instruction rationales, staff efficacy and administrative structures. I begin with a general analysis of community that was motivated by studies of geographic communities [Carroll et al., 2014], considering how these elements might be mapped into the landscape of online communities. Next I review a body of work that documented characteristics of online communities, leveraging community frameworks associated with members' identity, commitment, and practices, because it contains important insights concerning the nature of social connections in online networks [Ren et al., 2011]. Finally I present a community framework developed specifically to address the situation of distance education, where both teachers and students are geographically and temporally dispersed.

### 2.1.1 Facets of Community

**Table 2.1** Three Facets of Community [Carroll, 2014]

| **Facet** | **Description** |
|---|---|
| Community Identity (attachment) | Members experience ideal of community by sharing values, episodes, traditions, and experiences of local and world events; feel belonging and shared emotional connection; regard for local places and past events; experience community membership as part of who they are; are committed to and believe in the community's capacity to thrive and develop. |
| Participation and Awareness (engagement) | At least some members enact shared identity through collective activity, including leadership and innovation in community practices. Through this activity, members become known; their conduct and contributions are visible; all members see that one can have an impact on community decisions and initiatives, and come to regard the community as sustainable and effective. A less active form of participation is maintaining awareness of community activity. |
| Supported Networks | Community members typically play a variety of roles, and provide and reciprocate social and material support through a multitude of different tie types. The community is a relatively densely interconnected sub-network of the societal social network. |

Working from a broad range of field studies of community informatics for geographic communities, Carroll developed a conceptual model of community that includes the facets of community identity, participation and awareness and multiplexed support networks [Carroll et al., 2014]. Table 2.1 offers a description of each facet, which were created as abstractions to describe observations about communities that share geographic proximity. The three facets, despite being developed in research that was aimed at local communities, have inspired my investigation of the online community context, where I assume that analogous community activities, engagements and natural connections can help to bind together a virtual community.

*Community identity* distinguishes a member from outsiders and promotes the sense of belonging and attachment. However, it is not clear whether the characteristics of student community in online education settings differ from those of geographic communities. For instance, the local or cyclic events that are often characteristic of geographic communities (e.g., a summer festival, a football game, a nonprofit campaign) are less likely to occur online, due to the separation created by distance and time; at the same time there may be replaced with common educational activities, such as online discussions, question and answer sessions, or group projects, although often at the unit of course or group. Without bricks and mortar school, it is useful to learn about the alternative ways (if at all) of *Participation (and being aware)* as a schoolmate, a classmate, and a teammate, and how *Support Networks* are built and felt without in-person interactions in the learning context. Given the scattered distribution across a much larger geographic scope, and the diversity inherent in online learners in terms of age, culture, experiences and even disparity of time zones or schedules enrolled in online programs, fostering such a virtual community presents both challenges and opportunities.

A related body of work considers how a person's multiple identities may be faceted when participating in an online community. Farnham defined a faceted identity for a community member as *"one where different aspects of identity are performed depending on context, and expect that identity faceting will vary depending on the individual"* [Farnham & Churchill, 2011]. The extent to which one separates his or her multiple roles after crossing particular "online social boundaries" determines a person' inclination towards an integrated person, or faceted person in showing the different identities across different social contexts. In a survey of people who are relatively older than the median age in America, they found that many people have identities that are faceted across different areas of their lives to different extent: younger, working men without children reported the highest levels of incompatibility across facets. Accordingly, they reported that more incompatibility impacted technologies in raising worries about sharing, particularly in social networks, and more likely to lead to email usage.

Investigating a hybrid community with both online and offline activities associated with a local university, Zhang et al. [2011] also point to multifaceted identities (according to group and by role) drawing from scenario-based interviews that investigated Facebook activities. These researchers found the salience of online identity varied from one social interaction to another, and that these identities are activated contextually. Types of role identities, such as membership, fan (as follower), or event initiator vs participant, also defined patterns of engagement. Their study complements the time constraints and limited emotional resources discussed by [Granovetter, 1983] in a different context and technology era; it further emphasizes that stronger interpersonal bonds might be only a click away, even when the partners are large distances apart. They also found social affordances in Facebook that encourage people cooperate with others (e,g, course projects with classmates); these relationships may have a low group identity, but still fulfill a shared group mission. Such findings inspired my research to explore more the (multi-faceted) identity aspect of online learners.

### 2.1.2 Shared Identity and Ties in Online Communities

Ren and her colleagues classified affective commitment of online communities into two categories: identity-based attachment and social bonds-based attachment through the lens of common identity theory and common bond theory [Ren et al., 2011]. Common identity theory describes the causal relationships associated with people's attachment to the group as a whole based on a common social category [Abrams & Hogg, 2006], such as shared interests or identity, whereas common bond theory examines people's attachment to individual group members [Lott & Lott, 1965], and hence is based on based on peer interactions or personal ties (e.g. eating together or sharing a residence). In particular, they made a series of design claims that support the building of online communities oriented towards the two forms of attachments to groups respectively [Ren et al., 2011].

One investigation of these two bases for attachment studied university organizations (e.g., fraternities, clubs [Prentice et al., 1994]). Members of common identity groups reported feeling more attached to the group as a whole than to individuals; members of common bonds group felt the opposite.

This framework for online community has been productive in guiding research in multiple community domains, ranging from game communities [Farzan et al., 2011], to crowdsourcing [Tausczik et al., 2014], and online social networks [Brzozowski et al., 2015a]. For example, in a recent report that examined the two types of attachment, Tausczik and his colleagues found effect of group communication on community loyalty is completely mediated by attachment to the assigned work group, but there is no significant difference in the downstream effect in terms of community loyalty and retention between the two types of attachments [Tausczik et al., 2014]. As they pointed out, their manipulation on the condition may be not distinct enough to differentiate the two types of attachment types. Therefore, these findings again testified the potential difficulty to get acquainted with unknown others in a virtual context, either through identity or interpersonal bonds, especially in a general task context for only 25 minutes. Empirically, the issues they identified (such as time, task involvement, initiating relationships) suggest a need to consider similar issues when trying to promote such attachments among online learners.

Many of these researchers' design recommendations are consistent with my hypothesis about the benefits of enhancing visibility of community members' information, including the common identity categories shared among members, refreshing a stream of up-to-date members' statuses and activities, repeated exposure as well as the visibility of one's self-image (i.e. photos, and individuals' information).

With respect to shared identity, some researchers argue that common identities or goals are a primary driver for joining an online learning program. In one interview study, Brown found a theme of community tied to commonality in interests, experiences, goals, values or visions [Brown, 2001]; another interview study reported that distance learners recognize a boundary that separates members (who are in a *"different kind of world"* and know how it works) from nonmembers (family members and colleagues in physical proximity) [Haythornthwaite et al., 2000]. In a cohort-based online learning program, online students referred to the sharing of purpose, social experiences during an initial face-to-face meeting, and technology concerns when describing what community meant to them [Conrad, 2005]. Such studies suggest that some aspect of felt community may inhere in feelings of *"we are all online learners"*, the expectation that learning experiences are similar (e.g., mediated by technologies available across time and space), and possibly the challenges they all face.

These views of online learning community are similar to the Sense of Community (SOC) model developed by McMillan & Chavis [1986]. One element of SOC is group membership (a feeling of belonging); this seems to map to the feelings of commonality or shared purpose for online learners. Another element of SOC is needs fulfillment (members' needs will be met by the community); this seems to map to the expectation that staying together as a community will support learning across time and space, even though joining and maintaining one may also create its own challenges. SOC has been applied and validated in a broad mix of offline and online community contexts (e.g., local neighborhoods, livestream gaming communities and the community of YouTube contributors [Hamilton et al., 2014; Prentice et al., 1994; Rotman et al., 2009]), making it an attractive instrument for adaptation to our research goals.

While the feeling of shared identity emphasizes a group's commitment to a mission or purpose, community members may also experience feelings of connection that are based on personal ties with specific others [Ren et al., 2007]. Note that when learning takes place in an online world, it may be difficult to cultivate bonds and associated feelings of commitment [Farzan et al., 2011]. Yet, studies of degree-oriented online education programs have found evidence of strong personal relationships among online classmates even when face-to-face contact is unavailable [Haythornthwaite, 2000]. Beyond a single class, online learners' social capital within a cross-class social network is positively associated with their learning outcomes [Gašević et al., 2013]. The attachment to communities that comes through personal ties echoes the emotional support that is part of the SOC model, as well as other studies of classroom and school community [Nistor et al., 2014; Rovai et al., 2004].

## 2.2 Social Presence and Social Interaction

Social interaction are strong predictors of students' learning performance in CSCL [Xing et al., 2015]. For example, online learners who report higher feelings of social belonging also report better learning outcomes [Kizilcec & Halawa, 2015]. Furthermore, learners with higher social skills exhibit greater social interaction, which in turn mediates the impact of system functionality on learning [Xing et al., 2015]. Nonetheless, a number of researchers argue that without social presence, social interaction is not likely to occur, either through CMC, or other impression formation mechanisms [Kreijns et al., 2003; Lowenthal, 2010; Lowenthal & Dunlap, 2014; Tu, 2000]. Interaction or collaboration does not just happen [Northrup, 2001], but needs to be organized or designed [Brush, 2016; Johnson & Johnson, 1987, 1989]. These findings implied that support of interactivity does not always come along with increased social activities, or effective development of social relationships. Further, lack of social interaction hampers the development of social relationships. Other than more direct efforts from instructors alone, technology should not only support such, but also encourage this in a more explicit way —explicit but not coercive due to the possible high workload from adult learners given their multiplex identities. As a consequence, exploring and learning from design inquiries that consider such presence cues are essential for enacting meaningful social interaction.

Yet, social interaction among distance learners is often simply taken for granted [Kreijns et al., 2003]. A computer-mediated communication channel can create the possibility for social exchange online; however, the potential communication partners must be grounded in familiar social contexts to engage in social acts. That is, they need to know "who is there" and what social identities and norms are involved. Without some level of social transparency [Stuart et al., 2012], the cues that invite and guide appropriate social acts may be missing in online settings. Asynchronous online discussion —the most widely used collaborative learning activity [Appiah-Kubi & Rowland, 2016; Mustafaraj & Bu, 2015] —can be seen as forced or unnatural when social exchange is required [Biesenbach-Lucas, 2003]. Unless a contributor is careful to direct social content to a specific audience, free-wheeling discussions from large online classes may overwhelm learners and instructors [Palloff & Pratt, 1999].

Even real-time video meetings, which can offer rich social cues akin to face-to-face communication, are not always successful in distance education. If instructors do not require attendance at such sessions is (typical for distance classes, where needs for schedule flexibility are preeminent), student participation tends to be low; this in turn may suggest a lack of social interest and engagement to other class members or instructors [Sun & Rosson, 2017d]. In contrast, student-initiated social interactions (i.e., social behaviors that are carried out for context-specific reasons) can nurture feelings of community, as students experience the benefits of reaching out to their peers. A team project can evolve into a support group with social ties and emotional connections [Sun & Rosson, 2017d]; former team members may become useful resources for future education or professional activities. Over time the accumulation of social capital across shared classes can have positive effects on academic achievement [Gašević et al., 2013]. However, we know little about how to promote and foster the social ties that can underpin a learning community.

Considering the social characteristics of a group and the CSCL environment, Kreijns et al developed the concept of *social affordances* as an ecological approach to evoke, promote or otherwise reinforce social interactions [Kreijns et al., 2002]. Defined as *"properties of a CSCL environment that act as social-contextual facilitators relevant for the learner's social interactions,"* social affordances can be realized in at least two ways: (1) functionality that conveys socio-emotive information about users (e.g., real-time audio or video communication); and (2) features that invite social behaviors (e.g., directing a question to a peer).

In sum, a social context may be impossible due to the lack of social presence when a group of individuals begin to interact online. However, to build up the social presence depends not only on communication media quality, users' perception, and situation details, but also relies on the visibility of information as common grounds to inform any social initiatives. The visibility issue bridges the origin of social presence from Goffman's definition of co-presence on the mutual visibility in physical places and the evolving direction into the community context with respect to feelings of being connected [Goffman, 1959, 1963; Lowenthal, 2010].

### 2.2.1 Interpersonal Knowledge Among Online Peers

Uncertainty Reduction Theory suggests that strangers will be more likely to interact if they first grasp a bit of background information about each other [Berger & Calabrese, 1975]. Regarding online interaction, researchers have observed that increases in interpersonal knowledge increases interaction within globally-distributed academic communities; in these settings it appears that socio-emotional knowledge is a strong predictor for sense of community and knowledge sharing acceptance [Nistor et al., 2014]. More specifically in virtual learning situations, students who begin with greater interpersonal knowledge about other participants, take more initiative in shared discourse (e.g., critically evaluating contributions of others, asking questions and making new proposals) [Diekamp et al., 2008]. A recent interview study also revealed that getting to know one another is important for distance learners enrolled in online degree programs to feel connected, especially in the social context of small-group [Sun et al., 2019]. Unfortunately, without some amount of social transparency [Stuart et al., 2012], the social information that can help to invite and formulate appropriate social acts are likely to be missing in the online settings.

Researchers have shown that information visualizations can help to understand social media networks (e.g. friendship links), as well as the social patterns embedded in conversation logs (e.g. reply, comment) [Coffrin et al., 2014; Janssen et al., 2007; Lee et al., 1997; Schreurs et al., 2013]. At a higher level, visualizations of community structures can help users recognize acquaintances and discover new friends (e.g., Vizster [Heer & Boyd, 2005]; Network Awareness Tool [Schreurs et al., 2013]). Such awareness should be helpful in grounding requests for help or discussion of course-based content. However, we do not yet know whether visualization of online learners' characteristics (e.g., age, gender, interests) might also help to ground and evoke initiatives that are purely social in purpose. One of the goals of my investigations and design experiments was to determine whether visualization of peer's information may promote feelings of affinity and possibly inspire efforts for social interaction.

An important contributor to interpersonal knowledge is the *social identity* that people bring to a shared situation. Here I use the classic definition of social identity from [Tajfel & Turner, 1985]: "the individual's knowledge that s/he belongs to certain social groups together with some emotional and value significance to him/her of this group membership." Under this definition, a person has multiple social identities —as many as the groups he or she belongs to. In the real world, people regularly make identity inferences using people's physical characteristics and behaviors. But in online settings, these identity cues can be absent. We are interested in whether visibility of identity-evoking information can influence online learners' efforts and success at building interpersonal knowledge about their peers.

## 2.3 Designing for Community in Distance Education

### 2.3.1 Affect, Norms and Needs in Online Participation

Broadly speaking, prior work suggests three motivations for community members to participate online activities: affective commitment, which means that members like and identify with the community; norm-based commitment, which means they think it is the right thing to do; and need-based commitment, which means they derive net benefits that outweigh the cost of participation [Kraut & Resnick, 2011].

In the case of MOOCs, norm-based commitment (e.g., peer grading [Kulkarni et al., 2015]) has demonstrated considerable success in engaging online learners. Unfortunately, like forum interactions, such norm-based interaction may be perceived as forced and unnatural because *"social exchange"* is often required (e.g., learners must write mandatory posts for participation credit) [Mak et al., 2010]. In fact, a case study of inactive students in an online graduate course research found that the so-called "lurking" students view their "invisible" participation as part of their learning process using a low-profile approach [Beaudoin, 2002]. As Kreijns explained, social interaction in CSCL should not be taken for granted, just because it is technically possible [Kreijns et al., 2003].

In terms of affective commitment, a survey study found that feelings of common identity and friendship are two principle constituents of distance learners' sense of community, with the latter is relatively weakly perceived [Learning or lurking? Tracking the "invisible" online student, Sun]. Although it remains unclear as to what might promote feelings of friendship among online learners, patterns of communication that include emoji have been found to facilitate feelings of interpersonal closeness, perhaps because emoji add a flavor of lighthearted non-task related conversation [Sun et al., 2019].

However, little is known about the specific factors that might drive or facilitate students' needs-based commitment. One exception is an experiment that examines reputation mechanisms in MOOC forum participation; it indicated that extrinsic motivated elements (e.g. score, badges) facilitated faster response time and more responses in quantity, but still had no significant impact on participants' sense of community [Coetzee et al., 2014a]. Though communication channels may create possibilities for social interaction, the perception of community among online learners need to be grounded in their own intrinsic needs to engage in *"social acts"*. Therefore, we seek to explore what peer information they need to engage meaningfully with others in learning activities.

### 2.3.2 Fostering Connections Among Online Learners

*"Connections among peers"* is a complex notion given the variety of online educational settings. Some online learning takes place in well-defined groups (e.g., class projects), some comprise an entire degree program, and some is associated with a broadly distributed online course (e.g., MOOCs) [Learning or lurking? Tracking the "invisible" online student, Sun; Lin et al., 2014; Zheng et al., 2015]. Motivations and financial investment to learn at a distance, the source and number of potential peer learners, as well the time course for possible interaction, will vary across these settings and affect the nature of connections that can be built.

Researchers interested in cooperative learning in a class setting often focus on connections felt by group members or classmates [Du et al., 2012; Goggins et al., 2011], where small-group relationships are almost certainly an important source of social support when members collaborate to gain the course grades. Across multiple classes, students enrolled in formal programs at a distance exhibit more of a temporal pattern while balancing work and life constraints [Haythornthwaite et al., 2000]. Researchers studying an online master degree program reported that students' social capital accumulates across the sequence of courses, with higher social capital predicting better grades [Gašević et al., 2013]. A recent study of distance learners enrolled in multi-year degree programs [Learning or lurking? Tracking the "invisible" online student, Sun] revealed that online students felt efficacious in regulating their identities, coordinating with one another and providing support support [Bandura, 1997; Carroll et al., 2005]. Interestingly, this same study found that feelings of friendship – a peer relationship one might expect from face-to-face social settings – tended not to be experienced by distance learners.

At the other extreme, even some MOOC learners are socially motivated to study together and to make friends [Zheng et al., 2015]. Further, learning-at-scale researchers (e.g., MOOCs) found the connections that not only grow out of a specific class, but also those felt by alumni who lingered around after the end of a single class session. For example, MOOC researchers found that a persistent chat room encouraged course alumni to reach out and help new learners [Nelimarkka & Vihavainen, 2015]. This suggests that the alumni might feel a shared identity or connection "back to" an educational setting they had previously experienced.

Given the many forms of social connection that may play a role in online learning, I chose to investigate an online degree earning program that enables multiple types of peer groups as one instance of distance learning contexts. Specifically, I examine connections experienced in small groups, within an entire class, and beyond individual classes at the level of a university.

### 2.3.3 Technology-Mediated Social Engagement in Online Learning Environments

Prior work has studied both synchronous and asynchronous technologies as supports for online learning behaviors in different settings, ranging from formal education (e.g. classes taken in pursuit of undergraduate or graduate degrees [Sun & Rosson, 2017a]), to MOOCs (open access to large individual classes, e.g. Coursera [Zheng et al., 2015]) to social media platforms that appropriated for informal learning (e.g. Google+ [Brzozowski et al., 2015b]). Much of this work has investigated what types of remote interaction channels can enhance students' learning experience, as well as whether and how social behavior is embedded in learning activities.

For instance, researchers have observed that synchronous videoconferencing may raise levels of engagement in online learning [Kulkarni et al., 2015; Sun & Rosson, 2017d]. One study investigated the use of online class presentations that combined presenter audio with video streaming of the presenter and audience [Marlow et al., 2017]. The video was found to help the presenter assess audience engagement using facial and postural cues to enhance remote interaction. Unfortunately, the resulting composite presentation was so rich in content that it may have introduced issues with distraction.

Building on the oft-observed pedagogical benefits of peer learning [Smith et al., 2009; Springer et al., 1999], researchers have started to explore how online technologies can be used to encourage collaborations with other learners. One example is a crowdsourcing platform that emulates learning in a MOOC [Coetzee et al., 2015]. The researchers found that when learners participated in semi-synchronous discussion as members of small groups, they had better learning outcomes than those who worked alone. Another study assessed student performance and engagement in MOOCs, and found benefits for groups that use Talkabout, a synchronous, small-group video discussion platform [Kulkarni et al., 2015].

Asynchronous interaction with other learners – for example, in discussion forums – has also been seen to promote higher achievement in online learning environments [Coetzee et al., 2014a]. In comparison to synchronous technologies, asynchronous communication is easier to support across widely varying student populations and access situations. However, studies have shown that these more drawn out interactions are susceptible to the influence of learners who are willing to post or reply on a regular basis; if such individuals do not "step up", the forum may never develop a healthy level of activity and fade away [Huang et al., 2014].

Summing up, any technology that allows peer learners to communicate and work together on projects seems to enhance their learning outcomes. Synchronous interaction technologies seem to be preferred over asynchronous options, but can be difficult to implement and may be overwhelming in some situations. With this in mind, I took a broad approach in my early study of students' social needs and practices, asking open-ended questions about many technologies used to support distance learning.

## 2.4 Synthesis and Research Directions

I have used this chapter to summarize theories and empirical findings in the fields of distance education, social and organizational psychology, CSCW and CMC. As part of this I have identified a knowledge gap in how best to use contemporary technology and theory to enhance connectivity among peers and associated community building for distance education. In the following, I expand on these conclusions to lay out my research agenda.

First, despite the numerous theories and frameworks described above, there is a paucity of theories to characterize the holistic picture of community building for online learning contexts, and in particular the community-building power of the most prominent subgroup - students - in a distance education context. The anonymity and virtuality in computer-mediated environments diminish the exposure of one's presence, including static information that constitutes one's identity, or even the sole existence of others. In other words, the status quo of a virtual space curtails the visibility of virtual learners. In contrast to the previous theoretical models that concern what is missing compared against face-to-face communication as the reference, I argue that although direct observation from one's face-to-face cues is missing, greater possibilities are inherent in a cyberspace where social space is a constructed experience that emerges out of one's own needs and willingness to interact [Walther, 1996].

Second, I have surfaced the need to probe more deeply the meanings of "student community" for both instructors and students, specifically looking at their motivations, strategies and the relevant technology use in support of their community fostering or formation process, situated in their socio-technological environment. Researchers investigating the nature and impacts of social behaviors in CSCL have studied students' perceptions [Shelley, 2008], communication artifacts [Oliver Ferschke, Iris Howley, Gaurav Tomar, Diyi Yang, 2015] and activity logs [Soller et al., 2002]. For example, discussion posts and chat interactions are often used to enact social behaviors, such as lurking [Mustafaraj & Bu, 2015] and peer support [Appiah-Kubi & Rowland, 2016]. However, discourse analysis captures only part of the collaboration in an online environment, because the analysis of utterances ignores the larger social context within which a specific interaction takes place. In addition, class-based discourse lasts for only one course, whereas computer-mediated interpersonal connections typically emerge over longer periods of time [Walther, 2002]. My work adds further insight to questions about social behavior in CSCL by asking both online instructors and online students' experiences and observations about student interconnections as well as how they might be promoting such relationships over longer span of temporal period.

As central stakeholders for distance education programs, online learners are confronted with significant challenges, including multiple social roles as professional workers, family supporters, and other offline duties, and are very likely to face time constraints [Kazis et al., 2007; Rabourn et al., 2015]. Therefore, when considering the design environment to foster social learning environment, my dissertation work has also attempted to delineate the boundaries and tradeoffs in exposing social opportunities and the side effects that may come along with non-requested social interventions.

Based on the above reasoning and synthesis over multiple bodies of literature and design concerns, my high level goal is not to impose any type of communication media onto the online learners, but rather to present them with a broad range of choices that can be used to initiate any type of communication they want, and to build up the connections that they need to feel comfortable and confident in peer interactions. This aligns with the visual cues and information needs of social presence, and also grounds the possibility of social interaction. To facilitate social presence or even social interaction, I have experimented with novel designs that will enable online learners to project themselves into the virtual world, more broadly to enhance the visibility of community members, and to support meaningful peer interaction and relationships.

Towards this end, I now present four related investigations that I carried out to understand distance learners' needs and practices in fostering relationships and finding community in online degree programs (Chapter 3). These studies grounded my design explorations that I present in Chapter 5 and Chapter 6. In Chapter 4, I describe my work to create and validate two measurement scales that I later used to quantify community perceptions of online learners. The scales consider the two perspectives of feelings and group capacity; these were used to assess the impact of my final set of design interventions that are presented in Chapter 6. I present my evaluation of the final set of designs in Chapter 7. Finally, in Chapter 8, I synthesize the insights learned from my multiple user studies, design iterations and evaluations, and field observations, as a set of theoretical and design implications as well as ideas for future work.

# Chapter 3 Empirical Studies of Peer Connections in Distance Education

## 3.1 Chapter Overview

I have followed a user-centered design approach to uncover ways to foster community building for distance education; my approach has focused on understanding how I might increase the visibility of students within their shared interactive environment. In contrast to earlier research examining the communication media, discourse organization and users' perception of technology in the context of organizations or virtual communities, my research has focused on students who are enrolled in distance education programs using current learning management systems and related technologies.

I have been particularly interested in how information technology might affect the dynamics of community building, from both social and cognitive perspectives. By closely examining learners' needs and the current technology and behavioral landscape in which they operate, I sought to unveil the essence of community building from the angle of information technology design, and to inform the design with directions from my conceptual framework and related empirical studies.

In the balance of this chapter, I present several empirical studies of current WC students and instructors. Section 3.2 presents a study of instructors' views about student community and how to promote it; Section 3.3 presents a similar study but focusing on students' views. Section 3.4 followed up strategies and practices of peer connections with complementary methods, using surveys to probe WC students' beliefs about the types of information that might be most important to building community. Section 3.5 presents another follow-up interview study aimed to deepen the understanding of the highly rated peer information found in Section 3.4. Finally, I synthesize insights from the three studies in Section 3.5.4.

## 3.2 Instructors' Views of Student Community

As the central coordinators of class-wide activities in distance learning, online course instructors have the opportunity to promote exchanges among class members (e.g., through synchronous or asynchronous interaction during lectures or question and answer) over a considerable period of time, and even across multiple semesters. However, little attention has been paid to whether and how instructor-led activities might affect the sociality of distance education. One example is found in the reflections of two online professors [Mcelrath & Mcdowell, 2008] who offered suggestions about how to create a supportive environment, including course chats, interactive introductions, and student stories as the examples to illustrate the core concepts. Their work leveraged Brown's framework for community-building in distance education [Brown, 2001]; this work suggests a process of making online acquaintances, gaining community acceptance and experiencing camaraderie. However, this strand of work focused on text-based asynchronous learning in graduate teaching contexts that involved only experienced online instructors who attended to community building carefully. Along with the general expansion of online technology, I have been working to create a more expansive view of how instructors with varying experiences and disciplinary background may be doing to enrich online learners' social experience.

Many of these experiences are guided by the instruction team. For instance, research on distance education has shown that instructors' attitudes towards technology, teaching styles, and technology control will affect what students report about learning outcomes [Sun et al., 2008; Webster & Hackley, 1997]. These related studies prompted the following exploratory research questions [Sun & Rosson, 2017d]:

- *What sorts of social connections do instructors perceive among their distance* *learners?*

- *What techniques do online instructors use to promote such connections among students?*
- *What implications do such findings have for design —both of online pedagogy and technology support?*

### 3.2.1 Method

I conducted an exploratory interview study aimed at unearthing patterns for further research and discussion, not to test or generalize differences among courses or instructors. The semi-structured interviews were carried out at WC. I recruited the distance instructors through convenience sampling and personal connections within an iSchool at PSU, while also seeking a relatively diverse sample with respect to online teaching experience, subject areas and teaching styles. I analyzed the interviews through inductive thematic analysis [Braun & Clarke, 2006; Strauss & Corbin, 1998]. During the data collection, both authors took notes and discussed emerging themes after each interview. Once I did not discover new themes, I concluded that I had reached the point of theoretical saturation and stopped recruiting instructors (n = 11).

### 3.2.2 Participants

I interviewed 11 instructors in the spring of 2016; all teach both in residence and online at PSU, with a range of prior teaching experience (see Table 3.1; pseudonyms signify gender and the profiles are ordered roughly by teaching experience.). Eight are American; two are from outside the United States. Five are females. The courses taught online by these instructors mirror what they teach in residence; our organization within PSU has an interdisciplinary education mission, so the courses intermix a wide range of disciplines related to the information sciences, including computer programming, application and database design, security policies and laws, and enterprise architecture. Most of the online courses were for undergraduates; a few instructors teach both graduate (e.g., professional masters) and undergraduate courses.

### 3.2.3 Interview Process

I began each interview with questions about prior teaching experience (e.g. courses, class size, how long online). I next asked for general reflections about connections among online students in three different social contexts: a) among students in the class; b) between the students and the instructor; and c) within WC. I focused primarily on online teaching, but at the end asked for a comparison of online and residential teaching. In the process of gathering data, instructors often interleaved comments about their own connections to students with those that the students have with each other, without explicitly placing themselves as "outsiders" to the student milieu; this is not surprising given that I was asking for personal impressions. Therefore, there is considerable overlap in the first two contexts with respect to the presence of communities. However, perceived connections across PSU or WC were viewed as more distinct; these connections are inherently more abstract, diffuse and to a great extent invisible.

I recognize that the instructors' comments about student community serve only as indirect evidence, but no instructors found the questions odd and were quite able to answer, often in considerable detail. I also inquired about strategies and technology being used; student data they can access or would like to access; comparison between online and residential instruction; and visions or suggestions to improve sense of community. I interviewed 10 instructors in person; the other participated via a Google Hangout video call. I recorded 9.6 hours of audio data, with an average of 52 minutes per interview session.

### 3.2.4 Data Analysis

I first used open coding to obtain categories related to our research questions, organizing the resulting codes into a table that consisted of code names, code memos defining the codes, and sample quotes. I then asked a colleague to review this table using the method of constant comparison [Hallberg, 2006]. Coding issues identified by the colleague were resolved by discussion with me, with changes to categories as needed. I then worked with my research advisor to review the analysis in an effort to identify semantic themes and to examine similarities and differences. This led me

**Table 3.1** Instructors' Teaching Profiles

| Pseudonym | Teaching years (online/general) | Recent Online Class(es) | Class size |
|---|---|---|---|
| Lily | 3/26 | Undergraduate intermediate course in human-centered design | 35~57 |
| Kyle | 3/24 | Undergraduate intermediate course in programming | N/A |
| Matthew | 3/22 | Undergraduate introductory course & senior technical course in information security | 50 |
| Rebecca | 2/20 | Graduate project course in enterprise architecture | 44 |
| George | 7/16 | Undergraduate intermediate course in databases, project management, and human-centered design class | 50 |
| Eric | 3/15 | Undergrad introductory class in information security; graduate capstone class | 50 undergrads, 20 graduates |
| Scott | 1/15 | Undergraduate introductory course on programming | 35~55 |
| Kristen | 3/6 | Intermediate undergraduate classes in statistical methods | 50~60 |
| Amy | 1/5 | Undergraduate intermediate course in information security | 50 |
| Max | 3/4 | Undergraduate introductory and senior-level courses in information security concepts and cyber law | 50~60 |
| Julie | 1/1 | Undergraduate introductory course in information security; graduate course in social network analysis | 50 undergrads, 20 graduate |

to prune some codes that were irrelevant to the central issue of social connections. Finally, I organized the remaining themes – indicators of perceived connections and techniques to foster connection – into a thematic map. Each theme was articulated into the subthemes that form our primary findings. This analytic approach is not intended to yield conclusive arguments, but to provide exploratory insights that can guide further in-depth investigations and inform design thinking. In the following, I present each high-level theme as a separate section.

### 3.2.5 What Connections Do Instructors See?

All instructors reported social connections among students, though some offered more examples than others. In this exploratory study, I cannot be certain whether this is due to personal characteristics of the instructor, course design or content, or other course-specific variables. Thus I report a mix of general patterns and specific illustrative examples, in an effort to shed light on whether and how instructors find the presence and degree of students' connections. Instructors shared a wide range of "evidence" when they discussed how their students felt connected to one another. In the discussion of their comments I focus primarily on positive examples, as these help to paint a broad picture of what instructors notice about how their students interact with each other.

One type of evidence cited regarding student community was students' real-time engagement. Instructors who reported strong connections among students tended to see their students as highly engaged, for example participating in class-wide online chats. Rebecca once attracted ~40 students to a single evening chat and estimates that a large percentage of her students regularly attend her live sessions (e.g. 27 out of 28). George reported a similar success: "I invite them to come. Of course, I record everything so that they can watch it at a later time, but a significant number of them do come. That's part of building some of this community, too."

Instructors also reported that students' real-time engagement with one another could enrich the live session because their shared discussion might uncover some hidden knowledge gap that instructors would not otherwise know to address. The result is a benefit for a broader audience that includes not only the witnesses, but also the students who may not have been there but viewed the recorded session later.

> *Someone asked one question about the next assignment... It was an interesting question, and I explained it. The other student said, "thank you, Nelson, for asking this question. It was very good and helped me under-*

> *stand it." It is hard to know what students want to know, but the live session is an opportunity for real students asking real questions...It is not just my presentation; it has the student's question. I think it enriched the session a lot.* (Kyle)

Participating regularly in synchronous sessions gives students a chance to find out more about one another and feel more connection. Rebecca offers many such sessions in her online teaching, and in once class students displayed a strong attachment to the class at the end of the semester as evidenced by reluctance to end their semester together: *"I have students that say: 'It's Tuesday night and there's no class, I really looked forward to class'. There was real camaraderie... I think they really bonded. I think there are some good connections that... some good networking connections that have been made."* (Rebecca)

In contrast, Scott shared negative results for attempts to engage online students in real-time sessions. Reacting to low attendance rates (less than 5%) at "live with the professor" events, he questioned the need for fostering community, wondering whether people prefer to focus on connections in their real life settings:

> *I provide some fairly rudimentary mechanisms to be more connected, you know discussion forums, online live sessions, and they, and both have been, um very sparsely attended. So, that's my first reaction is that maybe the whole nature of, of at least some online learning is, 'you know I have enough community, at work. I have enough community with my family. Um, I don't want any more community with my online courses'* (Scott)

Some instructors reported students' efforts in sustaining or strengthening interpersonal ties as evidence of student connectivity. Examples of such efforts for continued collaboration included requests to join a team with someone; calling one another by name in a post; or referencing previous shared experiences. They also observed that students inquired about upcoming classes and anticipated future interactions: *"I've noticed that there can be almost cliques among students, where students that know each other very well and tend to take their classes in the same order, take them the same semester. They know each other already."* (Lily)

Interwoven within the normal rhythm of classwork, instructors observed evidence of mutual support groups, where student ties include emotional connections. Sometimes these social ties are born of team-based collaborations, but they seem to go beyond assigned projects: team members cover for others who may be suffering from personal issues at the cost of doing more than their own share of the work. At a broader scope, online students root for one another through hardships from their encounters in an online class setting, sending encouragement that addresses worries expressed in someone else's message. For example, Julie shared this heartening support offered in response to a self-deprecating post: *"A lot of people were rooting for him, 'I think you'll do good, I hope you'll get a lot out of this degree.' "*

Beyond the context of a class, students have also been seen to bond around and benefit from weak ties around a local university [Zhang et al., 2011]. Thus, I asked instructors whether and how their students feel connected with PSU. A majority (9 of 11) held a positive impression of students' feelings of connection to PSU, although not all offered evidence. With respect to institutional bonds, some instructors felt that online students may even feel more institutional pride than their residential counterparts. For instance, Eric said that his online students use Reddit (a popular forum) to publicize useful online offerings from PSU; other students use the university mascot as a profile picture, conveying a tie with PSU; still others drive over one thousand miles to visit the physical campus or attend commencement, suggesting that "they are very proud of" the institution (Eric).

Finally, George shared his own piece of evidence for connections among students at the university level: some of his online students have taken the initiative to create student clubs, reflecting an urge to experience social ties that extend beyond a class and are similar to those built by residential student populations: *"I'm also involved with the PSU Online Ed club. I'm just an adviser, so once again...I let them have their club. I'm just an adviser. They have a lot of pride in being PSU students. Overall, I get a sense of that".* (George)

### 3.2.6 What Techniques May Help to Build Community?

After asking the instructors about their perceptions of how student connections are present in their online classes, I asked what they do that may contribute to a sense of community. Not surprisingly, public introductions are a common technique for increasing the visibility of information about online peers [Mcelrath & Mcdowell, 2008]; some of our participants appropriated this as a socializing activity to find compatible team members. When used in combination with self-organizing their teams, students are encouraged to purposefully read information about their peers to select ideal collaborators. The self-organizing orientation instigates a social process in which students glean knowledge about their peers, find commonality and other implicit ties, and may thus build feelings of trust and community more quickly: *"As part of the discussion that takes place in this class introduction, I do suggest that they talk about their expectations for teammates and their general idea of their availability to meet as a team, if they're available in the evenings or weekends or whatever. That's a very important part of any World Campus class, is having a good team."* (Lily)

Some instructors award points for replies to peers' introductions, but students also exhibit voluntary interest in engaging this way. As one instructor explained, social exchanges that are oriented towards knowing more about each other seem to evoke reciprocal sharing, perhaps making them feel good about themselves and others: *"I require them to comment on each other's posts, but you could just see that they enjoyed it, that they were curious about what others were doing ... When they see the benefits of getting comments, there's some generosity in the way that they are giving them out. It's this culture of giving and receiving."* (Julie).

Another technique for raising students' visibility within the class relates to the work that they produce. For example, Rebecca publicizes bits of student assignments in the shared forum, inviting responses from the others. Later, she reviews the responses and generates themes as fuel for a live session, where a high percentage of students attend and discuss the topics. This reuse of content promotes attention to peer contributions; students regularly follow up on others' thoughts and experiences, often voicing gratitude that they are learning new solutions from peers who are working in different corporate settings. Similar strategies are employed by Kyle, who uses bits of student programs to demonstrate effective and ineffective coding techniques. Publicizing students' work products not only makes the online peers more socially visible (i.e. as seen in their work products) but also invites students to make comparisons to and reflect on their own work products.

Some instructors told me that they work to maintain a teaching presence to promote feelings of connection throughout their classes. For example, they hold one-to-many broadcast sessions to gather and attend to students' needs, or just offer general availability: *"I think the first thing to make sure that they understand that you're a real person ... I don't like doing pre-recording stuff. I like the live thing. I went to the live things more. I think that's the teacher connection, the teaching presence that helps students to relate to me."* (George)

I found two related approaches for enhancing one's teaching presence. Lily wants to make students believe that she is real, so she deliberately communicates using a rich communication channel (video conferencing); Max offers to take phone calls at any time, implying a real-time availability. Another approach is for instructors to provide personal cues about themselves. For instance, Rebecca injects chit-chat about ongoing university activities or her own life at the start of her live sessions (e.g., while students are "arriving"). This sets up a sort of casual socializing and can evoke student social chit-chat in return: *"... Oh gosh, hold on. Or the cat. Well, they love this. Then they'll say, "say nighty nighty to kitty," or "how is your dog"? "Is your son on the Xbox"?* (Rebecca)

Another technique that may enhance an instructor's presence is the personalization of their communications to students. George regularly sends out custom messages to individual students: he keeps a student roster and mines information from ongoing class activities to write a personal email for five students he chooses at random each week. His goal is to nurture connection with them: *What I try to do, and I don't do this all the time, it depends on my workload, my availability, I try to pick out about five students a week that I will send a personalized email by getting personal, but I know a little bit about them and I'll ask them, especially if they're working professionals, I try to connect with them. Actually, that's some of my success has been being able to do that, connecting with some of these adult learners."* (George)

Personalized communication might be experienced as a social overture that makes students feel cared for and embraced. Regarding student-instructor interaction, one instructor with expertise in sociology shared her happy surprise about reactions to simply asking "How's everything else going" in a class-wide email. Most students dismissed it, but some saw it as an opportunity to "unload"; they appreciated this gesture so much that this instructor was subsequently rated as a student's best professor ever:

> *... for some students apparently, a couple of small emails meant a lot. ... Some person reads it as being very casual, another person reads it as in I really care about their life and the way it was going. They write a long email and they start like, "Thank you so much for asking. I have this thing going on. I had another thing going on with this" ...* (Julie)

## 3.3 Distance Learners' Perception of Community

To investigate the nature of connections among distance learners, I wanted to understand what it takes to initiate, activate, or reinforce connections that respect the multiple competing roles of adult learners, who work full-time but also "go to school" [Compton et al., 2006].

In my investigation I drew from discussions of social affordances in computer-supported collaborative learning (CSCL), where researchers have pointed out that students will not always engage in social behavior simply because it is technically possible [Kreijns et al., 2003]. Many important interactions are not planned; impromptu encounters and informal non-task-related conversations can often form the backbone of a positive learning environment [Gilbert & Moore, 1998].

Even when purely "social" exchanges are seen as valuable, casual interactions may be difficult for distance learners, who often have little time to sset aside for peer interaction [Rabourn et al., 2015]. As a result, some researchers question the cost-benefit equation for community-building activities [Overbaugh & Nickel, 2011]. Any efforts to promote sociality in distance education should address the trade-off between the costs of time and the benefits of stronger social ties.

Researchers studying collaborative learning have pointed to the tight integration of individual learning with cooperative learning that may occur across multiple settings, including dyads, small groups, classrooms and communities [Rovai et al., 2004; Stahl, 2011; Vygotsky, 1980]. This view of shared knowledge construction underscores the importance of understanding the social structures that underlie online collaborative learning. This need is especially strong for distance learning, where the ways that peers can interact are radically different from traditional brick and mortar schools. My interview study of distance education students was a step in this direction and was guided by these exploratory research questions [Sun et al., 2019]:

- *How do distance learners perceive and develop social connections with peer learners?*
- *How do feelings of connection, and strategies for building connections, differ for different types of peer groupings?*
- *In what ways does the social and technology structure for distance learning influence the building of connections?*

### 3.3.1 Method

Using a combination of purposive and convenience sampling [Merriam, 2002], I recruited our target participants from seven summer online IST classes. Upon our request, six online IST instructors sent out invitation emails to seven classes (one was teaching two courses). I invited only students who were enrolled in an online degree program (undergraduate or graduate) and thus committed to a long-term educational goal. The seven courses covered legal issues, a capstone course, human-centered design, project management and programming. Some participants received extra credit for their time; others volunteered with no promise of compensation.

### 3.3.2 Participants

I used a screening survey to gather interview availability, education status, prior online courses, and demographic information. From this I selected students using two criteria expected to enhance the richness of the sample: 1) varying levels of experience with online education; and 2) variations in demographic background and employment status.

All participants resided in the U.S. at the time of the study, though some travel internationally while taking online courses. Most were 3rd year undergraduates or above, and had taken online courses within and outside of the iSchool; some were completing degrees in other departments but pursuing a secondary degree in the iSchool. This breadth in sample allowed me to ask more generally about different

**Table 3.2** Interviewees' demographics and background

| Pseudonym | Age/Race/# of Courses taken (Transferred history) | Work Experience |
|---|---|---|
| Becky | 31/White/15[1] (T) | 10 years in a major-related field |
| Judi | 24/White/22 (T) | 1 part-time job (major-related), job hunting |
| Gabriela | 23/African/Over 7 (T) | 2 part-time jobs (non-major-related) |
| Anne | 45/White/Over 7 (T) | 26 years in Navy; 13 years in a major-related field |
| Dan | 33/Hispanic or Latino/ 6 (T) | 14 years in military |
| Jeff | 31/White/26 (T) | Full-time (non-major-related) |
| Rafael | 59/White/Over 30 (T) | Self employed in a major-related field |
| Marshal | 35/White/ 18 (T) | 3 years in a major-related field; several years in a non-major related field |
| Elson | 44/Hispanic or Latino/ 9 (NT) | 21 years in a major-related field |
| Howard | 39/White/[N/A] | 14 years in a major-related field |
| Jeremy | 40/White/5 for this semester (NT) | 10 years in a non-major-related field; full-time position in major-related for 1 year |
| Charles | 38/White/[N/A] (NT) | 10 years in military; 10 years in a major-related field |
| Yolanda | 22/White/5 for this semester (T) | Internship |
| Ross | 26/White/4 for this semester (T) | 7 years in airforce; 9 years in a major-related field |
| Kevin | 24/Asian/13 (NT) | 1 part-time job (non-major-related), job hunting |

types of distance courses (including Math, Economics, Political Science, Philosophy, Spanish, English, Education, Music, Art, Nursing, Health, Meteorology, Statistics, Anthropology, Communication, Human Development, Astronomy, Architecture). In this sense, although students were drawn from a set of (seven) course rosters, their online learning experiences were much broader than the courses from which they were recruited.

As intended by my sampling strategy [Merriam, 2002] , the 15 participants represented a broad mix of age, ethnicity, life stages, and industry experience (Table 3.4). Eleven were White, two Hispanic or Latino, one Asian and one African American; the median age was 34 and five were female. A majority (11) had transferred credits from other institutions (e.g., community college), with ten already having obtained (2-year) associate degrees. On average, participants had taken 15.25 courses. Their current employment status also covered a wide range, from an undergraduate internship to more than 20 years in IT: 10 had full-time jobs at the time of the interview; one was self-employed, 2 part-time, one in an internship, and two job-hunting.

### 3.3.3 Study Procedure

Our (audio-recorded) semi-structured interviews were conducted remotely using the respondent's choice of communication medium. In one case the interview took place by telephone; in other cases I used video communication tools (Skype or Google Hangout). At the beginning of the interview, I asked participants to introduce themselves and reflect on their online education experiences with respect to their academic, career and social goals. I then probed in depth the students' experiences and felt connections with peers using three contrasting hypothetical three contrasting *scopes* for the felt community: project groups, classmates, and PSU as a whole.

After each interview, I reflected on points of interest and emerging themes, gradually refining my questioning to focus on social interaction in contexts that seemed to be yielding the richest reflections. Specifically, I added questions to probe the use of introductory forum posts to learn about their classmates at the start of a course. I stopped recruiting participants once I reached theory saturation and was no longer able to uncover new themes [Strauss & Corbin, 1998]. This happened after the 15$^{th}$

interviewee, yielding a sample size typical of remote interview studies [Caine, 2016]. Through the screening activity, I was able to construct a sample that contained rich variations of attributes such as age, gender, ethnicity, career stage, location, and family status.

### 3.3.4 Data Analysis

On average, the interviews lasted 61 minutes. The recordings were transcribed and examined through inductive thematic analysis [Braun & Clarke, 2006; Strauss & Corbin, 1998]. Specifically, I first used open coding to obtain categories emerging from the first few interview sessions. I organized these codes into a table of codes, with memos defining the codes, and sample quotes. I next work with two colleagues to discuss and refine the initial codes. After that, I carefully examined the codes to find semantic themes and to examine similarities and differences. This was followed by a pruning of any codes seen as irrelevant to my research questions. I applied axial coding to the remaining high-level themes to identify categories and relations among them. Finally, each theme was mapped into the subthemes that form my primary findings. In the following I present and discuss high-level theme, using sample comments as illustration.

### 3.3.5 Peer Relationships in Small Groups, Classes and the University

I first summarize how students responded to our questions about the three different scopes - project groups, an entire class, and Penn State in general. Following that, I discuss strategies or mechanisms that seem to promote feelings of connection at these various levels.

Not surprisingly, small groups (study groups or team projects) were the most tangible context for feeling connected to peers. All participants reported stronger feelings of connection when a course includes a group project as part of the assigned work. However, they also described feelings of connection at the level of a class and beyond.

#### 3.3.5.1 Group Members as (Almost) Friends

Working in a group was commonly described as *"how you get to know people"* (Ross). Four students (Ross, Elson, Gabriella, Judi) elaborated that group activities led to social interaction with their peers, enough so that these groups became the essence of the class: *"that (a group) is your class size right there, those three to five people whatever, tends to be your class size."* (Elson)

Some students reported that they had become "friends" with a few team project members, *"As we're doing group projects we don't only talk about schoolwork, we also talk about our personal lives, and our challenges"* (Dan). These friendships became evident when students felt free to reach out for help, relying on their shared interests and interpersonal knowledge, such as their peers' expertise domains and work ethics as part of a collaborative team. Over time, these "kind of friends" may offer other sorts of social support, such as help with course content, insights into professional practices and suggestions for future classes.

Even when group projects are not assigned in a class, some students seek a more intimate group structure to address the relative anonymity of an online class. They consider a small group as a source of help and information.

> *If I haven't had a group project in that class I've reached out to them and ask them, "hey I'm confused in this class, do you want to work together and try to learn together? I think it's easier if we can explain things to each other."* (Judi)

#### 3.3.5.2 Class Members Start to Seem Familiar

An online class can be a setting for interpersonal knowledge growth and nurturing of social ties. Typically, class-wide interaction consists of either synchronous sessions scheduled by the instructor (to answer questions about course content, grading, or other issues) or asynchronous discussion posts (for self-introductions and knowledge exchange). Through these interactions, students may at times notice names that are familiar from prior shared classes (this is particularly true for students pursuing the same degree). In addition, students who sign in personally for a class meeting or who make thoughtful posts make more of a personal impression.

> *We have been in the same cohort, and you see a lot of the same names over and over again because we're all on this major-related track We run into each other in the same classes.* (Becky)

When learners see a familiar name attached to a discussion post or other contribution, they may reply with something akin to *"Ohh hey, good to see you again"* (Becky). Dan said he is more willing to help a peer if he recognizes the name. It may be that the simple process of repeated exposure to student names in an online program is a way to create and reinforce social ties, wherein each viewing reinforces a feeling of connection [Stuart et al., 2012].

#### 3.3.5.3 University Identity is a Shared Identity

Feelings of connection to PSU were reported as diffuse and hard to describe, even though some students participate in online meet-ups of the online student club, or subscribe to notifications of the physical campus' activities. Similar trends have been reported for feelings of community in residential education, where students report a higher felt sense of community for classrooms than for the entire school [Rovai et al., 2004].

I found some evidence that connections with PSU may arise as feelings of *organizational commitment*, which relates to people's affinity to a group as a whole [Allen & Meyer, 1990]. Though they have no physical presence in campus-wide activities, the online students nonetheless said that they take pride in being part of PSU and feel connections to one another from their shared identity as PSU students.

I now turn to findings relating to mechanisms or strategies for building connections within groups defined by the three different scopes. I start with reflections about class-based connections, because these in a sense "set the scene" for more intimate small-group connections. I next discuss what seems to help or inhibit feelings of connections in these small groups, closing with the participants' reflections about connections that go beyond an individual online course.

### 3.3.6 Feeling Connected to Classmates

When students join an online class, they have little advance knowledge of classmates. They may be able to see a roster of other students' names, and may recognize students they have encountered before, but most of their connections emerge through activities that the class goes through together.

#### 3.3.6.1 Discovering Affinities in Introductory Posts

Most online classes begin with an activity in which students introduce themselves to peers. Students share basic information about themselves, such as hometown, family situation, profession, and perhaps a fun fact. These introductions are shared via different media platforms, for example VoiceThread, a cloud-based application that allows students to upload a short video or narrated PowerPoint presentation; other students can then comment. Howard said he uses these self-presentations to see where people have been *"because a lot of people put up vacation pictures or family pictures and I enjoy that."*

The information that helps students feel an affinity to a peer often relate to their own personal characteristics. For example, distance learners with a military background (e.g. Dan, Charles, Ross, Anne) said they are more likely to feel a connection with peers who also have military background. Judi saw herself as part of a minority group (females in Technology) and cited gender as a selection criterion for connecting with others. Note that most of distance learners share implicitly an interest in career advancement, which is what draws them to the online learning program. As a result, the learners are often seeking established professionals as sources of advice now or in the future.

#### 3.3.6.2 Live Sessions

Some participants reported that they build connections to their peers through live video or audio sessions, where students can pose or answer questions in real time, and connect with others at a more personal level. During a live session, the students felt that their peers were "present" with them. Howard liked these sessions because of real-time conversational nature of questions and responses. For him, getting an immediate response is better than having to *"send an email hoping I'll hear back today or tomorrow"*.

Reflecting more specifically on video conferencing, Yolanda said she felt more strongly connected to others when she can match a face with the person, rather than relying entirely on text conversations. Even when a student is not able to attend a live session, the recorded videos from class-gatherings evoked feelings of "being there": *"it's at least nice to sit there and hear it"* (Marshal). Jeremy said that live sessions made him feel like someone really cares:

> *[A live session] gives a more personable level. It's not like you're just reading a book, doing the content, or getting the grade. It's like a real person there hoping that you're well.* (Jeremy)

#### 3.3.6.3 Closeness after Facing Challenges Together

At times a shared struggle or challenge can create a feeling of connection. For example, several participants told me of rapport built during disruptions caused by teaching mishaps (e.g. an absent instructor), difficult content, or ambiguous instructions. Gabriella's Public Policy class created a Facebook discussion group to air confusions and exchange perspectives in a hard class. Likewise, Howard felt company when a teacher suddenly became severely sick increased the class' feeling of connection: *"We're all stunned, but there's a closeness in that at least we are all in this together. I am not suffering alone. I feel like that, we're veterans together: these people are suffering in the same way I suffer"*. While babysitting his sick daughter during the interview, Dan also cited a story to illustrate how supported he felt when he realized another peer student's similar situation: *"Sometimes she can't go to work because one of the kids is sick and she's trying to knock out the college before Sunday midnight. It really helps to know when things gets really stressful that there are other students out there that are having the same challenges as you are."*

Even when their life situations are not exactly the same, all distance learners share the challenge of balancing their course work with their many real-life roles. Seeing their peers succeed in handling these challenges helps them believe that they too can succeed: *"What stood out is that a lot of people have jobs, have kids, have to* *clean and cook and everything It's like they have to worry about so much more, and then they're in this class with me If they can do it, I can do it"* (Gabriela).

### 3.3.7 Building Connections in a Group

Group projects are common in PSU distance learning programs, and these are often the primary source of felt connections with peers. Because of the groupwork focus, students tended to view these sorts of peer relationships in an instrumental fashion; that is, they sought to anticipate and build connections that would help the group be successful.

#### 3.3.7.1 Vetting Candidates as Group Members

Some interviewees told me that they deliberately read their peers' introductory posts, so that they can assess a) how well they write (important for projects with substantial written work); b) how professional and responsible they appear to be (a predictor of team effort); and c) most importantly, evidence of some initial shared affinity: *"[if] you don't have something that connects you with each other, it makes it hard to bond as a team eventually; you're basically working with strangers."* (Dan) Kevin said he reads these posts to decide on whom to reach out to based on how similar he thinks that person is to himself in the introduction. As veterans, Charles, Ross and Anne looked for evidence of peers' military background, whereas Jeff, a retail store manager who hoped to move into IT said: *"Being able to find people in my area who have done the Mydegree and network with them is going to be the biggest benefit that I get out of Penn State."*

These vetting activities suggest at least two strategies. On one hand, students want to find group member with whom they can quickly and reliably connect, preferably someone with a similar background or similar professed interests. On the other, they are sensitive to characteristics predicting solid team contributions, hoping to connect with people who have these characteristics.

#### 3.3.7.2 Temporal Proximity

As a parallel to the well-known phenomenon of distance matters [Olson & Olson, 2000], seven participants mentioned time zones as a factor that influences their ability to connect with group members. For example, Elson explicitly called out the value of temporal proximity over physical proximity: *" [Teammates] don't necessarily have to be in CampusState but they have to be on the East coast so that there's no weird scheduling conflicts."* (Elson) Becky elaborated further that everyone can then (with the same time zone) be *"working at the same time"* when crunch time approaches.

Mismatches in preferred study time can work against efforts to connect with online peers. Gabriella was disappointed with one team collaboration, attributing the problems to being out of sync with her teammates: *"I've opted out of being in a group because I do my homework on the weekends and my group did this Monday morning. In the mornings I'm busy so I couldn't keep up with there."* This comment suggests that distance learners' feelings of connection with their peers might be mediated not only by geographic proximity (and corresponding time zones), but also to the days and times that these largely adult learners have set aside for study. Such scheduling constraints may affect how quickly a peer is able to respond, including times at which they are simply not available (e.g., work shifts in real life).

#### 3.3.7.3 Managing the Group's Common Ground

A group's early interactions establish a common ground that members use in managing expectations about each other. If successful in this grounding phase, the group can enjoy shared understanding, empathy and pleasurable interactions, leading to. As *"better product and grade"* and *"better feelings about working together"* (Becky). In contrast, resentfulness can build if a team member fails to contribute to the social fabric or does not carry their own weight in a project.

Beyond setting up appropriate expectations, some interviewees echoed Clark's theory that these shared understandings must be maintained with continuous effort [Clark & Brennan, 1991]. They were willing to accept an absence or missed deliverable from a peer if it had been announced ahead of time and if he/she had previously demonstrated a cooperative work ethic. Staying aware of team members' situations implies that important constraints or issues are shared and respected:

> *We try to keep abreast of what's going on in everybody's life, just so we know what to expect of them. We try to be understanding if something is going on that week that's going to prevent them from doing something, and try to be helpful in picking up the slack where we need to.* (Becky)

#### 3.3.7.4 Enjoying Purely Social Interaction

Judi noted that when a group schedules an audio or video call with a detailed agenda, the formality of the *"meeting"* can inhibit the casual and spontaneous interaction that often occurs in residential education (e.g. grabbing lunch). However, other interviewess told me that casual conversations do take place as a side discussion to online meetings, helping to *"lighten up the mood of the meeting"* (Jeremy). These low-stakes interactions may occur before everyone arrives, or at the end of a meeting:

> *When we'd have our group meetings usually we'll spend the first few minutes just sort of chitchat and see how everybody's doing, a little bit about the weather, sports, get to know each other.* (Rafael)

Another type of incidental social connection can occur when video meeting reveals an attendee's surrounding; this personal information may prompt feelings of closeness in surprising ways. For example, group members might see a child's birthday party in the background, or a baby held in the arms. As a single dad of a young kid, Dan felt empathy for a teammate who was holding baby during a live session:

> *The mother held the kid in the camera so we all got to see the baby too. We got to hear, like you're hearing right now my daughter just talking, that develops a little bit of closeness too.* (Dan)

#### 3.3.7.5 Having a Persistent Online Space for Group Work

To support project work, distance learners often create *persistent online spaces* outside of the online platform (i.e. Canvas) provided by the University (of course, this is also true for residential project groups, but distance learners have fewer options).

For communication, they used instant CMC tools such as GroupMe, Google Hangout or Skype; for collaborative work, they used Google Docs and GitHub to share and work on the same artifact.

One consequence of setting up a persistent space is a feeling of regular and consistent availability for collaboration and mutual support by group members. The interviewees told me that these spaces led to closer feelings of connections within their study groups. Judi, Gabriella, and Elson felt comfortable with dropping ad hoc questions about course content or assignments in their Google Hangout; they believed that these spaces produced a rapid response time:

> *It's a study group, so if you have questions they're always there, we've always used instant messaging like Google hangouts. Since that's always on if you have a question just drop it in there somebody is going to get back to you.* (Gabriella)

### 3.3.8 Connecting to Peers Beyond a Class

As I reported earlier, feelings of connectedness to peers beyond a project group or class were vague. But in some cases, there were rather specific and instrumental approaches to these diffuse feelings of connection. In other cases, the feelings seemed tied to the simpler notion of shared identity.

#### 3.3.8.1 Familiar Names in New Settings

Just as in residential education, distance learners may overlap with each other across classes, potentially bootstrapping peer connections, and easing the overhead of coordination during the course. For example, *"I have four group projects, and three of my groups have two of the same teammates. We're doing all of these projects together, and we know what each other's responsibilities are. We know the work load we're all carrying"* (Becky).

However, when asked whether they explicitly plan for overlapping groups across semesters, none of the students reported this as a deliberate strategy. Instead they said that they simply spot previous teammates when new classes start. As a comparison, Elson noted that it was difficult to reinforce connections when he knew that previous teammates were enrolled in the same class but a different section.

#### 3.3.8.2 Planning for Future Interactions

Another way in which peer connections were sustained beyond a course was to gather contact information at the end of a group project. As part of their group experience, students often rely on tools such as Google Hangout or Skype; they may also exchange personal contact information. These project-focused interactions may evolve into relationships that are maintained afterwards using general social media tools (e.g., LinkedIn).

> *Instead of just working on a project together and then just saying bye-bye at the end of it, I actually want to maintain communication and get their contact information, maybe add them on Linkedin and keep connected.* (Kevin)

Participants may be especially likely to invest effort in maintaining teamwork ties when they are entering a new field together. For example, Dan tried to stay in touch with teammates who evinced a strong work ethic in a particular shared project:

> *We've all agreed there's an importance in networking with each other. I'm staying in touch with them and we're keeping tabs to see how well we're doing with our online endeavors.* (Dan)

The shift from project-based communication to more general social media ties echoes observations about residential education, where the maintenance of course-based connections seems to depend on subsequent social media contacts [Barkhuus & Tashiro, 2010; Lampe et al., 2006]. However, these contacts may fade into inactivity even when friendships with other teammates grow and are maintained in real life. With regret, Becky, Judi, Ross, and Elson noted that they did not interact with their former team members, *"probably 95% of them I'll never speak to them again after the project."* (Judi)

#### 3.3.8.3 Celebrating a Shared Identity

As mentioned earlier, many students mentioned things that they do to reinforce their connection to PSU. Such connections are difficult to describe, because they are not ties to individuals or even to a group, but rather to the concept (campus) of the university in general. As an example more specific to distance education, Kevin said he feels connected with peers in PSU's online programs because they share the goal of distance education:

> *They've chosen Penn State for a reason. It's probably the same reason that I chose it. I know most of them are busy. Most are working adults.* (Kevin)

Individually, many of our interviewees also report that they collect PSU-related souvenirs and memorabilia, such as sweatshirts, car stickers, or previous student IDs. They also report that they they watch PSU sports events, even though they are not able to do in person together. Charles reported that he was once teased about how devoted he is to collecting everything about PSU:

> *It's just like, "If Penn State makes it, Charles probably has it." I'm like, "No, I don't. I don't have all of it." I like to show it a lot.* (Charles)

### 3.3.9 Summary

Looking across the three scopes for reflections about peer connections (small group, classes, virtual campus), I can see two general bases for felt connections. First, some early, relatively lightweight connections are formed through shared social identity. Distance learners seem to begin with an implicit bond that emanates from the challenging situation of being an adult learner who is juggling an active real world life with current education goals. In other cases, it is personal characteristics (e.g., former military, female in IT) that suggest affinities.

Second, some peer connections are formed and nurtured through steps that learners take to vet and invite collaborators of various sorts, including team members whose schedule and work ethic is a good match to theirs. These connections are strengthened through the processes of group collaboration, wherein members learn more about each others' expectations and work habits. This more instrumental connecting process can even extend beyond the class, as prior teammates may be recognized and recruited to new projects, or individuals with special characteristics may be cultivated as potential lifelong professional contacts. This type of connection can be seen as a natural side effect of organizing or carrying out collaborative activities.

## 3.4 What Information do Learners Need to Connect?

### 3.4.1 Overview

The interview studies with instructors and students revealed a rich picture of whether and how students might feel connected to one another. But prior to initiating the design of tools that might promote visibility of peer information (e.g., so as to promote initial feelings of affinity or set up the instrumental step of reaching out), I wanted to assess what kinds of information students thought would be most critical to promoting feelings of community. I chose to explore this using survey methods that I implemented in two phases [Learning or lurking? Tracking the "invisible" online student, Sun; Sun & Rosson, 2017b]. In the first small-N study I explore the information that might be useful; in a larger validation study I confirmed the initial results. The survey studies were guided by these exploratory research questions:

- *What types of peer information do distance learners believe would be important in building feelings of connection?*
- *How are these different types of information best organized and characterized as valid constructs?*

### 3.4.2 Preliminary Information Analysis

I was intrigued by learners' stories about how identity information invited social outreach to online peers (e.g. gender, military background, family situation). However, given the wide variety in students' and instructors views, I wanted a more systematic and structured view of the information that online learners might like to know about each other. I did this by creating an online survey distributed to the same general pool of students. The survey asked them to rate 19 information types from 1 (*Not at all important*) to 7 (*Extremely important*) in response to the following probe: *"Suppose you are able to view information about the other people enrolled in this class (e.g., as part of a profile description). Please indicate for each information type how important that information would be in creating a feeling of community within your class."*

I selected the 19 information types from comments made by our interviewees and from research on social identity formation and social presence [Hogg, 2006; Ke et al., 2011; Turner & Haslam, 2001]. The set included demographic characteristics (e.g. age, gender, job), online education experiences (e.g., courses, instructors, program); items that might help to recognize suitable collaborators (e.g., time zone, schedule, expertise, personality); and the more sensitive identity information that might imply differences in culture or value systems (e.g., nationality, political views, religion).

I received 59 responses to the survey (a response rate of 21.4%) and treated the rating scales as ordinal approximations of a continuous variable [Johnson & Creech, 1983]. As seen in Table 3.3, the mean ratings across information types varied: the highest rating was for Schedule ($M$= 5.5, $SD$ = 1.7); the lowest was Political Views ($M$ = 1.6, $SD$ = 1.3). Other highly valued information included Time Zone ($M$ = 5.4, $SD$ = 1.6) and Areas of Expertise ($M$ = 4.6, $SD$ = 1.6). This suggests that on average, students may value information relating to the logistics of collaboration (e.g., availability, proficiency in a topic) more than characteristics defining personal or cultural identity (e.g., politics, religion). This was somewhat surprising given the identity-related stories from the interviews, but consistent with the general emphasis on time as an obstacle.

To uncover the internal structure of distance learners' peer information preferences, I carried out an exploratory factor analysis on the 19 items. I found four factors that accounted for 69% of the variance in the ratings. After a Varimax rotation to enhance interpretability, I labeled the factors (Table 3.3 as: Avaiability; Learning Background; Profession; and Personal Details. (the Personality rating scale was dropped as a factor because it did not load strongly on any of the four factors). After confirming internal reliability (Cronbach coefficient $\alpha$ levels > .75), I averaged the item ratings under each factor to create four constructs. The table shows factor loadings for individual items, with means and standard deviations; the constructs are ordered according to average rating.

**Table 3.3** Exploratory Factor Analysis of 19 Peer Information Types

| |
|---|
| **Availability; $M$ = 5.50, $SD$ = 1.50; $\alpha$ = .791** |
| Time zone (M=5.43, SD=1.62, loading = .891) |
| Schedule for online work ($M$ = 5.54, $SD$ = 1.66, loading = .728) |
| **Learning Background; $M$ = 3.80, $SD$ = 1.51; $\alpha$ = .872** |
| Previous online courses ($M$ = 3.36, $SD$ = 1.75, loading = .894) |
| How long online student ($M$ = 3.46, $SD$ = 1.75, loading = .754) |
| Career goals ($M$ = 3.84, $SD$ = 1.79, loading = .711) |
| Previous online teachers ($M$ = 2.44, $SD$ = 1.67, loading = .710) |
| Major/level at university ($M$ = 3.56, $SD$ = 1.90, loading = .630) |
| **Profession; $M$ = 3.34, $SD$ = 1.44; $\alpha$ = .810** |
| Occupation ($M$ = 3.81, $SD$ = 1.93, loading = .752) |
| Military service ($M$ = 2.90, $SD$ = 1.82, loading = .745) |
| Areas of expertise ($M$ = 4.68, $SD$ = 1.59, loading = .618) |
| **Personal Details; $M$ = 2.17, $SD$ = 1.21; $\alpha$ = .881** |
| Nationality ($M$ = 2.05, $SD$ = 1.48, loading = .858) |
| Approximate age ($M$ = 2.31, $SD$ = 1.60, loading = .721) |
| Gender ($M$ = 1.97, $SD$ = 1.59, loading=.717) |
| Political views ($M$ = 1.62, $SD$ = 1.28, loading = .716) |
| Religion ($M$ = 1.76, $SD$ = 1.57, loading = .687) |
| Marital status ($M$ = 2.20, $SD$ = 1.64, loading = .648) |
| Number of children ($M$ = 2.41, $SD$ = 1.67, loading = .618) |
| Hobbies ($M$ = 2.95, $SD$ = 1.60, loading = .605) |

*Availability* has the highest mean ($M$ = 5.50, $SD$ = 1.50; $\alpha$ = .791), suggesting that time-related issues may be an important practical concern when connecting with peers for adult learners [Lundberg, 2003]. Interestingly, the construct is comprised of just two items, a person's time zone (note that this also implies some degree of shared region) and the days and times that students schedule for their online learning activities. The relative salience of this construct is also consistent with the repeated mention of time as a challenge in the Stage 1 interviews.

The *Learning Background* factor ($M$ = 3.80, $SD$ = 1.51; $\alpha$ =.872) integrates items relating to a student's online education (e.g., prior experience with courses and instructors, degree program, and career visions that motivate their online education).

*Profession* ($M$ = 3.34, $SD$ = 1.44; $\alpha$ = .810) captures information about specific expertise a student might bring to online interactions (e.g., occupation, military) and is reminiscent of interview comments from students in the military who were careful to notice others in a similar situation.

*Personal Details* ($M$ = 2.17, $SD$ = 1.21; $\alpha$ = .881) captures typical demographic items (e.g., age, gender), along with attributes that often reflect shared values (e.g., nationality, hobbies). It also includes personal aspects of familial situations (marriage, children). This category was consistent with what some of our interviewees' said they did when they had nothing else to go on, or when the characteristics (e.g., childcare) were especially salient. Note that several items in this last group are often seen as sensitive or private data (e.g., politics, religion). It may be that a sort of "political correctness" had a dampening effect on ratings for these personal attributes.

Summing up, the importance ratings and subsequent factor analysis suggested that I consider further how these four constructs of Availability, Profession, Learning Background and Personal Details might contribute to the creation of felt community. Note that that the survey was relatively small, and the factor analysis was of an exploratory nature only; therefore, I developed and deployed a larger survey to validate the stability and generalizability of these constructs.

### 3.4.3 Validating Peer Information Needs

In a much larger survey (n=740) [Learning or lurking? Tracking the "invisible" online student, Sun] distributed across the entire World Campus student population, I again asked about perceived importance for forming connections to others. However, in this case I asked directly about the four constructs discovered in the exploratory study.

I found a similar trend in responses, with Availability again rated most highly. A repeated measures ANOVA (with Greenhouse-Geisser correction for sphericity) revealed a significant difference in rated importance of the four types of information ($F(2.60, 460.42) = 73.443$, $p < .001$). A post hoc test with Bonferroni correction revealed that each pair of the means is significantly different except for the contrast between Profession and Learning Background. Participants rated other members' Availability information as most important in creating feelings of community ($M = 4.67$, $SD = 1.92$), followed by Profession ($M = 4.14$, $SD = 1.79$), Learning Background ($M = 3.97$, $SD = 1.76$), and Personal Details ($M = 2.70$, $SD = 1.71$). In addition, on a five-point scale from "Not at all interested" to "Extremely interested," participants reported that they were more than somewhat interested ($M = 3.50$, $SD = 1.25$) in response to "to what extent are you interested in feeling more connected within the community." In particular, 17.80% participants viewed availability information as "Fairly important," 21.87% as "Pretty important" (6), and 20.21% "Extremely important" (7). In sum, a majority of the 740 survey respondents (59.88%) viewed Availability as an important factor for their social and learning experiences in distance education, motivating us to look more in depth at this aspect of peer information.

### 3.4.4 Summary

As shown by these two survey studies, the WC learners were interested in peer information that included Availability, Profession, Learning Background and Personal Details. Peer availability was viewed as most important in creating feelings of community, but information about a peer's learning background and profession are also viewed as moderately important in feeling connected.

## 3.5 Why Does Availability Matter?

To understand why availability is seen as important with respect to online students' feelings of community in both surveys (in Section 3.4 and 4), I conducted an in-depth interview study. I sought to articulate different aspects of availability as a concept, and to understand why these aspects may be important for distance learning.

### 3.5.1 Method

Driven by the purpose of uncovering the meanings and values behind Availability, I decided to use a combination of purposive and convenience sampling [Merriam, 2002]. Specifically, I used the *Criterion-i* type of sampling strategy to identify and select all cases that meet some predetermined criterion of importance (in this case, high perceived importance of peers' availability) [Palinkas et al., 2016]. I reached out to students who had completed a survey (n = 740) aimed at assessing community perceptions in a online degree-seeking education programs [Learning or lurking? Tracking the "invisible" online student, Sun]. Specifically, I focused on student population who showed strong interests in viewing others' availability (that is, rating the importance of peer availability information as *"Pretty important" = 6* or *"Extremely important" = 7*[2] for connecting with others) in that survey. Although I recognize that this creates a highly self-selected group of individuals (i.e., previously willing to respond to a survey invitation, and from that set, those who felt strongly about availability), it is a useful approach for the *"deep dive"* into why availability matters. Our research goal was to explore the meaning and possible impacts of peer availability, not to test or generalize differences among varying types of online students.

#### 3.5.1.1 Participants

At the time of the interviews, all participants were enrolled in an online bachelor or master degree program, and I sought to recruit students who were enrolled in different programs (both graduate and undergraduate) across MyUni to maximize variation in our sample.

In the end, I interviewed 10 students; all are American living in different states, but some also live in different time zones : one works in Sydney, Australia most of the time, and others sometimes travel while completing their courses. Eight have full-time employment (working 40 hours or more per week); the other are working

[2] The availability item is a seven-point Likert item, as opposed to the Likert-scale described in Section 3.4.

**Table 3.4** Interviewees' demographics and background

| Pseudonym | Age | Enrolled education program |
|---|---|---|
| Jake | 47 | M.S. in Education |
| Louise | 49 | B.A. in Liberal Arts |
| Amy | 36 | B.S. in Public administration |
| Sophia | 29 | B.S. in Psychology |
| Adam | 25 | M.S. in Homeland Security |
| Rachel | 32 | M.S. in Human Resource |
| Kathy | 31 | M.S. in Statistics |
| Emma | 40 | B.A. in Liberal Art |
| Noah | 31 | B.S. in Accounting |
| Matt | 47 | B.S. in Information Science |

at a non-profit for 35 - 38 hours per week. Nine are Caucasian and one is African American. They come from a variety of educational background in terms of degree progress and major, with an average age of 36.7 (Table 3.4 provides individual details and the pseudonyms I use in our Findings).

#### 3.5.1.2 Interview protocol and analysis

I began each interview by introducing the study context, explaining how the interview is a further exploration of the previous survey outcome (I did not tell them why I selected them, but during the interview all confirmed that they felt availability was critical for feeling connections to other distance students). I then followed a semi-structured script, probing their perceptions of availability as it applies to their connections with online peers, and in particular how it might influence their shared learning and social interaction I asked in detail about their practices for knowing others' availability, including their rationale for attending to certain sorts of information. Finally, I asked them to describe their general practices and how they employ technology for sharing or discovering availability information.

All the recorded interview sessions were transcribed and analyzed using inductive thematic analysis [Braun & Clarke, 2006; Strauss & Corbin, 1998]. During the data collection, two researchers took notes and discussed emerging themes after each interview. In the meantime, the transcribed data was imported into MAXQDA software, where I applied open coding and axial coding to understand the data. Once I found no more new themes, I concluded that I had reached the point of theoretical saturation and stopped recruiting students (n = 10). On average, each interview lasted about 57.5 minutes. I turn now to a discussion of themes that I extracted from the interviews.

## 3.5.2 Availability Meanings for Online Learners

### 3.5.2.1 Temporal Availability: When is a Peer Available

Not surprisingly, an important sense of availability concerns the actual temporal alignment that online learners experience with their peers. More specifically, Ifound two subthemes that reflect two different aspects of temporal availabilty, both grounded in the learners' desire for real-time or at least rapid responses in interactions with online peers.

First, *I need someone to be present at the moment when I am online.* Sometimes, being available "right here, right now" for peer learners allows for instrumental reciprocity in online coursework. For instance, Sophia hoped to *"bounce an idea off a peer regarding an assignment or syllabus even if it's not a group setting"* (Sophia). As most of online class activities take place asynchronously, our participants reported hurdles in using the current LMS (Canvas) to find peers who might be available *on demand.* Adam found it hard to infer if someone is present online using forum posts in Canvas: *"you see that the person made a post to the discussion board one minute ago. But that is not to say that they didn't make that post and logged back off."*. In the meantime, some participants described workarounds using other tools. For example, Kathy found Google Hangout sessions helpful for her to receive real-time feedback, because *"anyone that's interested can drop in."* In this way, she would feel some degree of certainty that it was worth reaching out to people. Emma also appreciated peer discussions in Google Hangout sessions to clear up confusion over learning material, and she especially appreciated the real-time interaction as a way to enhance feelings of connectedness:

> *"going into Google Hangouts and typing to everybody is much more easier*

*and animated. You get to meet the people a little bit more than you do in the forums."*

It may be that immediate presence is sufficient in these cases to invite further social behavior; although friendship bonds among distance learners are difficult to build [Learning or lurking? Tracking the "invisible" online student, Sun], just being online at the right time may be a reasonable threshold for making a personal connection.

Second, *I need someone to be present during designated time slots*. Not surprisingly, as several participants argued, concurrent availability is not, and *should not* be, necessary for interacting with peers in the context of distance education. Distance learners often choose to pursue online degrees explicitly because of the herein flexible schedule. As Amy argued, *"it doesn't need to be available when I'm available. I don't think we need to have concurrent availability. It is just a need to know when someone will be available."* Because members of an online learning team (e.g., a group project) cannot always count on their members to be around when needed, they may schedule a time-slot when everyone agrees to join at the same time. In fact, all but one participant has used such group-level synchronous sessions to coordinate with others. Arranging a scheduled session avoids the problem of unavailability for ad hoc synchronous interaction, but these can still be problematic if students' general availability does not line up.

A scheduled synchronous interaction can be supported in various ways, ranging from text-based chat to video or audio conference calls. Jake and Adam prefer to rely on *"a conference, video call or Skype session,"* because these are effective in creating feelings of connection with others. However, Inote that video conferences require distance learners to be using a relatively high level of communication and network resources; this could exclude those with older or slower tools from participating (e.g., a slow Internet connection). For instance, Jake recalled how a peer without sufficient technical support was left behind during a synchronous video-conference, *"we tried to include her, but she would write these long explanations and post them. In the meantime, Iwere talking and moving on to different things. A lot of times, we would miss what she was saying."*

Sophia described how her classmates agreed on synchronous start times to collaborate using text-based communication, *"We set a Slack group where we said every Saturday at 8:15 in the morning, we are going to all sign on. And through that, we worked through the different milestones of our project."* Rachel also thought highly of setting aside a dedicated time frame for working together:

> *We don't have to worry about interrupting someone at work...and when it's in a chat room, it's easier to get that real-time information from your teammates. It is like that you are at a actual classroom environment when everyone has the same availability.* (Rachel)

#### 3.5.2.2 Social Meta-data and Expertise Availability: Who is Available

Sometimes just knowing that someone is around is not enough to see them as available for interaction. Instead, meeting and learning about a peer (often as a result of group work) makes people feel more connected and more willing to reach out for help. Louise found herself reluctant to interact despite knowing about a peer's temporal availability, raising the importance of other interpersonal knowledge regarding other learners' identity: *"it's just very hard when you can't see their faces. You don't really know who they are and you don't know where they're coming from."* Likewise, participants expressed discomfort in reaching out to peers if they had no prior social interaction or personal knowledge, e.g. *"I have limited interaction with other students for me to know who is around, because I would not feel comfortable reaching out to them because I don't know them."* (Sophia) Reflecting on his most successful group experience in the program, Noah believed that *"it is because we take the time out of our lives to really put forward the effort to get to know each other. Like text each other and call each other on the phone once a week to touch base."* Such conditions of personal knowledge about their peers to be "actually available" resonate with the Social Transparency framework in stressing out the importance of social-meta data surrounding information exchange [Stuart et al., 2012].

When motivated to reach out to others for instrumental purposes, our participants emphasized the availability of expertise, which in some way predicts the quality of the help s/he might receive. For example Sophia indicated that she might not want to ask a random classmate a question, *"because everyone is a student and I* *don't have that prior face-to-face interaction or background to know if this person is responsible, studious, or knows what they are talking about. So if they give me the wrong information, I don't have anything to base it on."* Hence, she feels most comfortable reaching out to group members who had previously exchanged emails or ideas with her.

As a more specific example of quality concerns regarding who to contact, Kathy notes that she is a straight-A student and wants to interact with peers who can match or raise her level of performance: *"Picking someone who is intellectually matched to me is challenging to tell in the first few weeks of the class. So I feel only after the class is finished do I see 'ohh this person offers helpful advice, and this person is not asking lots of questions.' "* This suggests that for some students, knowing which peers they should consider as available for help may not be possible until after sufficient social interaction or collaborative learning activities have taken place.

### 3.5.3 Coping Strategies for Assessing Peer Availability

#### 3.5.3.1 Finding Immediately Available Peers with Online Presence Indicators

The simplest approach to finding out who is available now is transparent communication system through which the distance students can perceive peers' current logged-in status, or find out when a person was last logged in Rachel used the example of Microsoft Link, a tool she uses at work to illustrate what is needed *"a technology enhancement to make it faster to connect with others."* As I have noted, these students use a range of similar tools that have these features, including Google Hangout, Skype, Facebook Messenger, and Google Docs.

These tools all have presence indicators, such as a green icon to show current status, or some time indicator that shows recent logged-in time. This allows peers to estimate whether and when others are online. Our earlier example of Adam who accommodated his peers' normal working rhythm depends on having real-time and historic information such as this. A presence indicator invites outreach by the viewer at an opportune time, and contributes directly to feelings of connection in the moment.

Interestingly, Matt noted a downside of using these communication tools as part of his distance learning (in his case, Facebook Messenger). He was concerned about potential interruptions during work or personal life; he prefers to keep learning-activity-related messages separate from other activities: *"but I hate it because you can't, it doesn't want you to turn off notices. It wants to ding on your phone every bloody time someone sends a message, which is crazy, drives me nuts."* This concern about interruptions, and more generally about the complexity of managing notifications (i.e., so that you only get the ones you really want!), is well known as a pervasive and challenging problem for collaborative systems [Adamczyk & Bailey, 2004; Tasse et al., 2016].

#### 3.5.3.2 Spending Time to Arrange Time

Communications aimed at establishing availability usually take place asynchronously over email or in Canvas discussion forums. Without prompt answers or in-situ responses, availability negotiations take effort, with the process consuming its own time resources and patience for completing a collaborative effort: *"We used the forums for MyUni's website, but it took a long time to get everybody to respond. I think it took almost a week to coordinate."* (Emma)

Without face-to-face class sessions or instructors' specification, the ability to schedule a synchronous session, or to create a timeline for assigned work, often depends on initiative taken by a group leader. Rachel, Matt, Emma, Sophia and Jake all shared the practice of using emails or forum posts to exchange everyone's availability based on a "rough" template:

> *Ask about your name, the days and time you are available, your handles for all of your apps, like Google Hangout names, Facebook Messenger, or Slack; what is your typical turnaround time. If you are sending asynchronously, then I know when to expect an answer from you.* (Sophia)

Without guidance from an instructor or group leader, the negotiation processes are unstructured. Matt complained that his meeting requests sometimes got neglected because other members would disagree with the need to meet until the last minute, even though he felt that it was crucial to do so. Rachel noted that a leader needs to manage scheduling and meetings as an extended process, because people tend not to want to engage until they know it is important, *"It is never set up at the beginning of the semester. Probably because people's general, innate fear people have of meeting new people, it doesn't happen until it has to."*

Scheduling conversations are examples of *articulation work*, the work that must be done in the background to allow productive collaborative work to take place especially when online learners might need to deal with varied priorities and multitasking [Su & Mark, 2008]. Opening an online space and seeing indicators of who's online can be a quick technique for judging immediate availability. But to schedule an agreed timeslot, whether for a one-time meeting or a regular check-in, specialized tools like Doodle may be needed, or collaborators may need to spend considerable back-and-forth in emails to choose a convenient time.

#### 3.5.3.3 Sharing Work Schedules to Cope with Time Management

Time management is a critical skill for online learners because they normally must juggle multiple obligations associated with different real world roles. For example, Rachel has a full-time job and young children, and thus must balance work with family and studying. As a result, her availability to peers was dependent on her kids' bedtimes. Interviewees described tensions that can arise between professional and personal activities and their involvement in collaborative learning. For instance, both Matt and Noah described their working schedule as "weird" or "odd." Matt works until midnight local time; Noah works 80 hours one week and has the next week fully available (works 0 hours) to interact with peers. These atypical work schedules are difficult even for same-time-zone peers to anticipate, making them more important to share in advance.

> *I have to be very very diligent with my time management skills when I'm at work, because I don't get a whole lot of time. It is solely on all my coursework so it's very trying on a person* (Noah)

When a team member does not show up for a meeting, or fails to contribute an expected work product (even if they submit this later), it may create bad feelings. As Rachel says, *"it comes across that person doesn't care enough to join at this time, or* *this person's work is not up to the standards of anyone else. Because we've done all this hard work, and they just popped on at midnight or 1am, go over everything and add one sentence or two to all the stuff that is already there."* Such types of unequal or late contribution in collaborative work often leads to resentment, pressure and breakdown of social connections; unfortunately, the phenomenon of unequal share of teamwork is somewhat associated with the challenges in contributing in the "right" time, which serves a key factor in determine the value of the contributed share of work, yet such constraints are the very reasons for online learners to choose online education in the first place [Kizilcec & Halawa, 2015].

When students study together remotely and rely on each other's contributions to earn course grades, one collaborator's unavailability could be the others' nightmare. This points to the importance of timely (non)availability notifications, which of course can be quite difficult in the case of unexpected emergencies. As Noah put it, *"I would love to know how available everybody is that I interact with. It helps me plan my day out if we have any coursework that we need to complete together. that's a big thing."* Adam gave a similar example regarding personal details about a team member:

> *Her schedule was gonna be a lot less flexible being a pregnant mom, having to go to different visitations along those lines. Just knowing that information like that upfront, rather than when it's too late to have to scramble. Maybe reassigning something, or rescheduling a call or something will be beneficial.* (Adam)

#### 3.5.3.4 Coordinating Time Zones

To avoid scheduling conflicts, all participants emphasized that their learning is taking place in a global environment, and some considered *time zone* to be a key constraint in team collaborations. They preferred to work with peers in the same time zone. This can be seen as an important piece of common ground, in that it implies a peer's general schedule will be similar to one's own, with respect to waking, meals, work, leisure time, and sleep. Interestingly, a shared time zone may also imply social elements of common ground, for example the normative culture or other characteristics of a shared geographic region. However, learners' individual work schedules and other personal commitments also constrain collaborative work. Most distance learners, especially adult learners, have off-line duties such as full-time employment. For instance, Rachel found it inconvenient to wait for a teammate's response from a different time zone; and Adam complained about missing the opportunity to influence a group-level decision making process, citing personal travel:

> *I am not receiving the notification till the next day...My input might be too late at that time because ultimately a decision would have had to be made before I could respond.*

### 3.5.4 Summary

In triangulation with the survey studies, I carried out a small follow-up interview study with students who had completed the second survey and had indicated a particular interest in Availability of peers. In this study – and in combination with the other interview studies – *time emerged in my dissertation work as a critical factor in feelings of connection.* I concluded that it should be a major component of visualizations designed to build community.

However, I also recognize that my follow-up study was limited in that I intentionally interviewed students who view availability as important so as to take a close look at the reasons and practices surrounding the phenomenon. Future work should look more broadly at students holding different value positions, and further investigate the causal impacts of peer availability. In addition, other factors, such as Learning Background and Profession, despite being significantly lower than Availability, also warrant further investigations in their roles to promote an online community of learners. Given these conclusions I moved on to explore possible design interventions, by gathering and visualizing availability and other peer information later (Chapter 5).

# Chapter 4
# Measuring Community Among Online Students

Research questions about community among online learners are gaining importance as enrollments in online programs explode. However, *what community means* for this context has not been studied in a comprehensive way. To support the evaluation of my design experiments, I needed an instrument that would yield quantitative measures of learners' feelings and behavior expectations about online community. To some extent this was achievable through adaptation of existing scales (e.g., Sense of Community or SOC [Rovai et al., 2004]); however I also wanted a measure of the felt community's capacity for managing itself – its community collective efficacy or CCE [Carroll et al., 2005]. This aspect of my dissertation was guided by the following methodological research questions [Learning or lurking? Tracking the "invisible" online student, Sun]:

- *How can Sense of Community be used to provide a useful measure of felt community for online students?*
- *What are the collective capacities of an online student community and how can they be assessed?*

In this chapter I elaborate on my measurement goals, methods and progress toward developing such measures of community for online students. I also discuss the implications of the measurements with respect to the student population used for instrument development. My analysis of students' responses to these scales revealed two factors that underlie SOC (shared identity and interpersonal friendship) and three factors that underlie CCE (identity regulation, coordination and social support). I used these factors to discuss contrasting definitions of community (shared identity versus ego networks). Exploratory data analyses also revealed relationships to other student variables that begin to articulate roles and mechanisms for online students' felt community, and raise design implications about what we might do with and for the community structure.

## 4.1 Developing Community Scales

Because one of my research goals is to provide a set of measures to characterize felt community among large populations of online learners, I developed a survey that leveraged my own and others' prior research and administered it to a large and diverse population of WC students. After analyzing and validating the internal structure of the community scales, I conducted exploratory regression analyses to investigate relationships involving the new constructs, with the intent to use such relationships to guide tool design.

### 4.1.1 Survey Design

#### 4.1.1.1 Sense of Community

One source for instrument development was the well-established Brief Sense of Community (BSCS) scale [Peterson et al., 2008]; this scale contains four constructs of the SOC model: needs fulfillment, group membership, emotional connection and influence [McMillan & Chavis, 1986]. For example, needs fulfillment is assessed with "The community helps me fulfill my needs" in a five-point Likert scale; group membership with "I belong in the World Campus community"; and emotional connection with "I have a good bond with others in this community". I selected six items from the BSCS instrument, dropping items about influence.

The influence construct[1] has a slightly different sense when considering learning

[1](The items for this construct would be *"I have a say about what goes on in my community"*,

communities that are hosted by a university, versus other communities that are self-governed. Thus we replaced the BSCS item assessing influence ("I have a say about what is going on in World Campus community") with an item from Rovai's Classroom and School Community Inventory scale (CSCI) that was developed specifically to address community in school settings ("I feel that I matter to other World Campus students") [Rovai et al., 2004]. To provide more coverage of the social connections learners might feel, we also added two CSCI items that assess the social aspect of school-based communities (e.g., "I have friends at World Campus to whom I can tell anything") and another CSCI item that probes feelings of trust ("I feel that I can rely on others in World Campus". The resulting 10-item scale is available by request; its psychometric properties are summarized in the Results section.

#### 4.1.1.2 Community Collective Efficacy

Collective efficacy is defined as *"a group's shared belief in its conjoint capabilities to organize and execute the courses of action required to produce given levels of attainments"* [Bandura, 1997, 477]. To develop a scale for community collective efficacy (CCE) in the domain of online learning, I began by doing a *capacity analysis* to consider what might be the responsibilities or "tasks" of online learning communities; this grounded the conceptual analysis needed to formulate collective efficacy probes. At the same time, I drew from existing research. For example, an earlier interview study of online learners found that they characterize their community as having commitment to a common purpose, adoption of normative standards of behaviors and emergence of hierarchy and roles [Haythornthwaite et al., 2000]. In my own study, online instructors shared views of community indicators among their distance students, including real-time engagement, repeated collaboration across classes (e.g. forming a team with a known school-mate), mutual support during "tough times", institutional pride, and membership in online student organizations [Sun & Rosson, 2017d].

With this prior work as grounding, I formulated efficacy items to probe online community capacities. Following Bandura's guidelines [Bandura, 1997], an important step was to identify meaningful challenges for success with a given collective

*"People in this community are good at influencing each other."*)

capability. I reviewed literature to identify challenges and obstacles for community development in online learning environments. For example, Brown attributes students' hesitance in forming community to low expectations for the presence or value of an online community, the reluctance of committing more than required work, and an assumption that community only grows through face-to-face or direct interaction [Brown, 2001]. Also, online learners often must juggle multiple social worlds (e.g. online degree pro-gram, work and home) such that a range of obligations and identities compete for attention and time [Kazmer & Haythornthwaite, 2001].

Inspired by the aforesaid literature and my analysis of the community's capabilities, I identified six areas of achievement and challenge for an online learning community: emotional support (establishing relationships and mutual help), developmental goals (academic and professional preparation), group management (creating subgroups, group dynamics), technological infrastructure (effective use of tools and techniques for online learning), managing crisis (resolving external and internal conflicts) and sustainability (community growth and well-being).

I then developed 24 items to probe these general capabilities in the presence of obstacles inherent to online learning. For example, one item assessing emotional support was "We can engage in activities that support or help other members despite the demands of our everyday lives (e.g. job, families)."; one for developmental goals was "We can leverage each other's professional connections (e.g. sharing of openings, tips, referrals), even though we do not usually meet one another in person"; for group management "We can self-organize ourselves into subgroups with more refined interests, even though it will take effort to manage a variety of groups"; for technological infrastructure "Our community can take advantage of [a specific learning management system], even though many members are not experts in online technologies"; for crisis "When a problem arises, we can coordinate amongst ourselves to gain help from outside entities, even when the problem is complex or multi-faceted."; and for sustainability "Our community can sustain long term associations despite our virtual forms of interaction". The full set of items is available by request; summary results for the most internally reliable items are presented in the Results.

#### 4.1.1.3 Other Survey Content

To examine other variables that might be associated with our primary measures of community, and to document the population of respondents, I also included questions about student demographics (e.g., age group, employment status) and other characteristics (e.g., perceived value of knowing information about others, club membership and current practices for sharing information about themselves online).

### 4.1.2 Definitions of Community for Online Learners

Because the literature is still unclear as to how best to define community for this larger collective (i.e. beyond an individual class or program to the level of an entire online "campus"), I specified two definitions of community that originated from the contrast between common identity and common bonds in online community [Ren et al., 2007]: an identity-based community scoped to all of WC was defined as "all students currently enrolled in any World Campus degree program" (I refer to this as the Virtual Campus Community definition; VCC). A ties-based community was defined as "World Campus students whom you have become acquainted with, regardless of whether they are now in a class with you" (Acquaintance-Based Community; ABC). Whereas the former defines community in the abstract, emphasizing a shared connection to a virtual organization, the latter implies an ego-network that is comprised of acquaintances. I introduced these two definitions as a between-group manipulation in the survey, with the general hypothesis that ABC might yield stronger feelings of community, particularly for measures of SOC and CCE that are related to interpersonal ties.

## 4.2 Survey Administration

Participants were recruited from the entire active population of students enrolled in one or more WC programs offered. This means that survey participants were recruited from a diverse population with a wide spread of demographic characteristics, drawn from different programs and colleges as well as varying employment statuses. Criteria for recruitment included (1) over 18 years old; (2) actively enrolled in at least one online course in spring 2017; and (3) in a WC major or pre-major that would lead eventually to an undergraduate or graduate degree. I excluded students enrolled only for course credit or for a certificate program, as well as those who opted out of the contact list. As compensation, I advertised a lottery of $30 gift cards for all participants who completed the survey, with a promise of 15 winners.

To test the contrasting definitions of community (VCC versus ABC), I distributed two versions of survey that used the same items but varied the definition of community. I used random assignment in blocks to ensure that the two group sizes would be the same. An a prior power analysis with significance level $\alpha = .05$ (assuming a small effect size power analysis) indicated a sample size of at least 290 in each sample. In the end (after eliminating incomplete data) I had sample sizes of 367 for the VCC condition and 373 for the ABC condition, an overall response rate of 6.45%.

### 4.2.1 Participants

I obtained usable responses from 740 online students; 59.9% were males, and 95.8% of them live in the U.S. (37.4% live in the same state as the host university). At an average age of 35.54 years old (SD =10.25), participants are enrolled across 15 colleges, with a majority from the three whose programs are most available in World Campus (Liberal Arts 21.9%; Business 12.1%; IT 17.4%). Nearly half are pursuing bachelor degrees (47.2%), followed closely by those in master programs. 26.1% of them report being members of the World Campus student club. 62.9% of the respondents have never visited the host university in person. 82.7% of participants currently work for a company, 14.6% are not employed, 7% are self-employed (some as a second source of income).

## 4.3 Results

My data analysis was done in three phases. First, I developed a set of robust constructs to use in assessing SOC and CCE for the community domain of online learning. Second, I examined differences that might have resulted from the VCC versus ABC definitions. And third, I conducted exploratory regression analyses to investigate how the community constructs are related to other student behaviors and attitudes.

### 4.3.1 Construct Development

I began by assessing the internal reliability for the over-all scales for SOC and CCE; both were quite reliable, with Cronbach  of .93 and .96 respectively. Because my research goal was to uncover the latent structure that comprises feelings of online learning community, I next conducted exploratory factor analysis for the two scales.

#### 4.3.1.1 Factors for Sense of Community

After employing a Varimax rotation to enhance interpretability of the factor analysis solution , I uncovered two factors from exploratory factor analysis on the 10-item SOC scale. Following the guidance in [Ferguson & Cox, 1993], I next examined items that "cross-loaded", in other words, that loaded with a weight greater than 0.4 on two or more factors, or whose loading on two factors is very close. This led to the elimination of two items ("I feel that I matter to other students" and "I have a good bond with others in this community"). I also deleted "I feel connected to the World Campus community" because its loading on both factors was well beyond 0.4, suggesting it maps to both constructs. The final scale has seven items across two constructs (Table 4.1).

**Table 4.1** Exploratory Factor Analysis for Common Identity and Friendship

| **Common Identity; $M = 3.65$, $SD = .93$; $\alpha = .886$ (46.95% variance)** |
|---|
| Fulfill my needs ($M = 3.48$, $SD = 1.12$, loading = .824) |
| Get what I need ($M = 3.67$, $SD = 1.08$, loading = .816) |
| Feel like a member of ($M = 3.81$, $SD = 1.11$, loading = .809) |
| Belong in the ($M = 3.89$, $SD = 1.09$, loading = .784) |
| I can rely on others in ($M = 3.37$, $SD = 1.17$, loading = .734) |
| **Friendship; $M = 2.01$, $SD = 1.09$; $r = .873$ (27.66% variance)** |
| Have friends at ($M = 1.79$, $SD = 1.13$, loading = .925) |
| Feel close to others in ($M = 2.22$, $SD = 1.19$, loading = .878) |

Table 4.1 reports the seven items using abbreviated phrasing that provides the essence of individual items, along with each item's mean (recall that these are Likert-type scales with values from 1-5), standard deviation, and loading on the factor (after removing the cross-loading items). Visual inspection reveals that the *Common Identity* construct is represented by five items, and seems to stem from the SOC concepts of needs fulfillment and membership [Peterson et al., 2008], along with trust [Rovai et al., 2004]. In contrast, *Friendship* appears to capture the emotional bonds one might expect in smaller groups such as projects, classes or social acquaintances.

As seen in the Table 4.1 headers, Common Identity has a higher mean value than Friendship ($F(1{,}680) = 254.37$, $p < .001$). It seems that online learners tend to experience community through a common identity with shared purposes, group membership and trust more than through interpersonal ties. However, this difference may also be an indication of how difficult it is to have "friends" in an online learning environment. Statistical factor assessment criteria [Hair et al., 1998] confirm that the factor loadings have convergent validity for each construct, with individual item standardized loading above 0.5 and average variance extracted (AVE) above 0.5.

An analogous exploratory factor analysis with the 24 CCE items, again using Varimax rotation, uncovered three latent constructs: *Identity Regulation*, *Coordination* and *Social Support* (the complete scale and factor loadings is available by request). As I had done for SOC, I removed items with significant cross-loading, yielding constructs that were more interpretable and non-overlapping [Ferguson & Cox, 1993]. The final scale (Table 4.2) includes 13 items distributed across the three constructs.

#### 4.3.1.2 Factors for Community Collective Efficacy

*Identity Regulation* appears to represent foundational characteristics that ground a large group of online students to operate together as an online learning community. In particular, the items comprising this category are tied to the special circumstances of online education, such as earning an online degree in the midst of many individual challenges; using provided tools appropriately; and recognizing that online learning is largely self-directed. The strongest item in this construct refers to the social norms that must be in place for such a community to be successful, underscoring a belief

**Table 4.2** Exploratory Factor Analysis for Identity Regulation, Coordination and Social Support

| |
|---|
| **Identity Regulation; $M$ = 4.15, $SD$ = 0.77; $\alpha$ = .813 (21.27% variance)** |
| Follow social norms without supervision ($M$ = 4.46, SD=.84, loading=.800) |
| Facilitate self-directed learning without full awareness of others ($M$ = 4.04, $SD$ = .96, loading = .700) |
| Encourage us to pursue a degree despite individual challenges ($M$ = 3.97, $SD$ = 1.02, loading = .652) |
| Leverage CMS despite technical complexities ($M$ = 4.14, $SD$ = .97, loading = .628) |
| **Coordination; $M$ = 3.67, $SD$ = 0.87; $\alpha$ = .879 (25.51% variance)** |
| Sustain associations despite virtual interaction ($M$ = 3.54, $SD$ = 1.16, loading = .763) |
| Convey mission to outsiders without being forced ($M$ = 3.71, $SD$ = 1.01, loading=.731) |
| Coordinate to get help despite complexity of problem ($M$ = 3.80, $SD$ = 1.02, loading = .719) |
| Have individual influence despite speaking with a united voice ($M$ = 3.69, $SD$ = 1.05, loading = .692) |
| Self-organize into subgroups despite the management cost ($M$ = 3.62, $SD$ = 1.08, loading = .656) |
| **Social Support; $M$ = 3.68, $SD$ = 0.92; $\alpha$ = .848 (21.71% variance)** |
| Help others despite everyday demands ($M$ = 3.62, $SD$ = 1.13, loading = .773) |
| Inform one another of news despite separation in time and space ($M$ = 3.90, $SD$ = 1.09, loading = .764) |
| Build interpersonal relationships despite turnover in courses ($M$ = 3.15, $SD$ = 1.23, loading = .734) |
| Welcome new members despite difference in joining times ($M$ = 4.04, $SD$ = 1.00, loading = .685) |

of the sort "we know what we are doing and how to do it." This construct can be seen as a collective version of the common advice that online learners should be able to take the initiative and engage in self-directed learning [Moore & Kearsley, 2011], but here the emphasis is on the expectation that this is a collective responsibility of all students in a community.

A second construct is *Coordination.* The five items loading on this factor suggest a community that is able to engage in united action despite a variety of challenges (e.g. computer mediated communication, a lack of top-down organization, or the complexities of problems that may arise). In particular, the community of online learners can manage and resolve conflicts (of interests or benefits) through coordination that allows individuals to express views and have influence. As a consequence, the community is able to act in a united way as one entity. It is also noteworthy that while sustaining associations loads most strongly on this factor, it has a rating barely above the midpoint of a 5-point rating scale ($M$ = 3.54, $SD$ = 1.16); this may indicate some difficulty in sustaining associations for online students.

Finally, *Social Support* is quite intuitive as a community capacity. The items that comprise this construct relate to the interpersonal help, support, and maintenance of ties. It may be that some interpersonal "jobs" are easier than others (e.g., compare welcoming new members with building interpersonal relationships), but this sort of capacity for helping one another seems also to be an element of collective efficacy.

Again, the CCE factor loading indicates construct validity based on the average variance extracted related criteria [Hair et al., 1998]. I used three established indices to assess adequacy of model-data fit in a confirmatory factor analysis: *Comparative Fit Index (CFI)* $\geq .90$ [Hoyle, 1995], *Root Mean Square Error of Approximation (RMSE)* $\leq .08$ [Byrne, 2013] and *Standardized Root Mean Square Residual (SRMR)* $\leq .08$ [Hu & Bentler, 1999]. I confirmed that the three-factor model has acceptable levels of fit ($CFI = .95$, $RMSEA = 0.07$ and $SRMR = .04$).

Comparing means among the three constructs, a repeated measures ANOVA (with Greenhouse-Geisser correction for sphericity) revealed a significant difference among the constructs ($F(1.94, 1314.95) = 194.57$, $p < .001$). Post-hoc tests with Bonferroni correction showed that the mean for Identity Regulation is significantly higher than that for both Coordination and Social Support, which do not differ. This mean difference parallels the earlier finding of a higher mean for Common Identity versus Friendship in the SOC constructs, and helps to reinforce the importance of shared identity experiences and associated community responsibilities for the community of World Campus students.

### 4.3.2 Contrasting Different Definitions of Community

Before examining possible differences between participants who made their ratings using the two definitions of community (VCC and ABC), I conducted confirmatory factor analyses on the two groups of survey respondents (n = 367 for VCC and 375 for ABC). Based on the criteria of model-data fit indices ($CFI \geq 0.90$, $RMSEA \leq .08$, $SRMR \leq .08$ [Byrne, 2013; Hoyle, 1995; Hu & Bentler, 1999]), I found that the three-factor model for CCE has an acceptable fit for all three criteria regardless of community definitions; the two-factor SOC model satisfied two of the three criteria for the two sub-populations. Given the exploratory nature of our measurement instruments, I found this promising as it suggests that the constructs are operative under different instructional sets.

**Table 4.3** Community construct means for VCC and ABC

| | **Common Identity** | **Friendship** | **Identity Regulation** | **Coordination** | **Social Support** |
|---|---|---|---|---|---|
| **VCC (367)** | 3.72 | 2.00 | 4.23 | 3.70 | 3.67 |
| **ABC (375)** | 3.74 | 2.03 | 4.06 | 3.66 | 3.67 |

Next, I compared the means of the community constructs for the VCC and ABC groups (n = 367 and 373 respectively; see Table 4.3). A 2 (ABC vs VCC, between) × 2 (Common identity vs Friendship for SOC, within) ANOVA revealed no interaction effect between the two SOC constructs and community definition ($F(1{,}738) = .03$, $p = .87$). This is not surprising given that the means are virtually identical for both constructs. However, I did find a main effect for these two SOC constructs, with Common Identity having a higher mean than Friendship $F(1{,}738) = 984.69$, $p < .001$); this replicates the earlier finding from the entire dataset. There was no main effect of the community definitions $F(1{,}738) = .08$, $p = .78$. Put in other words, instructing respondents to focus on WC in the large, or on acquaintances within World Campus, has no effect on feelings of either Common Identity or Friendship. I was surprised to find this, as I had expected to see higher ratings for scales relating to interpersonal ties when respondents were instructed to focus on acquaintances. It may simply be that students' ego-networks in World Campus are too small or loosely connected to play much of a role in feelings of community.

I also analyzed the three CCE constructs (Identity Regulation vs. Coordination vs. Social Support) as a within-subjects factor across the two community definition groups. In this case, the ANOVA revealed a significant two-way interaction between the three constructs and the two definition conditions ($F(2{,}676) = 4.60$, $p < .05$. To probe more deeply, I also examined simple main effects separately for the two definition conditions, and found that for each sub-group there were statistically significant differences among the three constructs ($F(2{,}676) = 73.05$, $p < .001$ for ABC and $F(2{,}676) = 130.44$, $p < .001$ for VCC). A post hoc test with Bonferroni adjustment for multiple pair comparisons indicated that only the contrast of Identity Regulation was significant ($p < .001$). In other words, for both ABC and VCC, Identity Regulation has a higher mean value than Coordination and Social Support. At the same time, the simple effects of community definition were significant only for Identity Regulation ($F(1{,}677) = 9.14$, $p < .01$); respondents given the more identity-based VCC definition reported higher collective efficacy on this construct than those given the acquaintance-based ABC definition. Again, this is easily seen in a visual comparison of the means, where Identity Regulation differs by 0.17 while the other constructs' means are virtually identical.

In sum, only the CCE construct of Identity Regulation appeared to be sensitive to these two contrasting ways of defining community. The differences were as one would expect, with the VCC definition evoking higher ratings of what is expected by WC community members. Recall that our sample sizes for the between-subjects variable were 367 for VCC and 373 for ABC; the comparison has a power value of 0.88, indicating sufficient sample sizes [Faul et al., 2009]. In reviewing the observed pattern of means, it seems intuitive that Identity Regulation was rated more highly as a collective efficacy under the definition of community as all WC students. However, it is surprising that neither the SOC construct of Friendship nor the CCE construct of Social Support, receive higher ratings when the community is understood to be acquaintance-based. It may be that one's ego network is a mix of individuals, only some of whom can contribute to feelings of friendship or social support.

### 4.3.3 Predictive Validity for the SOC and CCE Constructs

Beyond establishing a set of constructs for the online learning community, I also wanted to explore the usefulness of the measures in studies of online community. Thus, I examined the constructs' predictive validity with respect to other student variables that have some face validity as indicators of community-oriented attitudes among online students. Two of these are the self-reported behavioral measures of club membership and degree of profile completion. The other two are value judgments concerning how important they feel it is to have information about their online peers, and to what extent they want to feel more connected to others. I first introduce these variables, then consider their relations to the SOC and CCE constructions.

*Student club membership.* Among various steps a student might take to seek

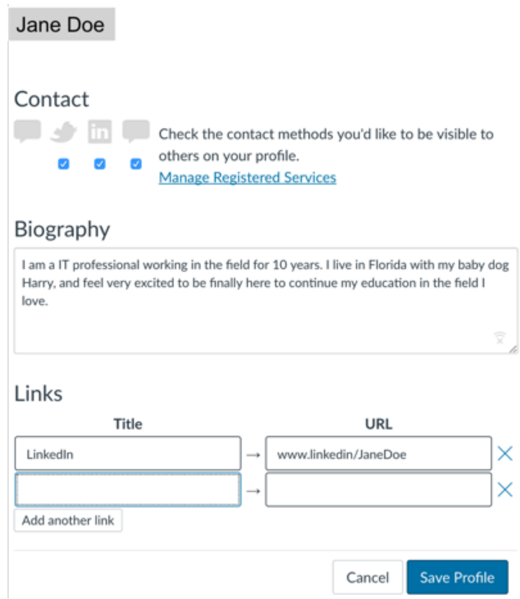



**Figure 4.1** Canvas Profile Editing User Interface

and develop community, joining an online student organization comes with gaining member-ship to a subgroup of students (with an entry fee of 15 dollars). 30.3% of responding participants reported to be a member of an online chapter affiliated with the host university's alumni association. The online student club holds a monthly meeting via Adobe Connect, with an organization goal to establish a lifelong commitment to the university.

*Profile completion.* The online learning technologies used by World Campus include a Profile module whereby students can "introduce" themselves to the instructor and other students. For example, students can post a biography in which they describe their interests, hobbies, goals, and expectations, or even provide external social media links (e.g., Twitter, LinkedIn, see 4.1. Such self-introductions are one way to present oneself and invite outreaching behaviors from others in World Campus.

Use of the Profile tool is normally discretionary, but seems to be a designed feature that corresponds to the introduction technique used for building community [Sun & Rosson, 2017d]. Because completion of a profile suggests some interest or willingness to share personal information, I gathered self-reports about the use of the module. The relevant survey question included a screenshot of the Profile tool (in case the student was not familiar with it or had forgotten; 4.1), and asked for percentage estimates of how much the student had completed in his/her profile. On average, students said they had completed about half of the profile fields ($M$ = 0.49, $SD$ = 0.31), with an evenly distributed range across the sample. In a separate question, I also asked for an estimate of how many World Campus students the participant had come to know via profile pages; the mean was 3.12 ($SD$ = 8.59, after removing an outlier value of 2000). However, the distribution was rather skewed: 375 out of 635 responses confessed that they have come to know no one via profile information. Both the completion rate of profile and the social use of profile to know others indicate that profile is limited in enhancing interpersonal knowledge.

In addition to asking about Profile use, I invited open-ended comments about whether this tool might help in building community and if so how. Many students admitted that they were not even aware of the tool. Even though they are sometimes asked by instructors to fill out this information, these students have not done so, even while acknowledging that this could have been a good place to get to know one another. However, other students explained that there are no contextual factors that could enable meaningful social acts. It may be that a generic student information-sharing tool is simply seen as extra and unnecessary work, unless it brings with it some intrinsic or extrinsic benefits.

*Perceived importance of peer information.* I probed the perceived value of having information about other learners in building feelings of community. Uncertainty Reduction Theory [Berger & Calabrese, 1975] suggests that strangers will be more likely to interact if they first grasp a bit of background information about each other; reduced uncertainty helps them to tailor their communication content, level of intimacy and information-seeking behaviors. Researchers have also observed effects of interpersonal knowledge on interaction within globally-distributed academic communities, where socio-emotional knowledge is a strong predictor for sense of community and knowledge sharing acceptance [Nistor et al., 2014].

For this survey, I adopted the four information categories extracted in my earlier study of information preferences [Sun & Rosson, 2017c]: other students' availability; professional background; education status; and personal details (e.g. family situation, gender). A repeated measures ANOVA (with Greenhouse-Geisser correction for sphericity) revealed a significant difference in rated importance of the four types of information ($F(2.60, 460.42) = 73.443$, $p < .001$). A post hoc test with Bonferroni correction revealed that each pair of the means is significantly different except for the contrast between professional background and education status. Schedule availability was rated most highly, followed by professional background, education status and personal details. The ordering of these value judgments echoes that reported in [Sun & Rosson, 2017d].

*Desire to be more connected.* To assess the degree to which students want to feel more connections to others, I inserted a 5-point scale asking about interest in feeling more connected, ranging from "not at all interested" (1) to "extremely interested" (5). The median selection was 4 ("rather interested"; $M = 3.52$, $SD = 1.25$), which is moderately promising with respect to a general goal of enhancing online learner community. If I divide the participants into two groups using 4 as the median, I find 366 who are fairly interested in feeling more connected (55.2% of 663 responses to this specific item). A post-hoc correlation analysis with demographic information reveals that online student club membership (yes or no) is significantly related to self-rated desire to be connected ($r = .14$, $p < .001$).

#### 4.3.3.1 Relationships to the Community Constructs

To investigate the predictive validity of the five new constructs for online learning community, I calculated bivariate correlations among all four student variables and the five constructs (Table 4.4).

**Table 4.4** Correlation (Pearson) between student variables and SOC and CCE constructs

| | **Clubs** | **Profile** | **Peer Info** | **Desire Connect** |
|---|---|---|---|---|
| **Common Identity** | .10*[a] | .19**[b] | .20** | .12** |
| **Friendship** | .18** | .20** | .22** | .21** |
| **Identity Regulation** | .10* | .13** | .20** | .19** |
| **Coordination** | .13** | .18** | .25** | .25** |
| **Social Support** | .15** | .16** | .19** | .19** |

[a]items are significant at the 0.05 level (2-tailed).
[b]items are significant at the 0.01 level (2-tailed).

As indicated in Table 4.4, all correlations were positive and significant at least at the level of 0.05. That is, higher levels of each SOC and CCE construct is associated with higher levels of the other student variables. Of course, I cannot assign causality to these relationships; it may be that people who join clubs feel greater community but it could just as easily be that people who feel community are more likely to join clubs. Similarly, someone who already feels high SOC or CCE may wish to connect with others and know more about them; or this desire could be a motivator to take actions toward creating community. Future work is needed to tease apart these alternate accounts of the correlations. However at present I am satisfied to note that the new constructs demonstrate good predictive validity with respect to these various indicators of community involvement.

### 4.3.4 Summary

With two constructs underlying ratings of SOC (Common Identity and Friendship) and three for CCE (Identity Regulation, Coordination and Social Support), I have produced a multi-faceted and reliable basis for assessing feelings of community; it integrated behavior potential (CCE) with the social lens of shared identity and bonds (SOC). These constructs relate in expected ways to other indicators of online learners' orientation toward community activities (e.g., club participation, sharing information). Although the constructs were discovered through exploratory factor analysis, my extensive refinement activities (e.g., dropping cross-loading items) and the constructs' robustness and utility in other analyses give me confidence that they can be adapted by researchers in future studies of online learning community.

# Chapter 5 Design and Evaluation of Interactive Prototypes

Based on the research activities and results reported in Chapter 3, I developed a series of interactive prototypes and evaluated them to iteratively refine my understanding of online community among WC students and corresponding design goals and solutions [Sun, 2017]. In particular, I first explored design options to visualize common identities and common bonds among students, using a combination of technology probe and survey responses to assess these initial design ideas. Next, I improved on these ideas based on the users' feedback and re-integration of empirical findings regarding the importance of peer availability. I used the new design ideas as a probe for an in-depth interview study, both to uncover explore more etails about the need to know peers' availability and to elicit reactions from students on the design approaches.

## 5.1 Visualization Design

After analyzing the gap between online learners' desires to connect with others and the difficulties in gathering relevant information, I considered visualization approaches that could externalize such information. To create an information base for these visualizations, I first administered a survey to the WC students, so as to gather the information that could be visualized. I focused particularly on collecting information that related to what we had learned about the students' practices and strategies for building connections with others. These included the common bond of being part of the same education program(s), information that could provoke feelings of shared identity, and current residence:

- Common bonds: major, enrolled online courses, and year of first enrollment
- Shared identities: age, gender, employment status, military history, number of children
- Residence: current state of residence, time zone.

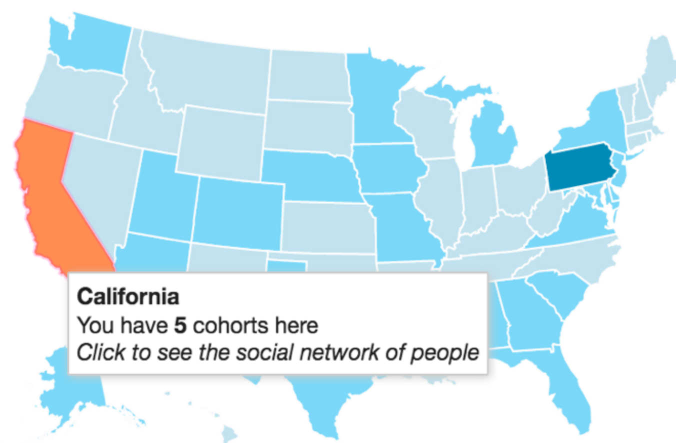
Working with instructors who sent out email invitations (offering extra credit and an opportunity for a gift card drawing), I distributed the survey to classes in same program that I had approached for interviews. I received 91 responses: 76% of them were male; 80.9% were advanced (junior or more with 20% masters); 54% had full-time jobs; 79% reside in the U.S.; their median age was 30.

With respect to design ideas, I drew primarily my earlier interview findings concerning situations in which WC students typically reach out to their peers (e.g., setting up projects or study groups). I generated two preliminary designs, with the goal of offering flexible views and approaches to interacting with the information I had gathered. One of these was a general purpose network graph, where students are represented as nodes linked to hubs that represent shared information (see Figure 5.1 (a)). The second visualization was a map-based view that used students' current residence as a clustering technique (see Figure 5.1 (b)).

### 5.1.1 Connecting through Common Hubs in a Network Graph

The network graph was designed as the primary view for visualizing and exploring peer relationships (Figure 5.1 (a) implemented with D3.js [Bostock et al., 2011]). I chose a network representation because it is flexible and dynamic and offers a simple way to explore connections to different sets of peers (by changing the definition of the hubs). As I had learned during the interviews, the ways students "connect" with each other vary and depend on factors associated with different facets of one's identity and personal experiences. Because I began with a broad set of information about


(a) Network by course number




(b) A map of participants' distribution in the U.S.

**Figure 5.1** Visualization design for externalizing shared identities and clusters by place

each student in the graph, this network view allowed me to create a malleable view that supported successive exploration of different types of connections.

In this visualization, nodes and arcs define a given network, and these elements can be mapped to student information in a variety of ways. Students can choose what will count as a link (e.g. a course or a major); by default the initial network uses common classes as the broker to connect students. The hub nodes hold the various defined values of the information type selected (e.g., a specific class would link to all students who had taken it).

This simple network framework allows users to specify and explore different "connection" criteria (via menu selection) to define different networks. For instance, Figure 5.1(a) visualizes a network that has "courses" as the linking criterion. The link between the orange node "IST440W" and "John Attia," (a pseudonym) indicates that John has taken the course. Students who are also linked to IST440W can thus see that they have this in common with John. The force-directed layout represents overall connectivity from a third-person viewpoint, but individual users also have the freedom to grab and drag a leaf or hub node to more or less central positions.

If the learner hovers over a leaf node, a brief summary of the corresponding online student appears, offering more details about the person (e.g. contact info, as well as an opportunity to message the student). The hope was that presentation of these individual details might promote feelings of more specific connections to the peer learner in focus. Hovering over a hub node produces a summary of how many learners are linked through that node, providing a sense of the "connective power" provided by that hub.

My working hypothesis is that the network visualization can be used to discover relationships among students that might otherwise be invisible. While students may have a sense of who else is in a particular course, they might be much less likely to know that person's typical time for studying, geographic region, history with online education, employment status, and so on. Researchers often see social network representations as indicators of available social resources [Lin, 1999; Seibert et al., 2001]. I hypothesize a similar affordance in our prototype: knowing more about a peer learner may make a student more comfortable reaching out with a request for help or an invitation to work together. I am particularly interested in studying how learners view different network structures and what they see as valuable in the connections.

### 5.1.2 Clustering Learners in a Geo-coded Map

The network visualization was designed to offer a flexible view for exploring possible connections among students. However its flexibility means that it is also abstract, representing relationships among students as shared connections to a hub node. As a more concrete alternate view - and a design concept that responds more directly to interviewees' interest in distance and time - I also designed a map visualization (Figure 5.1(b); for simplicity and because most of the 91 respondents were U.S. citizens, I included only students living in the U.S.). This representation conveys less information than the network graph but leverages location (and secondarily time zone) as a primary feature for selecting groups of peers.

The primary affordance of the map is the color-coding that signals the relative number of students from a given state. Hovering over a state provides more details, such as the number of students who reside there; clicking on a state (e.g., California) sets that state as a filter on the network graph (i.e., on return to the network visualization, the learner would see that it now visualizes only the subset of students from California). I wanted to know whether students are interested in the geographic distribution of their online peers and what meaning if any they would attribute to the sharing of geography. For example, my interviewees' comments had led me to wonder if they would find such a map useful as a way to identify and explore regional sub-communities.

### 5.1.3 Summary

In sum, I fleshed out two design concepts for presenting and exploring information about online peers. The network visualization was intended to support broad access to different types of possible relationships to peers; whereas the map visualization emphasized place-identity with its potential corollary of overlapping time zones.

## 5.2 Prototype Evaluation

To explore the potential of the network and map visualizations, I designed a feedback collection survey that presented the prototype visualizations as design probes. In addition to asking for general feedback, the survey sought to contextualize users' reactions with six different scenarios of use; I hoped that these would help me assess whether and how the two design concepts might be useful. The evaluation responses included a mix of open-ended and Likert-scale questions that probed the respondents' experiences with and reactions to the design concepts.

The survey was comprised of two pieces: the first introduced a user-guided exploration with the two visualization designs; the second guided respondents through a more concrete set of scenario-probed reflections. In the first part, I encouraged users to simply explore the tool on their own; when they had done so we asked them to reflect on their experiences and suggest ways that they might use such a tool.

In the second part, I introduced six scenarios that were inspired by the interviews and our own design brain-storming. I asked about the tool's usefulness in these contexts: 1) help with homework; 2) recruiting for an online club; 3) seeking peers for off-line extracurricular activities; 4) forming a virtual study group; 5) getting advice about classes to take in the future; and 6) forming a project team. For example, the homework prompt said: "You are working at home and baffled by a homework problem. You would like to find someone who has the expertise to help you out. To what extent do you think the tool ..."

Rather than sending out a general invitation to students in the summer program, I decided to reach out to only the 91 students who had provided the information that was in the visualizations, hoping that this would make the exploration more meaningful; unfortunately, this produced only 13 responses. Eleven of them were advanced students; two were in their first year of a degree program. I was surprised that only 13 students responded to my invitation, but later realized that the summer semester had ended and many students had likely moved on to other responsibilities. Nonetheless, my qualitative analysis of their comments and suggestions provided many insights into our two designs' strengths, weaknesses, and future directions. To analyze comments, I applied thematic analysis to the open-ended questions about their experience and reaction [Braun & Clarke, 2006; Strauss & Corbin, 1998]. Specifically, I first generated initial codes and then refined the code names and relationships with axial coding.

### 5.2.1 Discovering Shared Learning Experiences in Network View

Traditional classrooms are bounded by physical space, whereas the "boundaries" defining groups in an online class are inherently flexible; learner-specified basis for network visualizations was an effort to leverage this flexibility. Thus, E5 *"like(d) the visualization and dynamic changes as I choose different settings."* Similarly, after viewing online learners' shared classes as well as other classes their peers are taking, E6 was *"impressed by the granularity of the connections by classes taken together"*. The network view made it *"a lot easier to see connections of similar classes"* (E2) or *"majors"* (E5). E2 mentioned that he could see value in knowing the various classes his peers are taking, and that he might try to organize teams using the same persons across multiple classes.

Furthermore, E12 recognized the value of the network views *beyond a single class*:

*"I think it might be most useful in efforts to gather info about other classes... Every class I've taken has a class roster or list for email, so I already know exactly who to get in touch with for the class at the time I am taking it."* In contrast to the familiar view of a class roster, the network visualization allows students to discover (and potentially reach out to) students who have taken a particular class in the past, for example a student who is trying to decide whether to take a class in the coming semester can send a query to someone who is currently enrolled.

E5, E6 and E9 were quite interested in learning about peer learners' previous classes. The utility of such information comes in two ways. First, students are able to seek recommendations based on prior class-taking experiences. this use case was expanded in E9's proposal to add a professor tag: *"speak with students who have had experience with a particular professor (kind of like if you could speak directly with students who reviewed professors on ratemyprofessor.com)"*. Second, students can use this information to *"get in touch with students who have previously taken a particular class"*; as E2 put it *"past cross-over in classes would be nice"*.

### 5.2.2 Using the Map to Select Peer Sub-Groups

Unlike the network visualization, the interactive map affords the finding of peers who live nearby. Half of the participants discovered they could drill down to peers in a given state by clicking on the map and returning to the network visualization. Leveraging "state" as a filter, the sequential views from map to network allowed students to create a more specialized sub-group.

The visual summary of peer distributions across a map seemed to suggest a "place identity" cue that may contribute to personal connections: *"There's no concrete connection there, but people identify with their states, so I found it interesting to feel more connected just because we live in the same area"* (E12). This learner further explained that visiting the same places contributes to feelings of connectedness. Meanwhile, looking ahead into the future, E5 foresaw the value of geographically proximal peers for possible meet-ups, group projects or friendship.

Driven by the goal to find good candidates for collaborative pursuits, E1, E5 and E9 responded positively to the use of a geographic filter that would let them discover students who had the same major or past classes but also lived in the same area: *"I liked how it is broken down by states and then you can go even further and do it by classes that you took online."* (E9) Even without the similar residency experience, E1 thought it would be interesting to see that his former teammates are living in different locations. Applying multiple filters "brings forth" a group that is suitable for a given learner's current interests that involves more than one dimension.

The map allowed the learners to use geographic location (a state) as a filter on the network visualization. Without using the map, the learners could only specify one type of hub connection at a time. However, some participants suggested other combinations of information that would be useful to explore. For example E5 and E8 proposed to use age group as a first-level filter followed by other network explorations; E5 also suggested filters by language or culture, as well as food or hobbies, arguing that such clues *"will allow for more meaningful connections and similarities."* As another nod toward shared cultural identities, E1 thought it would be valuable to include international locations on the map.

Some students were interested in defining a specific group of people within the large entity of online community, or sub-groups: *"Creating sub-groups, connecting through classes, majors, cohorts and goals. These elements enable students to develop interpersonal relationships thus enhancing a learning environment."* The coordination between the two forms of visualizations also allows for accumulative refinement into a body of focused interests: "*I liked how it is broken down by states and then you can go even further and do it by classes that you took (online).*" (E9)

### 5.2.3 Design Suggestions for Improvement

More relevant to participants' motivations for being in an online degree program in the first place, some participants commented that offering employment status as a tie highlights one core benefit of peer networking, that is, getting advice and suggestions for career advancement. For example, E12 wanted a filter related to job types, not just whether full-time, part-time, etc: *"It would have been interesting to break out careers, such as military, education, service, fine arts, etc. I think that would also highlight a good starting point for networking"*. E6 currently uses LinkedIn to connect with alumni but wanted to use our prototype tool for similar purposes. Others expressed a wish to connect to peers who have experience with a particular professor.

Along with the utilitarian values students care about, distance learners further pointed out the need to reach out to a specific individual. As an add-on to the network visualization, students wanted some kind of sortable structure, such as alphabet ordering of names; others wanted to search for names. More oriented to the direct use of such information, students desired for more *"personalized encounters"*, such as a link to even more complete information about a peer, or at least a photo that would convey a more real world sense of the individual in the network graph. At the same time, the disclosure of personal details raised concerns about privacy: students want control over who can see their information, as well as explicit articulation of willingness to be contacted: *"We need a place to enter our interests and willingness to be contacted"* (P3).

### 5.2.4 Summary

My preliminary evaluation of the two different visualization designs points to two mechanisms of connecting distance learners that current learning technology should support: some representation of prior learning experiences; and filtering mechanisms for specialized social groups. In general, my participants reported that they were able to gain new insights about the online population represented in the visualizations. In some cases, these explorations enabled students to see their peers in new ways while still supporting their efforts to coordinate coursework. Knowing about others' educational experiences (courses, degrees, and so forth) provides students with an additional source of informal learning and support, which may help to promote social behaviors. Considering students' use of the network graph versus the map view, it seemed that as expected the network was seen as a way to discover different ways of being connected to other students, whereas the map evoked the more instrumental goal of setting up a good team.

The strategies and use cases reported in this design-probe user study align well with my previous interview study with distance learners on their social needs and practices. I refined my understanding and design trials based on students' information needs for initiating and sustaining social activities in online education programs. Specifically, I recognize that the goal to invite social interaction creates a tension between social visibility (of each others' information) and personal privacy (of one's own information; see also other work in socially translucent design [Erickson & Kellogg, 2000]). Design efforts to activate social identity processes that connect people within a virtual space might increase the risk of unwanted information disclosure. Future work should address the meta-information surrounding personal data (e.g. willingness to be visible or to be contacted, and in what scenarios and by whom) and differentiate social contexts through which learners can manage their online peer knowledge structures for different purposes. Additionally, users kindly offered suggestions for new features and activities, and shared their excitement and difficulties in utilizing such tools, which also informed my ongoing design work. I turn next to a more comprehensive design and evaluation of tools intended to facility online community.

# Chapter 6
# WeAre! and WeConnect: Community-Enhancing Tools for Online Learners

The needs analysis I reported in Chapter 3 pointed to the importance of being aware of other online peers' availability; it also emphasized the information gap of getting to know one another with regards to common bonds and common identity. My initial design exploration and evaluation reported in Chapter 5 suggested that I should develop design concepts that communicate the notion of availability and common identities, as well as convey information relating to students' interests in availability, profession, educational background, and personal information.

As I considered options to either rely on existing tools or build a new tool from scratch, I found that current social network sites do not meet my initial requirements: for instance, LinkedIn only emphasizes professional impression management, and Facebook is too broad to be used as a platform for learning community. At the same time, building a tool from scratch often leads to unpredictable errors introduced by the system and a corresponding lack of trust, clearly a detriment to community growth. Instead I was inspired to adapt Slack Workspace tools [sla, 2019]; Slack is a communication software platform designed to support collaborative work. In Slack, not only can users communicate in different channels associated with different purposes, projects or units, but also they can integrate external Apps from an growing ecosystem of Slack Apps to add special features or to simplify collaboration (e.g.

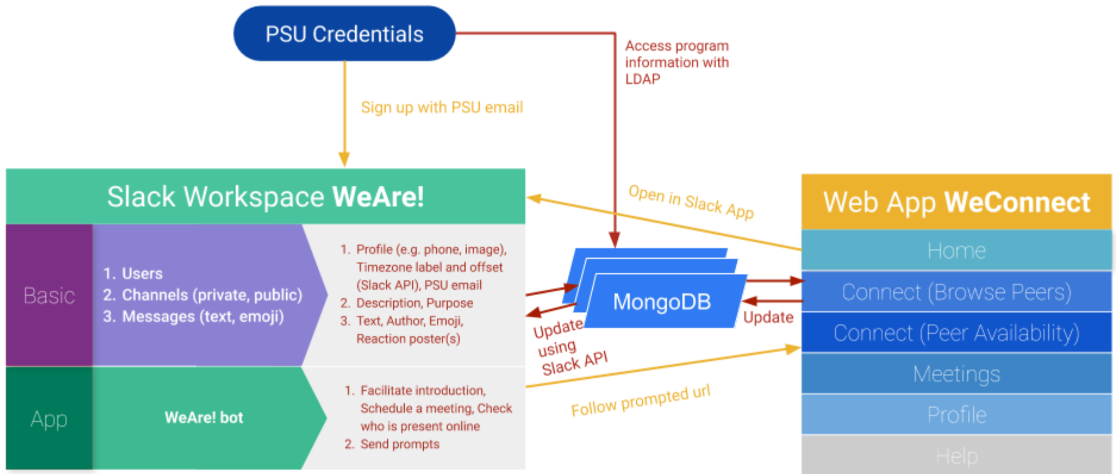



**Figure 6.1** WeAre! and WeConnect: System Overview

Google Drive).

In this chapter, I first introduce the information architecture as an overview of the complex technology context within which I was designing and implementing tools; this is comprised of a communication workspace supporting possible interaction among peers and a web application that visualizes aspects of members' profiles and activities using the Slack API. I follow this with a more detailed design description of features and intended user behaviors.

## 6.1 Information Architecture of WeAre!

To provide a reliable interactive environment to support diverse social interaction (e.g. public *vs* private, in-depth threaded conversations *vs* light-hearted reaction) while also increasing the information awareness among distance learners, I designed two complementary components in my overall suite of community-building tools:

1. *WeAre!* is an enhanced Slack Workspace with basic Slack channel and messaging features and custom App `WeAre!` (a bot I developed to facilitate social

interaction)[1].

2. *WeConnect* is a web application that consists of multiple views in support of different design intentions.

As shown in Figure 6.1, the two components share the same data stored in a non-SQL database MongoDB. The yellow lines show the anticipated user navigation across the two components, and the red lines show the data acquisition and update across different sources. I used PSU email accounts as the foreign key to associate all the data from a member's Slack account (the PSU account is used to log into the *WeConnect* web application) and the student educational records (e.g. program, campus). Slack users can follow a URL[2] sent by Slackbot `WeAre!` or by administrator to visit *WeConnect*. Note that Slack users need to log into the web application *WeConnect* through their PSU email and to grant *WeAre!* authorization in the web browser even after users sign in Slack Workspace *WeAre!*[3] (as shown in Figure 6.2 (b) and (c).

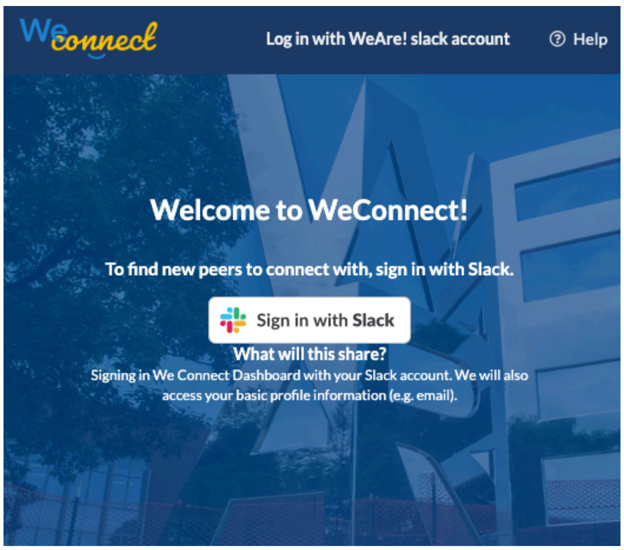
From the web application context, in other words within *WeConnect*, multiple hyperlinks are provided to enable users to return directly to the dialog window inside the Slack Workspace *WeAre!* (e.g. Open in Slack in the homepage, Open a direct dialog with someone inside the *Table View* or others' profile view, as shown in green button on the top right corner in Figure 6.3).

As an illustrative scenario, a new member of the Slack Workspace *WeAre!* might set up his or her profile information in the Slack environment, but then shift to the *WeConnect* web app to elaborate the profile, or view other peers' information to spot commonality (see Figure 6.3). Members' profile information across the two components is stored in MongoDB once a user signed up as a member in the Slack Workspace *WeAre!*. Mediated by the same database table, I also synchronize profile information between the basic user profile settings inherited through the Slack API (e.g., their Slack (or PSU) emails used as identifier) and the more elaborate custom

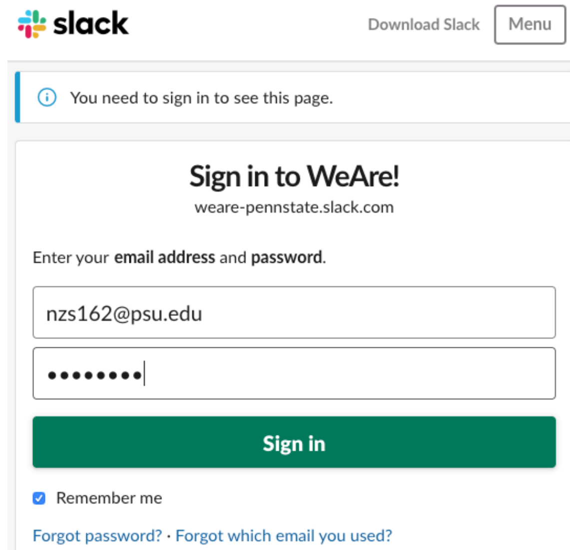
[1]In order to differentiate the Slackbot and the Workspace in this writing, I will use different fonttype: *WeAre!* denotes the entire Slack Workspace, whereas `WeAre!` denotes the Slack App, or more specifically, Slack bot.

[2]The URL to web app *WeConnect* is `https://weconnect.ist.psu.edu:8443`

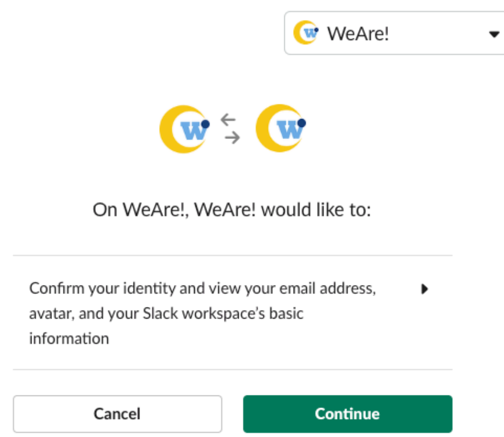
[3]The Slack Workspace URL is `https://weare-pennstate.slack.com`.




(a) Land in the *"WeConnect"* and sign in




(b) Sign in to *WeAre!* Account to complete log-in for *"WeConnect"*




(c) Confirming to login in to the Slack Workspace *WeAre!*

**Figure 6.2** Sign in to *WeConnect* using Slack Account and Authorization as a *WeAre!* Member

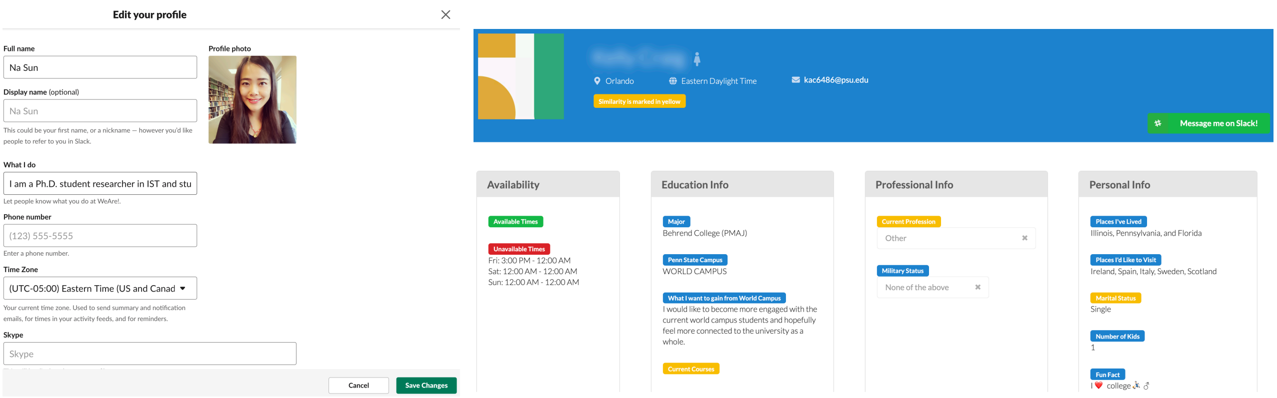



**Figure 6.3** Edit profile in WeAre! and View others' profile in WeConnect

profile supported by the web app *WeConnect*. As a result, any changes made in in one component will also be reflected in other views.

Finally, I use the Slack API to access Slack users' activities in public channels, thereby obtaining extra information about the Slack users' presence, chat participation, and time zone distribution. In addition, conversations and interactions with the Slackbot `WeAre!` are logged using the Slack API.

### 6.1.1 Student Data

The users' profile information comes from a mix of different sources. First, any Slack system requires students to register in Slack as the initial step to be an active member of the Slack Workspace *WeAre!*. Additionally, users will be prompted to enter their full name as well as display name (i.e. the nicknames for fellow members to recognize). After a user joins the Slack Workspace *WeAre!* successfully, I executed an automated call of the Slack API to access the member's user profile information [4], including their time zone and custom avatar image if available. Second, the LDAP NodeJS packages is used to access student information in the public Penn State Directory [5]. By querying this public directory, I am able to gather and store basic

[4]The information can be accessed via , which can retrieve a range of user attributes, including `email`, `first_name`, `is_bot`, `is_custom_image`, `last_name`, `last_updated`, `local_area`, `phone`, `real_name`, `status_text`, `title`, `tz_label`, `tz_offset`.

[5]Students' directory information, such as school campus, major and affiliation status is public at `http://www.work.psu.edu/ldap/`, and the information is obtained by making a query with the

information, such as Affiliation status with the university (e.g. Grad student, Staff), Campus and Curriculum Majors. All of this information is stored in the database `Users` along with Slack UserID, PSU email and full name.

### 6.1.2 Calculating Similarity Among Peers

I used Gower distance, a metric for computing distance between mixed types of variables, because I need to consider variables that are ordinal or binary (e.g. gender), along with other variables that are continuous (e.g. latitude and longitude in the geolocation information decoded from IP address). Gower distance is a technique for combining similarity metrics across different types of data [Kocher & Savoy, 2017]. Most mixed-data similarity metrics use the same approach –compute the similarity between each variable individually and sum the weighted average to get an overall distance metric. Gower distance uses both Manhattan distance and Kronecker delta. This works well with my data because Manhattan distance does not increase distance greatly for outliers. Gower distance in general has the advantage as to not take into account the length of the vector (i.e. the length of the student attributes). With my data, having a larger vector has no meaning, whereas in other applications like natural language processing a larger vector might be more meaningful (e.g., indicating that a text document has many words).

More formally, given vectors $\vec{a}$ and $\vec{b}$, the Gower distance between the vectors is defined as (6.1) where $d(a_i, b_i)$ is the distance between values $a_i$ and $b_i$ and $w_i$ is the weight given to attribute $i$. With this formulation, the distance value can be decomposed into contributions made by each dimension (or feature). When only the rank of the different distances is required, both the Manhattan and Gower measure return the same ordering. More broadly, properties of Gower similarity include:

- When all variables are quantitative or ordinal, Gower distance is equivalent to range-normalized Manhattan distance;
- When all variables are binary, Gower distance is equivalent to Jaccard distance;

students' PSU email or full name.

- When all variables are nominal, Gower distance is equivalent to the dice matching coefficient, obtained from recoding nominal variables into dummy variables.

$$Gower(\vec{a}, \vec{b}) = \frac{\sum_{i=1}^{n} w_i \times d(a, b_i)}{\sum_{i=1}^{n} w_i} \tag{6.1}$$

$$d(a_i, b_i) = \begin{cases} M(a_i, b_i) & \text{if variable at index i is numerical} \\ \\ K(a_i, b_i) & \text{if variable at index i is categorical.} \end{cases} \tag{6.2}$$

Where $M(a_i, b_i)$ is the range normalized Manhattan distance:

$$M(a_i, b_i) = \frac{|a_i - b_i|}{max_x(x_i) - min_x(x_i)} \tag{6.3}$$

And $K(a_i, b_i)$ is the Kronecker delta:

$$K(a_i, b_i) = \begin{cases} 0 \quad if \quad a_i \neq b_i \\ \\ 1 \quad if \quad a_i = b_i \end{cases} \tag{6.4}$$

To calculate Gower Distance for each vector of student user, I first encode ordinal attributes by converting them to numerical data, and then apply the aforesaid formula to generate the similarity vector (or distance index) for each student, which is then used to sort students in the *Table View* (see Section 6.4.1).

## 6.2 The WeAre! Workspace

As Figure 6.4 shows, there are different sections for user interactions in the Slack interface (the screenshot shows a desktop version). In section 1 (annotated with yellow color), there is a list of channels the workspace users are subscribed to. By clicking "Channels" at the top, participants can browse an existing channel; by clicking the plus icon on the top right of section 1, users can create their own channels. By

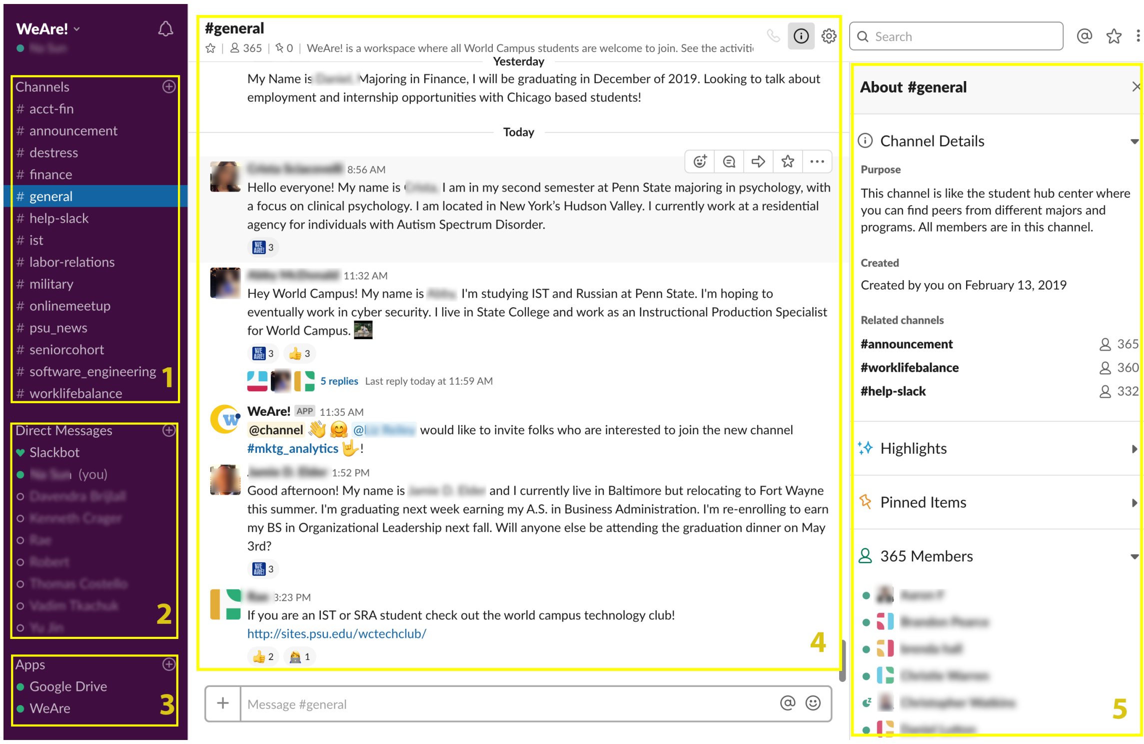



**Figure 6.4** Slack Interface Overview

default, users are enrolled in four channels immediately after they sign up as a Slack Workspace member in *WeAre!*. The four channels are `#general`, `#announcement`, `#worklifebalance` and and `#help-slack`[6]. In particular, I changed the default `#random` channel in any new Slack Workspace into `#worklifebalance`. Initially, I set out the purposes and topics of the channels as listed in Table 6.1.

In section 2, there is a list of direct messages the user has opened with individual members of Slack Workspace *WeAre!*; likewise, users can initiate a new direct message window by clicking the plus icon.

In section 3, users can see what Slack Apps are installed as part of this Workspace; they can also add their own apps with the plus icon on the upper corner [7].

[6]There are by default two channels automatically enrolled by users in any Slack Workspace. However, workspace admnistrators can customize the channels, names and purposes of any channel (except for `#general`), and configure more channels to be *automatically enrolled* channels for members.

[7]For the purpose of experiment control, I set the Workspace to require pre-approval by me before members can install a new app.

**Table 6.1** Purposes and Topics in four default channels

| Channel | Purpose | Topic |
|---|---|---|
| `#general` | This channel is like the student hub center where you can find peers from different majors and programs. All members are in this channel. | WeAre! is a workspace where all World Campus students are welcome to join. See the activities in the dashboard: `https://weconnect.ist.psu.edu:8443/` |
| `#worklifebalance` | A place for non-work-related flimflam, faffing, hodge-podge or jibber-jabber you'd prefer to keep out of more focused work-related channels. | Type /intro to introduce yourself with the help of our Slackbot `WeAre!`, and win a lot of heart and We Are!. Talk about the fun or confusions you encountered today. |
| `#announcement` | This channel is used to announce system updates/maintenance, feature revisions, research explanations and other activity related guidelines. | Announcement: We will announce important software update, and share the tutorials, such as GIF materials to help you use the tools here. |
| `#help-slack` | Getting technical help of using Slack from the administrators or your peers | To initiate a discussion with your peers, use slack command /wediscuss |

In section 4, users can interact with other members of the Slack team by plain text, or text with emoji decorations; In addition, users can react to others' messages (either in text or emoji) with emoji (as shown in Figure 6.4). In the screenshot, section 4 shows the content of channel `#general`. Note that the conversation structure is organized with threads, with replies to a message being folded in a stack, and can be opened when user clicks the [number] replies (3 replies in the case of the Figure 6.4 section 4). Each message is marked with number of emojis that it receives as reactions from peer members. Also, Slack members can upload images as they like as their custom profile pictures.

Finally, section 5 shows the detailed information about the opened channel, such as the purpose of the channel, creator of the channel, time when it was created, and a list of members who subscribed this channel.

In general, Slack offer a variety of customization for the Workspace or individual status. As I initially laid out the space, I customized the Workspace Icon with a logo I designed for *WeAre!* that include the W initial surrounded by a moon sign (symbolizing reunion) . I also added a customized response so that when someone says "WeAre!", the Slackbot will respond with "Penn State!". And, I added custom emoji, including a Nittany Lion [8], a PSU avatar , PSU love , and "WeAre!" .

## 6.3 Functions of Slackbot `WeAre!`

In my interview study of WC students, I found that the phase of introduction in a class setting is critical for peer learners to get to know one another virtually, and that this activity can also establish an initial affinity among peer learners based on shared identity [Sun et al., 2019]. Therefore, I decided to leverage this practice to encourage *WeAre!* members to get to know one another. I designed support for this use case with the Slackbot `WeAre!` (also customized with  as its bot avatar).

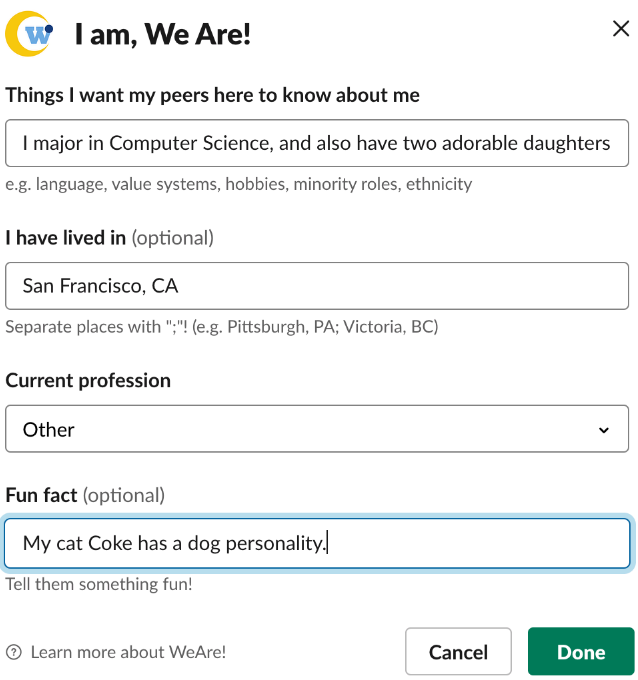
As shown in Figure 6.5 (a), I designed both an interactive Slack command and a custom button, offering two ways to initiate a introduction form; the form is presented as a modal dialog within Slack. The form includes four input fields: "*Things I want my peers here to know about me*(required field, 150 characters at most)," "*I have lived in* (optional)," "*Current profession* (optional)," and "*Fun fact* (optional)". The user submits the form by clicking the "Done" button (colored in green).

Once a user submits the introduction form, Slackbot `WeAre!` sends a private message to the user in the current channel (see "Only visible to you" in figure 6.5 (b)), encouraging the user to authorize Slackbot `WeAre!` to post a public message in this channel that will introduce himself or herself. In the meantime, a second direct message from the Slackbot `WeAre!` is sent to the user (see Figure 6.5 (c)). This second message includes a prompt that invites the user to fill out more information about himself or herself and a URL that will connect them to the web application *WeConnect*. (see further details in Section 6.4).

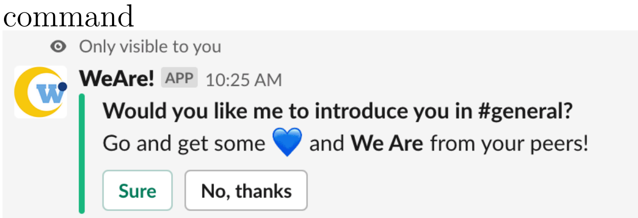
If a user agree to be introduced, Slackbot `WeAre!` sends a public message visible to all members of the channels (see Figure 6.5 (d)[9]). This message includes two options for others in the channel to respond: they can either click the We Are!

[8]Nittany Lion is a mascot of Pennsylvania State University.

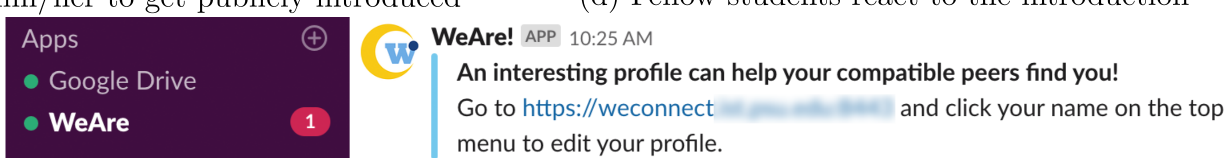
[9]I blurred the actual names and user names to keep anonymous for our participants.



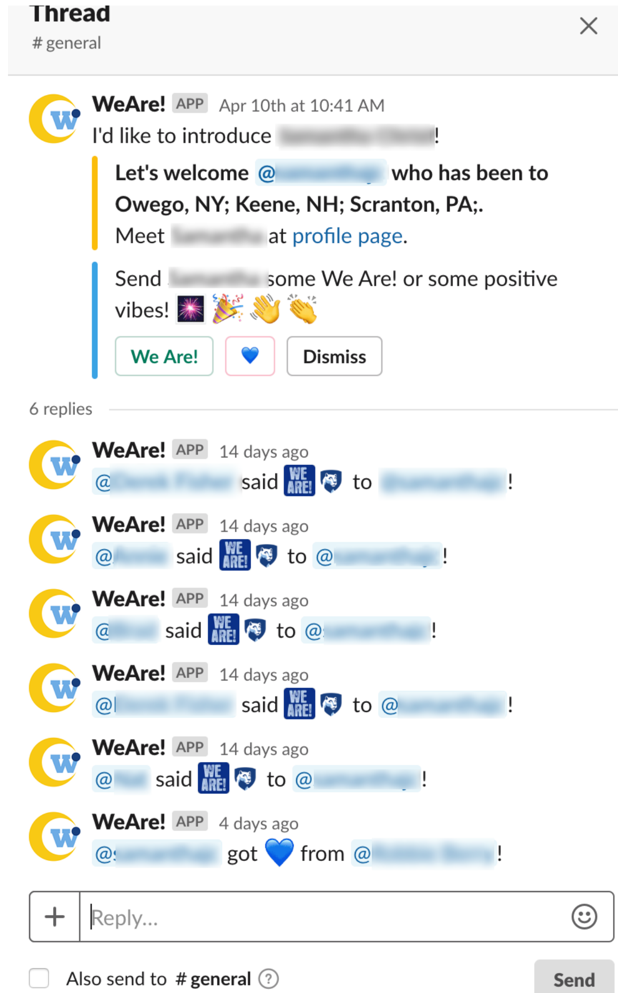

(a) Dialogue interface after initiating the /intro command




(b) User will receive a private prompt message that invites him/her to get publicly introduced

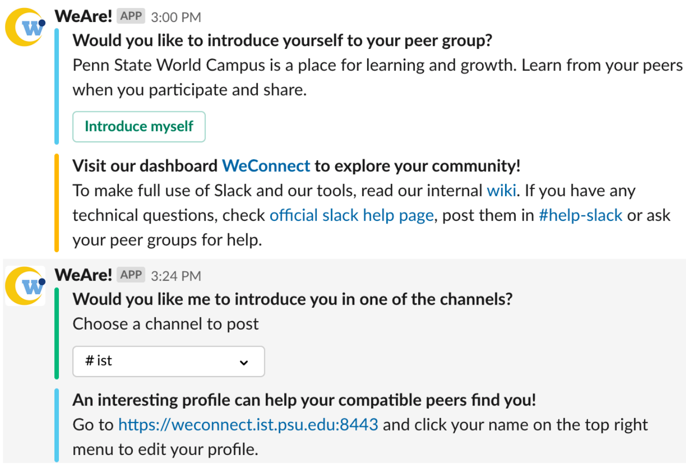



(c) User will receive a direct message from *WeAre!* after /intro

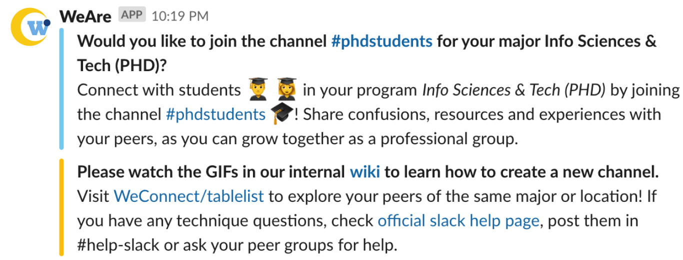



(d) Fellow students react to the introduction

**Figure 6.5** Slackbot `WeAre!` help the new members introduce themselves and receive welcome from their peers




(a) `WeAre!` prompting users to introduce oneself




(b) `WeAre!` prompting users to join or create channels

**Figure 6.6** Slackbot `WeAre!` prompts users to take actions to socially engage with others

button, click the "blue heart" emoji 💙, or simply click "Dismiss". The selection of "Dismiss" serves as a signal to log users' attitudes towards public introductions of this sort; it does not trigger any change in content. If a user clicks "We Are!", both the sender and recipient are mentioned by the Slackbot, remarking on this friendly gesture. Note that these bot-generated conversations are prefaced with the "WeAre!" slogan and a Penn State Nittany Lion icon ; they appear as replies to the public introduction message. Further, students can follow up with one another in text or with emoji under this introduction also as replies.

In addition, I sent out prompts on behalf of Slackbot `WeAre!` for users to introduce themselves every other day only if new users have not introduced themselves and also have never received such prompts (in other words, each user got such prompt once at most). As shown in Figure 6.6 (a), students can be triggered by Slackbot with introducing themselves even without using slack command to initiate the process. Further, `WeAre!` guides users through the steps of being publicly introduced by the Slackbot, so as to share the information the user just entered with fellow students. I also sent prompts to invite students to join or create channels of their majors or interest groups (also for once at most).

## 6.4 The WeConnect Dashboard for Visualizing Peer Information

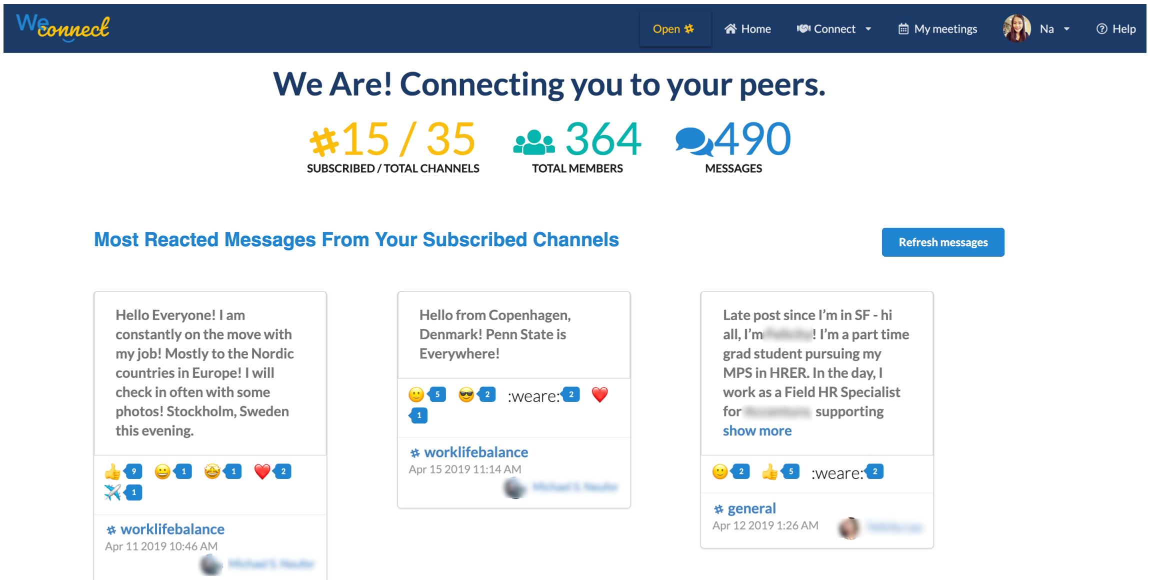



**Figure 6.7** Slack Team information and popular posts

*WeConnect* is a visual dashboard that I designed as an external portal to visualize students' information as members of the Slack *WeAre!* workspace and as WC students. The web application is built with Nodejs as a backend server, and MongoDB as the database to store users' information and logged activities.

The homepage of *WeConnect* highlights information and activities of Slack Workspace *WeAre!* (as shown in Figure 6.7). This includes the number of subscribed channels of the currently active user versus the total number of channels; the total number of members in the *WeAre!* workspace; and the total number of messages. The most popular messages (those receiving the highest number of emojis or other reactions from members) and the most recent channel activities are also shown in the homepage. To make the interface clean and organized, I truncate Slack messages longer than 6 lines to display "show more" as an option to unfold a message's full content.

Users can navigate to other pages of *WeConnect* through the menu bar choices at the top. "Connect" offers a visualization of peer information that may help the students find other peers based on their educational interests; “My meetings” provides meeting schedule support. Users who select their name and avatar can complete or revise their *WeConnect* profile information. To gain further instructions to use the website, users can click “Help” button, which offers information about how to use Slack Workspace *WeAre!* as well as the web application *WeConnect*. I describe the data sources and information structures in detail in the Section 6.1.1 and Section 6.1.2; I also characterize the design of other two main visualizations *Table View* in Section 6.4.1 and *Availability View* in support of peer information exploration in Section 6.4.2.

### 6.4.1 Browsing Peer Information in a Table View

Because I found earlier that students would prefer a structured representation of peer information (see Section 5.2.3), I visualized students’ commonalities with other peers as sortable table structure, as shown in Figure 6.8. All members in Slack Workspace *WeAre!* appear in a table with four columns, including (full) Name, Major, Location and Channels. The table headers can be used to sort the records. As an example Figure 6.8 shows the situation when members are sorted alphabetically by degree program name. If a person shares majors with another active user, that major will be color coded with orange; if a person shares Location with another active user, the Location will be color coded with red (as shown in the top of the legend). Similarly, channels shared with another active user will be color coded with purple. The table is paginated with 10 records per page, and supports search on specific fields (e.g. major, name) or across all fields.

### 6.4.2 Visualizing Time Zones and Presence Status

Another representation of peers’ information is provided by the *Peer Availability View* (Figure 6.9), which emphasizes on the temporal dimension of the students’ information. Two aspects of peers are represented here: First, all the Slack members of the Slack Workspace *WeAre!* are clustered into spatially-arrayed groupings based on timezone. Each group is marked with corresponding time in that timezone at the moment of visiting the page; the corresponding time zone for the active user

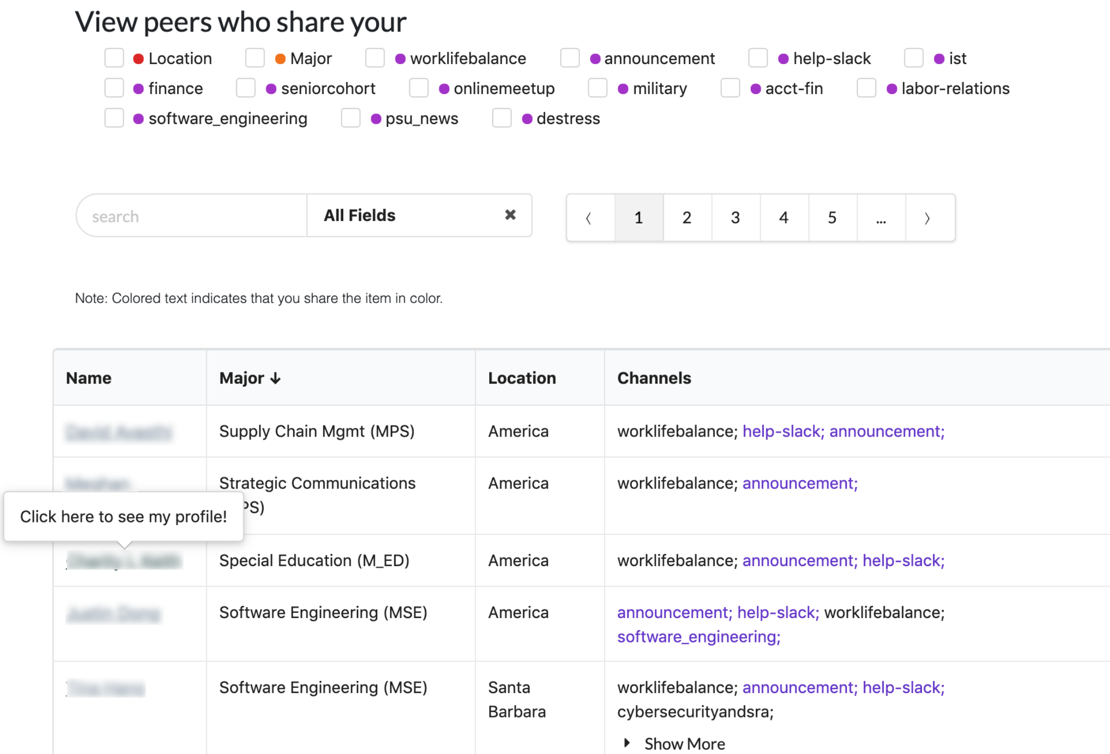



**Figure 6.8** Browsing Peers' information in a Table view

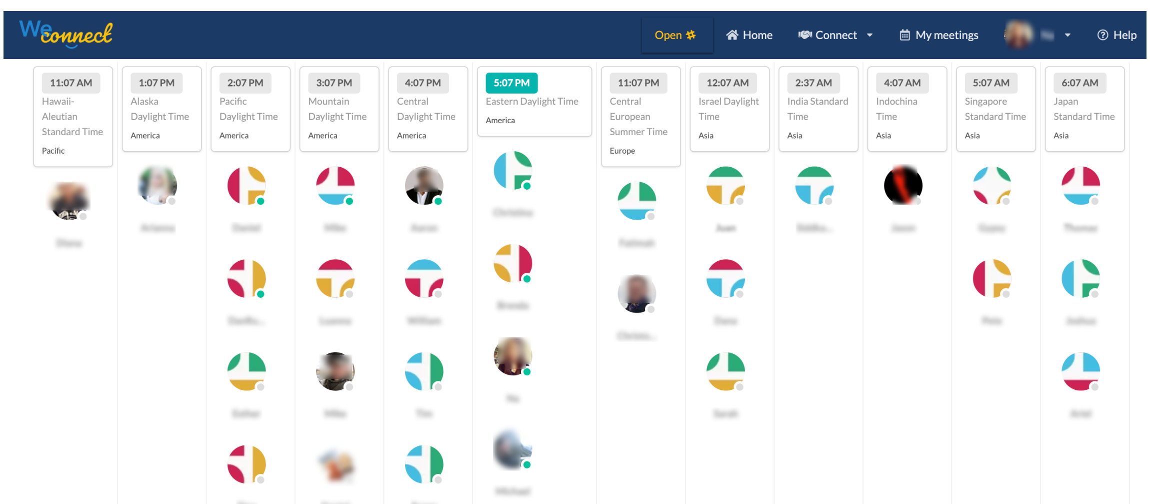



**Figure 6.9** A visualization of peer students' time zone distribution and presence status

is marked with Green. Second, Slack members who are currently present online are decorated with a green dot at the right bottom of the avatar. A scenario this particular view hopes to support is when a student wants to ask a question at the moment and wants to first to check who is available. Moreover, individual users can find out other fellow students' availability information in others' profile view if the information is filled out. In addition, users can also go to the "Peer Availability" tab that is part of the Connect menu in the navigation bar. In the Availability view, users can easily see other peers' temporal availability based on their current online status, and their local time in different time zones.

## 6.5 Design Summary

Now I have described the information architecture of *WeAre!* (including both basic and custom functions provided by Slackbot `WeAre!` in the Slack Workspace) and *WeConnect*. I also characterized the design and customization of Slack Workspace *WeAre!*, Slackbot prompts, and social features in Slack and the major views supported in *WeConnect* in detail. The features and information flow described in these tools delineate the technological background, especially the social affordances for the tools I deploy for field study. The affordances and constraints of using such tools set up the scenes for further evaluations of such major functions in Chapter 7.

# Chapter 7 Evaluating WeAre! and WeConnect

In this chapter, I describe my evaluation of the WeAre! and WeConnect tools, which I carried out as part of its deployment to the WC student population. My research goals for this deployment and evaluation phase were multiple:

- *Characterize the WC student population as a community and determine how these student characteristics predict their interest in WeAre!*
- *Assess the self-reported impact of using the new tools*
- *Document and reflect on how WeAre! and WeConnect are used by Workspace members*

## 7.1 Evaluation Methods

Most of the data I gathered regarding deployment and evaluation came from surveys distributed before and after the tools were launched I also gathered some information through the logging of activities in *WeAre!* but for technical reasons these data files were not robust and have not been analyzed to a great extent.

Before launching the tools, I invited all WC students to participate in a baseline survey (implemented in Qualtrics); I did this by sending a mass email invitation. The survey began on on March 30th, 2019 and was open for 15 days (closed April 14th, 2019). During that time I collected a total of 509 survey responses from total email list of just 12,245 active WC students [1]. The list was obtained from the Outreach Analytics & Reporting group of World Campus.

I offered the opportunity to join a WC online community space (i.e. Slack Workspace *WeAre!*) and the chance of getting a $30 Amazon giftcard as an incentive for completing the pre-survey[2]. Participants were invited to complete the survey with the promise that they will see the invite link of joining Slack Workspace for World Campus at the end of the survey.

After this first survey was closed (12 days of data collection), I again distributed a survey to the entire contact list, but in this case simply included the invitation to join the Slack Workspace *WeAre!*.

Finally, I distributed a post-usage survey on April 27th; this survey went to the union of participants who had responded to the pre-usage survey or signed up in the Slack Workspace *WeAre!*, leading to a total of 703 students as the target population. The post-usage survey was designed to provide data for a pre- and post-usage comparison; these comparison data were collected for both the group of users who had used Slack Workspace *WeAre!* and those who had not. In the case of students who had used the tools, the survey also contained questions collecting feedback about the system. To incentivize students' participation in the post-survey in the later of a spring semester, I provided a lottery of 15 winners of Amazon giftcards (each is worth $50).

I configured the survey flow with logic such that only students who previously took the pre-usage survey were able to complete the community scales again, and also such that only students who registered their PSU account in Slack Workspace *WeAre!* were able to see questions related to Slack Workspace *WeAre!*. All participants of

[1] The list includes students who are in a major offered through WC, and who were enrolled in spring 2019 in a WC course. Non-degree seeking students, students in a certificate program, students under the age of 18, and students who show a permanent or local address in a European Economic Area (EEA) member country are excluded. Students on the WC "Do Not Contact" list have been removed from the list as well.

[2] Student participants will be entered into a drawing for a $30 Amazon gift card; I promised to make 15 awards in total for all participants. I also promised that student participants of the post-survey will have the chance of getting an increased value of each gift card ($50), again among a total of 15 awards.

the post-usage survey were asked about their experiences of using other tools to support their social experiences, and how they wish to improve on these tools. The post-usage survey was open for 3 weeks (closed May 19th, 2019)[3].

## 7.2 Pre-Usage Community Perceptions

### 7.2.1 Participants

The preliminary survey yielded responses from 248 females and 197 males[4]. Of these, 320 are Caucasian, 36 Hispanic, 35 Asian, 22 Mixed race, 20 African American, and 14 others. Tree fourths of the participants are not a member of Blue and White Society (referred later as *BWS*, n = 332), and 118 students reported part of the BWS. 236 enrolled in bachelor degree, 178 in master degree; 23 in Associate degree, 8 others, 4 doctoral.

When asked about proximity to Penn State local campus, 246 participants self-identified as living "in the same country with Penn State", 109 "live in the same state with Penn State," 52 "live in the same town/city with Penn State", 16 "live in the same continent with Penn State," and 27 chose "None of the above". This suggests that only a small proportion of my participants live far away from the residential campus.

Instead of distributing the surveys separately with distinct definitions for community of learners separately as I did in Chapter 4, I offered the two definitions as options in a single choice question before asking any other questions about community. Participants chose between *"All students currently enrolled in any World Campus degree program"* (Virtual Campus Community or VCC) and *"World Campus students whom you have become acquainted with, regardless of whether they are now in a class with you."* (Acquaintance-Based Community or ABC). In case neither of the definitions seemed correct, participants could also choose a third option *"Other (specify)"* to capture other variants of community definitions.

[3] See Appendix D.2) for reference to the full description of the survey content and display logic.

[4] Note that not all participants answered every questions, especially for demographic information, which is placed at the end of the survey. The same goes for the other variables, where the sum of participants reported may not add to the 509

### 7.2.2 Pre-Usage SOC and CCE

Participants who selected the VCC definition (n = 382) outnumbered those who chose the ABC definition (n = 108). Interestingly, in contrast to what I observed in [Learning or lurking? Tracking the "invisible" online student, Sun], a between-subjects comparison of participants who chose VCC or ABC had differing community perceptions. Namely, ABC participants reported stronger community perceptions, across all factors for SOC and CCE measurements ($p < .05$; see Table 7.1.)

Only 19 participants chose *"Other (specify)"*. With a closer examination of the 19 responses, I found many of this small group mentioned the inclusion of faculty and staff who are also involved in the online learning and teaching activities, as well as alumni who might not be currently enrolled (as I explicitly specified in the predefined option). One participant even said that the community should include students at the physical campus, as long as they take some online courses. Only very few described a definition of community as only close partners with whom he or she interacts (e.g. teammates). Given the definition of community on the spectrum from very close and acquaintance-based members to very broad and identity based members, I can see that World Campus students indeed perceive community in different ways, and that a large proportion consider the community concept as an overarching entity, including individuals beyond current World Campus students.

**Table 7.1** Community construct means for VCC and ABC

| Type (#) | Common Identity | | Friendship | | Identity Regulation | | Coordination | | Social Support | |
|---|---|---|---|---|---|---|---|---|---|---|
| | M | SD | M | SD | M | SD | M | SD | M | SD |
| **All (509)** | 3.47 | 0.88 | 2.19 | 1.04 | 4.15 | 0.71 | 3.53 | 0.88 | 3.69 | 0.88 |
| **VCC (382)** | 3.40 | 0.87 | 2.05 | 0.97 | 3.80 | 0.69 | 3.15 | 0.86 | 3.32 | 0.86 |
| **ABC (108)** | 3.73 | 0.84 | 2.65 | 1.14 | 3.86 | 0.71 | 3.51 | 0.83 | 3.61 | 0.86 |
| **Other (19)** | 3.16 | 0.97 | 2.32 | 1.12 | 3.82 | 0.92 | 3.39 | 1.02 | 3.50 | 1.00 |

### 7.2.3 Time at will

Bean & Metzner [1985] extended Tinto's persistence model, making it more applicable to nontraditional students by considering environmental variables that can account for the added complexity of nontraditional students' lives. Persistence is thought to be a function of background characteristics (demographics, etc.), academic and environmental variables (study habits, financial resources, work and family obligations, etc.), and academic and psychological outcomes (GPA, satisfaction, etc.). Therefore, to examine the moderating role of time deprivation, a common explanation for adult learners' situations, and the effect of other distractions in life, I included a few survey questions probing how the students perceive their free time availability, and time used for job and personal or family matters [Ackerman & Gross, 2015]. On average, participants reported 14.74 hours of free time each week ($SD$ = 12.16, n = 450), and spent 38.21 hours on work ($SD$ = 18.35), in addition to 26.5 hours with families ($SD$ = 21.94).

The survey also asked students to rate agreement with a set of time-related Likert items that explored the perceived effect of time on their school performance and peer connections on a 5-point scale. In particular, on a range of *"Strongly disagree"* to *"Strongly agree"*, I asked for ratigs of *"I feel as if I need more time to do school work."*, *"I need more time for my classes."*, *"I could perform better in school if I have more time."*, *"I could build connections better with my peers at Penn State if I have more time."* Although these items all relate to the perception of time for online students, I am particularly interested to see the mediating effect of such perception on the association of time and the outcome of social connections.

## 7.3 Joining Slack Workspace WeAre!

As of April 23 2019, 301 students had joined Slack Workspace *WeAre!*. Among these, 144 had completed the pre-usage survey (i.e. 365 students completed the pre-usage survey but did not join the workspace; 167 students joined the workspace but did not complete the pre-usage survey). As Table 7.2 indicates, the SOC ratings for students who joined *WeAre!* and those who did not were quite similar. The CCE factors seem to vary more across the two groups, but there was no reliable differences in any of these means (none of the Mann-Whitney U tests were significant).

In a secondary analysis, I found that students' community perceptions did not predict the likelihood of joining *WeAre!*; none of the corresponding logistic regression models was statistically reliable. Controlling for participants' learning background, progress in the online degree program, proximity to the main campus and other demographic information, I explored the usefulness of other student variables –time-related variables, community definitions, and five constructs of community – in predicting whether they joined *WeAre!*. I found four variables that had significant predictive power: belief that more time will allow for better connections they can build with their peers; [5] whether the student is Hispanic (one level of the ethnicity variable); to what degree s/he has finished the current degree, and age (i.e. the year they were born as listed in the Appendix) (Pseudo $r^2 = 8.80\%$, $p < .05$)) I assessed a reduced model that included just those four variables, in addition to the community definition variable, and found that all the variables except for the age have a significant effect in predicting the likelihood of joining *WeAre!* (Pseudo $r^2 = 6.615\%$, $p < .001$).

**Table 7.2** Community constructs for WeAre! users and non-users

| Type (#) | Common Identity | Friendship | Identity Regulation | Coordination | Social Support |
|---|---|---|---|---|---|
| **WeAre! (144)** | 3.45 | 2.12 | 4.13 | 3.54 | 3.66 |
| **Non-WeAre! (365)** | 3.47 | 2.22 | 3.69 | 3.12 | 3.28 |

## 7.4 Using Slack Workspace WeAre!

In the post-usage survey, I repeated the questions about community perceptions (if the respondent had participated in the earlier survey); the perception of time deprivation; physical proximity with Penn State; and affiliation with the online student club. In a second survey block I also probed experiences members had with the *WeAre!* tools [6]. As a complement to self-reports of usage patterns and tool reactions, I also collected user log data in Slack *WeAre!* and the visual dashboard *WeConnect*. I consider all of these sources of data in the next few sections.

[5]See Appendix B, Table B.2 for the item: *"I could build connections better with my peers at Penn State if I have more time."*

[6]See Appendix D.2) for reference to the full description of the survey content and display logic.

### 7.4.1 Post-usage Survey Participants

Note that although only 144 pre-usage survey participants joined *WeAre!*, I extended the definition of users to include students who joined the Workspace after the pre-usage survey closed. Through this extended recruiting effort, I collected post-usage responses from 81 members of *WeAre!*, 49 of whom had contrbuted pre-usage survey data. In parallel, I collected responses from 73 non-users; these participants serve as my control condition, as the only "intervention" they experienced was completing the scales assessing community perceptions and general questions about tool use.

In the end, I received 154 responses to the post-usage survey, 88 females and 62 males. A majority are Caucasians (70.20%), followed by Asian (8.61%), Hispanic (7.28%), Mixed race (5.30%), and African American (5.3%) (3 individuals chose Others). Their age ranged between 18 to 68, reflecting great diversity in this variable (Median = 34). Table 7.4 offers a more detailed description of the post-usage survey participants, including a breakdown by users and non-users of *WeAre!*.

For students who participated in both the pre- and post-usage surveys, a majority (n = 95) had chosen the VCC definition of community; 22 had selected ABC. To further probe community-like experiences, I also included a question that listed other community platforms in response to *What other tools have you used as part of your World Campus activities to interact with others?*. The options include *World Campus Student Center* [7], *LionLink*, *LinkedIn*, *Facebook*, *Twitter*, *Yammer*, and *Other*. As seen in Table 7.3, most participants use World Campus Student Center, followed by Facebook and LinkedIn.

**Table 7.3** Post-usage survey participants' use of other community venues

| Venue options | WC Students Center | LionLink | LinkedIn | Facebook | Twitter | Yammer |
|---|---|---|---|---|---|---|
| # of participants | 43 | 8 | 36 | 37 | 5 | 14 |

[7] World Campus Student Center is a discussion board within Canvas. It is open to all Penn Students to actively enroll, but the topics are all about World Campus students and activities.

**Table 7.4** Post-survey participants' demographics for both Slack users and Non-slack users

| **Type (#)** | **Gender** | | **Ethnicity** | | **Age** | |
|---|---|---|---|---|---|---|
| **Conditions** | Female | Male | White | Others | *M* | *SD* |
| **All (154)** | 88 | 62 | 106 | 45 | 35.66 | 10.23 |
| **Slack (81)** | 42 | 35 | 53 | 23 | 35.12 | 10.04 |
| **Non-Slack (73)** | 46 | 27 | 53 | 18 | 36.25 | 10.47 |

### 7.4.2 Experiences Using the WeAre! and WeConnect

Among the post-usage survey participants who had used *WeAre!* (n = 81)[8], 31 reported that they accessed *WeAre!* primarily through a Web browser (38.75%), 28 through a Mobile app (35.00%) and 19 through the Desktop (23.75%). In response to *"how competent you are with use of Slack prior to exposure to WeAre!"* on a 5-point Likert scale (with 1 being *Extremely incompetent* to 5 being *Extremely competent*), participants reported that they were somewhat incompetent ($M = 2.60$, $SD = 1.45$).

In terms of how frequently they used *WeAre!*, 14 participants reported that they visited at least once a day (either *"Every hour"*, *"Every few hours"*, *"Once a day"*); 20 visited *"A few times a week*; 24 visited on a weekly basis (i.e. *"Once a week"*, *"Every other week"*; and 22 participants reported few if any visits (*"Only when registered"*, *"Never"*).Figure 7.1 summarizes these responses. Given this distribution, I collapsed the eight frequency levels into four, labelling them from 0 –3, so as to simplify their use in other analyses.

During their visits to *WeAre!*, a large majority of users spent 1 –10 minutes (n = 54); 10 participants reported visits of less than 1 minute; 11 said they stayed for more than 11 minutes. Most participants reported to have never used private messages in *WeAre!* (n = 59); 13 reported *"Sometimes"*, 2 *"Half of the time"*, and 1 *"Most of the time"*. As for visit frequency, I collapsed these responses into a binary variable (1 means *Have used*, 0 means *Never*).

17 participants (23%) reported that they had visited *WeConnect*; 57 said they had not. If a respondent chose *"Yes"* when asked about visiting *WeConnect*, she or he was also presented with three follow-up questions probing reactions to the three

[8]these individuals were identified as users by matching their PSU email associated with the survey response to the PSU email they used to register in Slack. Note that even though they registered for the Workspace, they might not actually have used it.

different views of this tool (homepage, table list, temporal view), as well as one more question asking about the extent to which they think *WeConnect* helped them to connect with others. After examining these open-ended responses about their thoughts of these designs, I found that most students explored any views beyond the home page. However, most participants did understand the intent of the *Table View* and *Temporal View* in terms of highlighting commonality among students and the filtering mechanisms (screenshots were provided and they were able to respond to these in their comments).

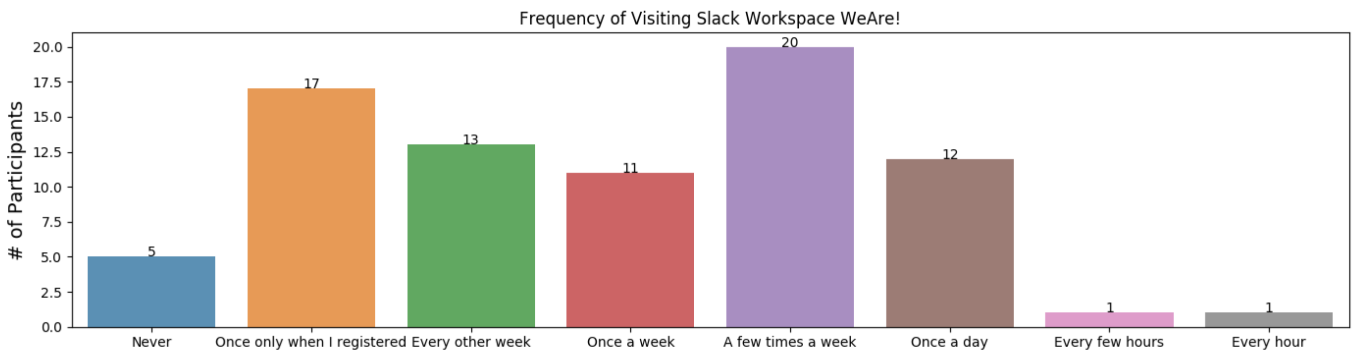



**Figure 7.1** Self-reported frequency of visiting Slack Workspace *"WeAre!"*

#### 7.4.2.1 Perceived Effects of Using WeAre! for Community

Several survey questions solicited feedback about using *WeAre!* for the purpose of connecting with wc peers. To measure the helpfulness of *WeAre!*, I asked a single question *"To what extent do you think Slack Team Space 'WeAre!' helps you connect with your peers?"*, using a 5-point Likert scale ranging from *"None at all"* (0) to *"A great deal"* (4). On average, the reported helpfulness of *WeAre!* was $M = 1.45$, $SD = 1.08$. See Figure 7.2 for the frequency count of participants in reporting the helpfulness.

To uncover the reasons behind these ratings of helpfulness, I used a follow-up question: *"Please explain why 'WeAre!' fails to connect with your peers."* for participants who chose *"None at all"*; and *"How did you use 'WeAre!' to connect with your peers?"* for those with a higher rating. After analyzing these responses for themes, I found several general reasons for either believing or not believing that *WeAre!* is

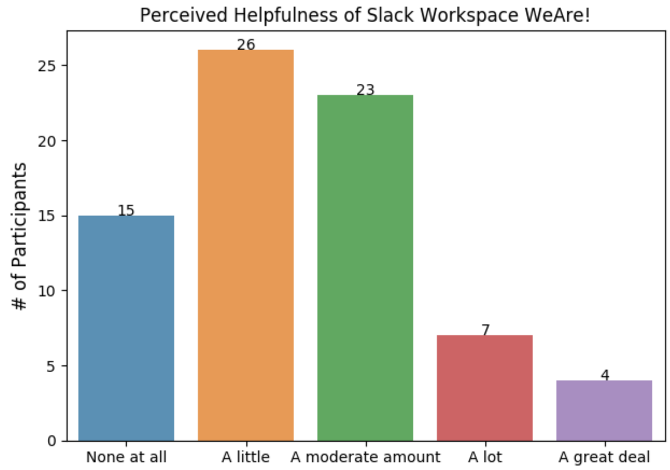



**Figure 7.2** Self-reported helpfulness of *WeAre!*

helpful in making connections. With respect to those who saw its usefulnes, the themes reflected different sorts of community engagement:

1. *Conversing.* Quite a few participants reported to have developed connections by engaging with others through initial greetings (e.g. introduction), asking questions, participating in the discussions in the channels, and occasionally exchanging private messages [9].

   Some participants asked questions about professors or recommended classes to take, whereas others asked general questions about work-life balance strategies, and ways of developing their careers. Likewise, some other participants actively posted information to share with other fellows, or commented on others' post out of interests or out of acknowledgement of others' efforts (e.g. *"I just commented on other students messages within the same program as me"* (P56) (for more details, see the content analysis in Section 7.5.3).

2. *Consuming the content.* The most common but least visible way of engaging with the community was found to be reading the content from *WeAre!* conversations. Some participants read the content to keep up with their peers:

[9] Although only 16 participants reported to have ever used private messages, compared with 59 student Slack users in *WeAre!* who had never used private message, there are some users who reported benefits of privately messaging with others.

*"I used 'WeAre' to try and keep up with what classes my peers are taking and how far along they are in their academic career."* (P59) In particular, many students reported to enjoy the initial greetings during the introduction phase. For example, happily lurking in the Workspace without actively engaging in conversations, P130 reported to have *"enjoyed seeing students from various programs and regions introduce themselves."*

Other participants browsed the feeds to check for useful information, such as *" textbooks for sale."* Still others enjoyed lurking as content consumers, because it contributes to their community feelings yet also avoids the fears of speaking in public: *"I enjoyed reading the comments by others and it made me feel closer to them. I haven't posted anything, but I would be afraid to either."*

Interestingly, while not all participants check *WeAre!* often, some are satisfied with a daily email summary to catch up with the community conversations. For example, P143 complained that learning to use Slack was difficult, but enjoyed reading emails daily: *"Signed in and was confused what it is. Haven't used it since, but have been reading the daily emails."*

3. *Reaching out and reaching in.* Some members of *WeAre!* created their own channels to reach out to others based on their personal interests. For example, P88 shared how she oriented herself among the different channels or discussion groups: *"You can start a chat and others with interest in that topic [and] can post messages to others on the board. It is an easy way to connect to others regarding any variety of interests and topics, school related or extracurricular, or job."* Specifically, many participants created channels to foster a sub-community around their majors or education programs.

   Other participants simply appreciated these options for more focused communication. For example, *"I like that channels for individual concentrations appeared."* (P130) Accordingly, participants also reported that they intentionally joined channels that fit their interests and information needs. For instance, P20 said *"I joined the military, senior cohort, and energy and sustainability policy groups in addition to the general and work/life balance groups I was automatically enrolled in. I have had a few interactions thus far, but it seems as* *if 20% of the participants are doing most of the posting."*

I also examined the explanations from participants who did not find the Slack Workspace *WeAre!* to be helpful in making connections with others. These explanations tended to revolve around usability, mismatch of interests, or simply not having enough time to get started.

1. *Confusing interface.* Some participants complained about their confusions with the Slack user interface, for example that it is hard to navigate through numerous posts from others. P30 failed to connect with others simply because he was *"not sure how to use it"*. P52 also expressed the difficulty of following up on a conversation, and prefesr to use *"the WC Psychology Club's page on Facebook which is a lot easier for me to use/keep track of"*. The need to read messages over long conversations or very active posts might also drive some Slack users away, as P29 shared: *"Slack seems to be overwhelmed with information and people talking. I don't want to continually scroll and scroll just to read things and find information, so I find myself uninterested."*

2. *Lack of interesting conversations.* Some participants found the information in *WeAre!* to be barren, considering their focused interests in their own level of education or their specialized areas. For instance, two graduate students (P74 ad P75) cited their identity of "graduate students" and emphasized their interests to see more of such comments coming from those at the same "graduate" levels, although one of them did like *"seeing the notifications and what others were talking about though."* Likewise, there were also lots of comments about *hoping to connect with peers of their major or program.*

   Other participants felt lost because of lack of clear purpose in connecting with others using these tools, or explicit incentives for doing so. In addition, the fact that the tools were being launched at the later stage of a semester might also affect the passion to fully engage with other students, as P27 argued below:

   > *"At least in this trial, there wasn't a lot of context or purpose given, and it was partway through spring semester when life and assignments were taking priority. Possibly if launched at orientation week,*

> *or with more engaging activities than webinars, it could be different. Or if rolled out as a tool for students to use to work on assignments instead of selecting between text, email, canvas, box, google docs, and zoom."*

Without such purpose, participants might hesitate to comment as shared by P24, *"There was not much incentive to do so. There were many students with a large variety of interests but few reasons to engage in conversation."*

3. *Short-time use and lack of time.* Some participants were still planning to participate: *"I have not utilized it to connect with my peers but intend to do so."* (P90). On the same tone, many participants ascribed the fact of having not fully engaged with the platform to the time constraints in living a busy life, as P90 shared in her explanation: *"I was very busy with family life, work, and school work so I did not feel that I gave this experiment my best shot. However, I think it is a wonderful idea to connect with peer and feel like a part of the Penn State community."*

In alignment with the perceived helpfulness, only 13 participants reported making new connections through *WeAre!*; 59 reported to have not made such conections (this was in response to *"Did you make any new connections using Slack Team WeAre?"*). However, given the short period of deployment and intentional choice of not intervening too much, it was still impressive to hear some affirmative answers about new connections built with the help of *WeAre!*

Through the follow-up *"How did you establish and follow up with that connection?"*, I found that *WeAre!* users often took their conversations from public channels or live sessions into private conversations, sometimes even to the state of exchanging phone numbers. Occasionally, they also followed up with previous classmates using a brief greeting. For instance, P19 shared that *"It made me think about some of my classmates in the courses that I have taken and I followed up with them about how they were doing."* Some other students found new connections by creating a new channel to create a specific focus, although sometimes further interaction at a meaningful level might require consistent maintaining efforts from both channel creators and the participants, which could be a challenge, as P25 disclosed, *"Established [the connections] through a new group I created; I did not adequately follow up. The process feels barren; the interaction is unnatural."* P88 noted that such efforts may require *"people consistently participating in this forum."*, echoing the well-known issues of critical mass and associated activity.

In addition to perceptions of community, I asked participants about whether they might introduce the *WeAre!* tools to their peers (a 5-point Likert scale, 0 being Extremely unlikely, and 4 Extremely likely). On average, participants reported their likelihood to share to be in neither likely nor unlikely, $M = 1.93$, $SD = 1.30$, suggesting a rather neutral attitude.

#### 7.4.2.2 Preferences for WeAre! Features

61 participants responded to the ranking question of six major features provided by *WeAre!*, including basic features inherent in the Slack platform and enhancements I built with Slack API. Specifically, I asked *"What features in Slack Workspace WeAre! do you find useful in connecting with this particular community? (From Most Useful to Least Useful);"* Six features were listed in a ranked-list question: *"Introduction and welcome messages from the bot WeAre! "*, *"Prompts of creating or joining Slack channels from the bot WeAre!"*, *"Public channel messages from members of the Slack Team WeAre!"*, *"Private messages among members of the Slack Team WeAre! "*, *"Entry reminder to visit the dashboard WeConnect (e.g. filling out profile, visit the time zone distribution of peers)"*, *"Sending and receiving emoji reactions to Slack messages"*. The rankings provided by each person were translated into an ordinal scale of 1 (highest) to 6 (lowest).

Table 7.5 summarizes the mean ranks and standard deviations for the six features; Figure 7.3 shows the percentage of users who chose a given feature as the most useful (rank 1). As can be seen in the Table and Pie chart, most participants gave a high ranking to the introduction provided by the Slackbot `WeAre!` (designed by me); followed by public messages (basic Slack feature), and the *WeConnect* dashboard prompt.

To further unravel the reasons for the feature usefulness rankings, I asked them to explain their rationale for their ranking decisions. Again, based on thematic analyses,

**Table 7.5** Usefulness ranking for six Slack features

| Feature types | Intro by Bot | Emoji | Private message | Channel prompt | Dashboard prompt | Public message |
|---|---|---|---|---|---|---|
| *M* | 3.02 | 3.98 | 4.11 | 3.62 | 3.54 | 2.70 |
| *SD* | 1.79 | 1.66 | 1.59 | 1.72 | 1.59 | 1.52 |

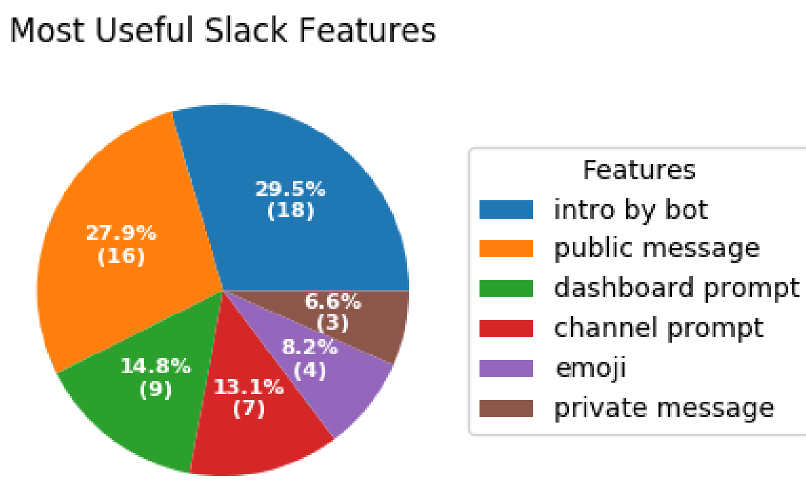



**Figure 7.3** Percentage of most useful Slack features

I learned about both the benefits and risks of using a custom Slackbot for community building, especially at such an early phase. Before going to the details, I cite P59's comment as an overview:

> *The welcome message helped to introduce each person to the channel, being able to receive and send emojis helps to send and receive quick feedback, the prompt regarding joining Slack channels is useful to promote joining other subgroups that we may not have been aware of previously.*

1. *Bot-facilitated social orientation.* Indeed, many participants appreciated the aid of Slack prompts sent from the Slackbot `WeAre!` and the access to public messages for them to orient themselves to the community context in the beginning of their experience of using Slack Workspace *WeAre!*. Note that this community space has been built from the ground up; routines and practices were being established and developed as it attracted new members and

matured. Yet, it is tremendously important for newcomers to learn the expectations and technological mechanisms to interact with others. Slackbot `WeAre!` plays an important greeting role to welcome newcomers with prompts about self-introduction. Such welcoming gestures, as intended by our design, not only shows hospitality from the tool, but also guides users toward something they can do right away. P19 elaborated on the usefulness of Slackbot `WeAre!` and public messages for him to learn about the tools, and to learn from their peers about the technical possibility and the practical routine:

> *"First I needed to figure out what type of student community this was and the introductions are important for that. It is nice to receive prompts as it feels that you get invited. Public messages are important for understanding what Slack was intended for and what would be possible with this tool."*

In addition, after introducing oneself as a newcomer, users are able to engage with Slackbot `WeAre!` regarding other new members' joining activities. Beyond a simple greeting to the newcomers, the `WeAre!` welcoming message invites other users to engage in social interaction (e.g., using the 'WeAre!' button or 💙 as shown in Figure 6.5) by offering useful contextual information about the newcomer ([the new member] comes from [the new member's city]). As P124 commented about the utility of introduction from the Slackbot: *"It's great to know who else joins the channel and if they are in your field."* Such opportunities of discovering commonality allow users to decide on whether or not to reach out to particular individuals.

2. *Information overload with notifications.* While some participants, such as P25, ranked *"Introduction and welcome message from the bot WeAre! "* and *"Prompts of creating or joining Slack channels from the bot WeAre! "* as most useful features in connecting with this community, commenting that *"Being aware of who was in the channel and what channels existed was helpful."*, other participants showed annoyance with `WeAre!` messages, noting that these might have drawn too much attention. P101 explicitly called out the importance of content-based messages over auto-notifications: *"Member messages are more*

*important than join notifications.”*. P43 also complained about the frequency of receiving notification emails of joining a new channel:

> *“Personally, I found the @ channel invites annoying. I don’t see why there wasn’t just a centralized list with all the channels available. Didn’t seem practical and was a fairly big turn off for me to get 5 emails a day with pings to join new channels.”*

These sorts of comments reflect the inherent tradeoff between offering awareness information to keep those interested ’in the know’ but also not overwhelming those who hate notifications. An obvious goal for future work would be to enable and encourage customized use of `WeAre!` notifications.

3. *Informal interaction.* Emojis, increasingly popular for communication in a social media context, attracted good positivity because they appear to add fun and entertainment. Some participants showed favor in the emoji features as a light-weighted way of collecting feedback from the peers, as P39 articulated: *“it’s enjoyable to see others react to the content presented.”* As another perspective, participants also found the easy use of emojis to be convenient for recognizing someone else’s post while keeping their own posted content modest, as P119 put it, *“Using the emoji reactions is an easy way to show you read a message.”*.

## 7.5 Activities in Slack Workspace WeAre!

Not all students who joined *WeAre!* completed the post-usage survey. Therefore, I also investigated the actual behaviors of all *WeAre!* users. In this analysis, I am drawing from conversations in *WeAre!*, all logged data, including each member profile *WeAre!* (n = 409[10]), and specific behaviors that I was able to infer from messages posted or replied. In this section, I begin with a quantitative analysis of the log data in Section 7.5.2, then characterize notable threaded conversations in Section 7.5.3.

[10] The users’ log was exported from MongoDB of *WeAre!* Web Server in May 24th, 2019.

### 7.5.1 WeAre! Member Characteristics

The WC students who joined *WeAre!* come from 83 different programs.[11] The number of students from each of these many programs vary, ranging from 1 (e.g. *Architecture (PhD)*, *Health Human Dev (PMAJ)*) to 19 (e.g. *Liberal Arts (PMAJ)*). They consist of distance learners distributed across 13 time zones: not surprisingly, for online students enrolled in a university located in north eastern America, a large majority (n = 265) live in Eastern Daylight Time (n = 265), followed by Pacific Daylight Time (n = 49), Central Daylight Time (n = 47), Mountain Daylight Time (n = 15), Cuba Daylight Time (n = 5), Japan Standard Time (n = 4), Central European Summer Time (n = 2), Moscow Time (n = 2)[12]. The greatest timezone difference is 19, underscoring the temporal aspect of "distributed" learners.

### 7.5.2 Quantifuing the use of WeAre! and WeConnect

#### 7.5.2.1 WeAre! Customization

Some *WeAre!* members seemed to be very tech-savvy with respect to Slack, for example able to install their own apps (e.g. Simple Poll[13], Google Drive). One user even set up a custom response fo the Slackbot `WeAre!`; another uploaded her own emoji of a *Nittany Lion*. At one time, a user accidentally uninstalled the Slackbot `WeAre!`, and posted a *"Oops"* in the `#general` channel, indicating this unintended error. After that point, I restricted the app installations to be approved by the administrator first.

As of this writing, there are 38 channels in *WeAre!*; 28 were created by WC student members[14]. As is common in Slack practices, the channel creators often

[11] I used LDAP to access the educational record from the public PSU directory. Here different levels of education of the same major are treated as different programs (e.g. bachelor degree vs master degree).

[12] The full table of time zone distribution is in Appendix E

[13] Simple Poll is a Slack app that can be installed in any Workspace for the purpose of initiating and collecting polling. See more at `https://simplepoll.rocks`.

[14] The remainder of the channels were create by me as a community facilitator, both to create some initial structures for the community, and to foster sub-communities that are most likely to be interesting for them based on my previous research.

announced the purpose of their new channel, and added some descriptions. On average, the participants joined 4.60 channels ($SD = 1.40$).

On the individual level, the *WeAre!* members can choose to complete their profiles to the extent that they want; this involves the basic profile settings from Slack as well as the extended profile editor provided by *WeConnect*. 82 users posted introductory information (*"Things I want my peers here to know about me"* in the generic introduction dialogue). 74 users uploaded a custom image as a user avatar; 335 were satisfied with a random image created by Slack platform. 36 users contributed phone or Skype information in the generic profile. 31 users shared cities they have visited.

#### 7.5.2.2 Visits to WeConnect

In contrast to the low numbers of visits to *WeConnect* that were reported in the post-usage survey (n=17), the log data showed that 73 different users visited *WeConnect* at least once. By decoding geographic locations during these visits, I found that these visit were associated with 38 distinct cities (e.g. *Philadelphia*, *Cincinnati*, *San Diego*).

Using the extended profile supported by *WeConnect*, 14 users shared their gender information; 6 users entered the number of children they have; 5 entered WC courses they have taken; 4 shared their military status. However, due to the low rate of completion, I did no investigation of these student characteristics.

### 7.5.3 Analysis of WeAre! Conversations

*WeAre!* users often posted their own self-introductions in free-form text (e.g. "*Hello Everyone! I am [name] from [place], with two daughters*"); others relied on bot-facilitated introductions (as shown in Figure 6.5 (d)); many also reacted to others' posts. As shown in Figure 6.7, *WeAre!* users leveraged a rich variety of emoji in responding their peers' introduction, including 👍, ✈️, ❤️.

Members also created channels related to their educational background (such as major, program or its acronyms, e.g. `#ecom`), location (e.g. `#westcoast`), purpose (e.g. `#textbook-swap`) or interests (e.g. `#wctechclub`). The activity within each channel varied greatly, ranging from 0 to 511 messages (counting only messages composed by the Slack real users, as opposed to by the Slackbot). From the total of 38 channels, 28 channels were 'active', with at least one message. Across these 28 channels, I observed a median of 7 messages. See Figure 7.4 for an overview of active channels with at least 7 messages.

Most of the channels created by *WeAre!* members were not very active; one exception was `#plsc` (political science). Low activity levels in general might be due to lack of a critical mass, or lack of peer availability for particular topics. In fact, 87 messages across all channels are from me as a facilitator (e.g. initiating a poll for selecting the date and time for a live session, guiding users' attention to the channel of their interests (e.g. their major/program), or general reminders of functions, which most often occurred in `#help-slack`.

On an individual level, each member contributed 7.41 messages across all channels ($SD$ = 29.61, this excludes posts by the Slackbot, other non-WC members and my own posts as a facilitator). I used open coding to analyze the content of these free-form messages, focusing on just the four most active, This message set included 73 posts and replies related to major, 71 related to location, 46 related to job or career, and a broad range of other categories (e.g. family, hobby, future plan, social media exchange).

Besides the 96 user-composed messages used for introduction, there were a total of 47 introduction messages created by Slackbot `WeAre!`[15], along with many other notifications sent on behalf of `WeAre!` for the purpose of advertising newly created channels for channel creators, and for reminders of introducing themselves or joining a channel. In the following subsections, I characterize three patterns of channel conversations associated with different community activities.

#### 7.5.3.1 Facilitated and Custom Introductions

For a newly-initiated community, the phase of self- or other-introduction is without question a critical element in setting the tone for the nascent community. The *WeAre!* members relied on two mechanisms for this step (see Figure 7.5):

[15]Among the 47 introductions, 10 introductions were confederates' introduction to educate the Slack users of getting introduced by Slackbot, with the help of my colleagues at the CSCL/HCI lab of IST.

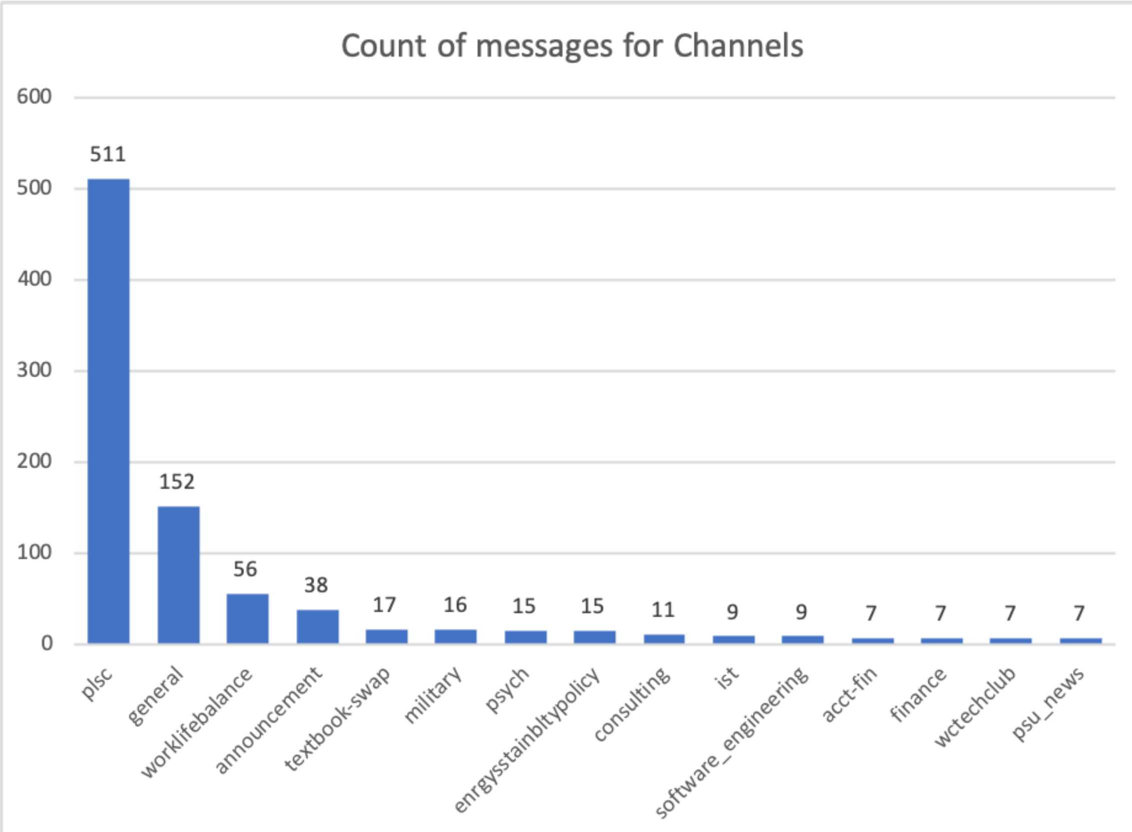
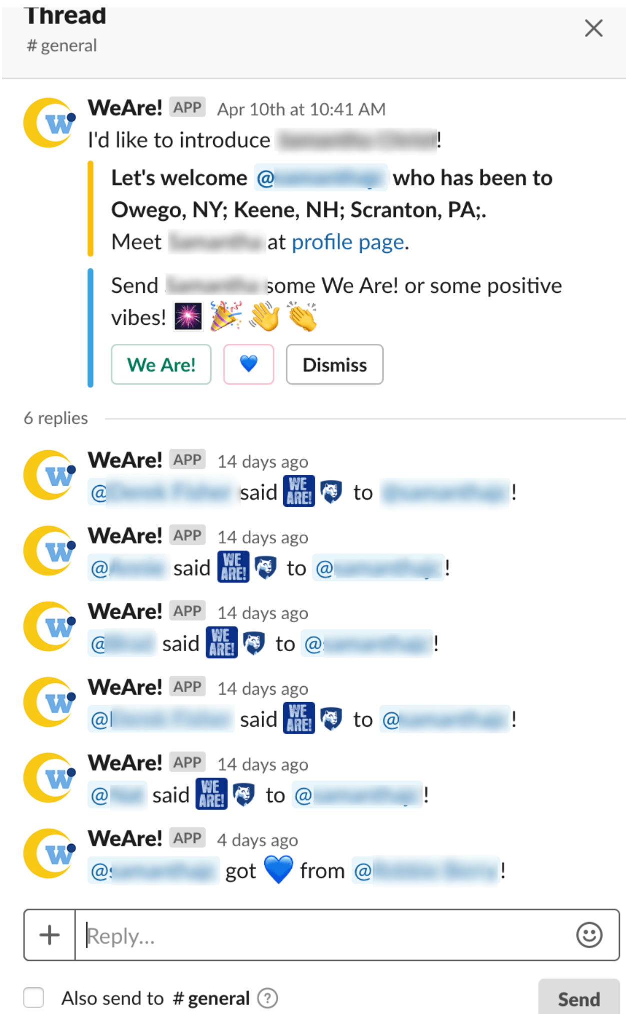



**Figure 7.4** Frequency Count for Messages

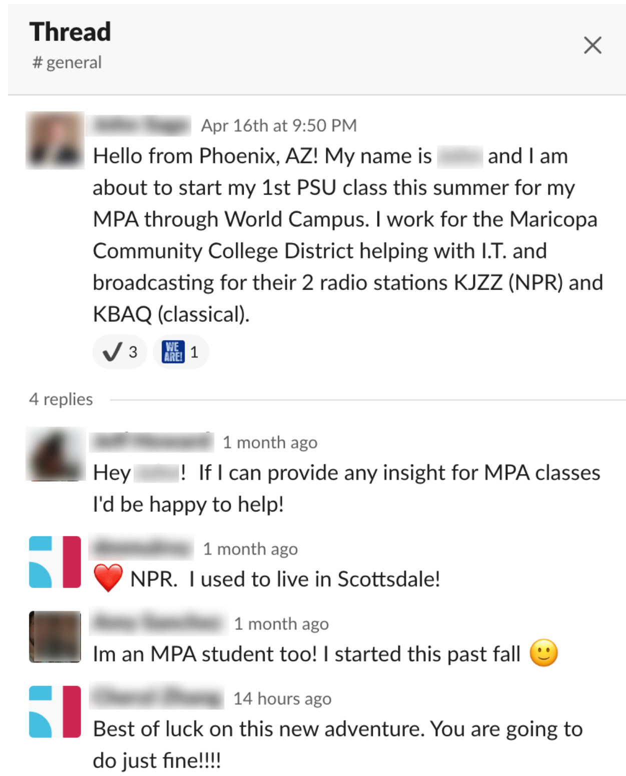
1. *Bot-facilitated introduction.* This was triggered by Slack command `\intro` or by clicking the `Introduce myself` button in the welcome stage (n = 37)

2. *Messages in free-form text.* These were custom-built and often leveraged the option to decorate the message with emoji (n = 96)

The bot-facilitation approach to introduction helps overcome the anxiety of introducing oneself out of nowhere, and offers a template for students to disclose something about themselves, such as *"Things I want my peers to know about me"*, *"I have lived in (optional)" "Current profession"*, and *"Fun fact"* (as shown in Figure 6.5 (a)). One advantage of this facilitation is the convenience for other peer students to signal their welcome intention in an invited and salient manner. The buttons of 'WeAre!' and 💙 are also very convenient as responses, in contract to responding to someone's




(a) Bot facilitated introduction




(b) Free-form self-introduction

**Figure 7.5** Two ways of self-introductions in WeAre!

custom message. Another benefit of a facilitated introduction is the corresponding notifications about members' friendly gestures in a dedicated discussion thread (as shown in Figure 7.5 (b)). Both the newcomers and welcome-providers receive notifications that symbolize the friendly act of welcoming. Although I did not show the example in 7.5 (b), users often posted several messages with a single intention of introducing oneself (e.g. separating *"hi there"* and actual introductory content, along with others' lack of response or less salient response (usually in the form of emoji reaction).

However, note that even if the introduction form was completed, only the field specifying location was shared by the Slackbot `WeAre!` (and only after approval). This may be one reason that such messages were not always seen as informational or useful: *"I found the messages and post from the members the most useful. I thought the prompts to be less useful and take up to much space."*. This is an example of another pervasive tradeoff that we encountered throughout this project, namely that people value personal information about others but do not always want to share such information themselves.

In comparison, free-text introductions often contained a personalized version of friendly greeting and enthusiasm about joining *WeAre!*, perhaps promoting to a more authentic sense of sharing and natural interaction. Interestingly, the information shared in this way often overlapped considerably with that we had found to be relevant, but importantly it was shared by the member directly rather than being collected or inferred. As shown in Figure 7.5, CP48 shared bits about his current city, working status and also the exciting education journey of starting his 1st PSU class. In response to his introduction, other members gave him three heavy check marks ✔️, one , and three friendly welcoming messages with offers of help, similarity and good wishes (these replies spanned over a month).

Furthermore, many introductions not only followed conventions of introducing one's major or educational background, but also included personal disclosure about identity management issues, pressure and sometimes call for peers of specific characteristics, which could be many other facets of their life (e.g. profession characteristics, job status, marital status, the number of kids) and physical location. To illustrate the disclosure about one's life, I draw from the `#worklifebalance` channel.

**A picture draws a thousand words.** In general, pictures posted in the channels received many responses. For example, CP2 posted a picture of sunshine and beach with a message sharing a moment of achievement and relaxation: *"Finished work for the week now enjoying the weather here in Hawaii #worklifebalance"*. Her message along with the picture received six emoji reactions, such as 😍, ❤️, 👍, from her peers. She also got comments from a prior classmate suggesting that they reconnect.

CP21 is one of the most active embers of *WeAre!*, often posting pictures he took around the world when he travel as part of his current nomadic lifestyle [Sutherland & Jarrahi, 2017]. This vivid depiction of his different lifestyles drew many reactions, comments and conversations. As shown in the Figure 7.6, his post of a European sunset evoked 14 reactions from his peers. Some fellow students became curious about his career, posting questions such as *"What do you do?"* (even repeatedly). As seen in the thread details at the right of Figure 7.6, several students were interested in types of professions; others were more interested in the photo, for example remembering a hockey game in Stockholm, or even an interest in location-specific professional development (see the reply in the end of the Thread in the Figure 7.6). Interestingly, although CP21 did respond to the question about his work functions and company, another member asked the same question nearly a month later:

> **CP108**: *"Good morning [CP21]. I love the picture of Stockholm you shared on April 11th. It sounds like you have a fun career that allows you to see the world. I apologize if it's a repeat question, but what line of work are you in?"* [05.07.2019]
>
> **CP21**: *"Hello [CP108], I do work in Cybersecurity."* [05.07.2019]
>
> **CP108**: *"Sounds like an interesting career field. Thanks for sharing the image, it reminds me of my time overseas."* [05.07.2019]

The phenomenon of repeated questions is reminiscent of the general challenges of digesting and following up conversations, especially in popular channels, and leads to dialogues that expanded a expanded span of time range for conversation participants, especially considering the newcomers who join *WeAre!* at different points in time.

**Geographic locations still matter.** Not surprisingly, distance matters even for community of distance learners. However I did observe a somewhat more nuanced meanings of distance, namely content that reflects attachment to a place, and the affinity that can emerge around a shared place. There are at least 71 messages about locations in `#general`, `#announcement` and `#worklifebalance` channels. Aside from location-evoked memories and shared experiences of visiting some places (as shown in the previous section), I also found questions specifically related to a member's interests in local neighbours: *"Anyone else from Ohio here?"*, *"Anyone here in Rome?"*

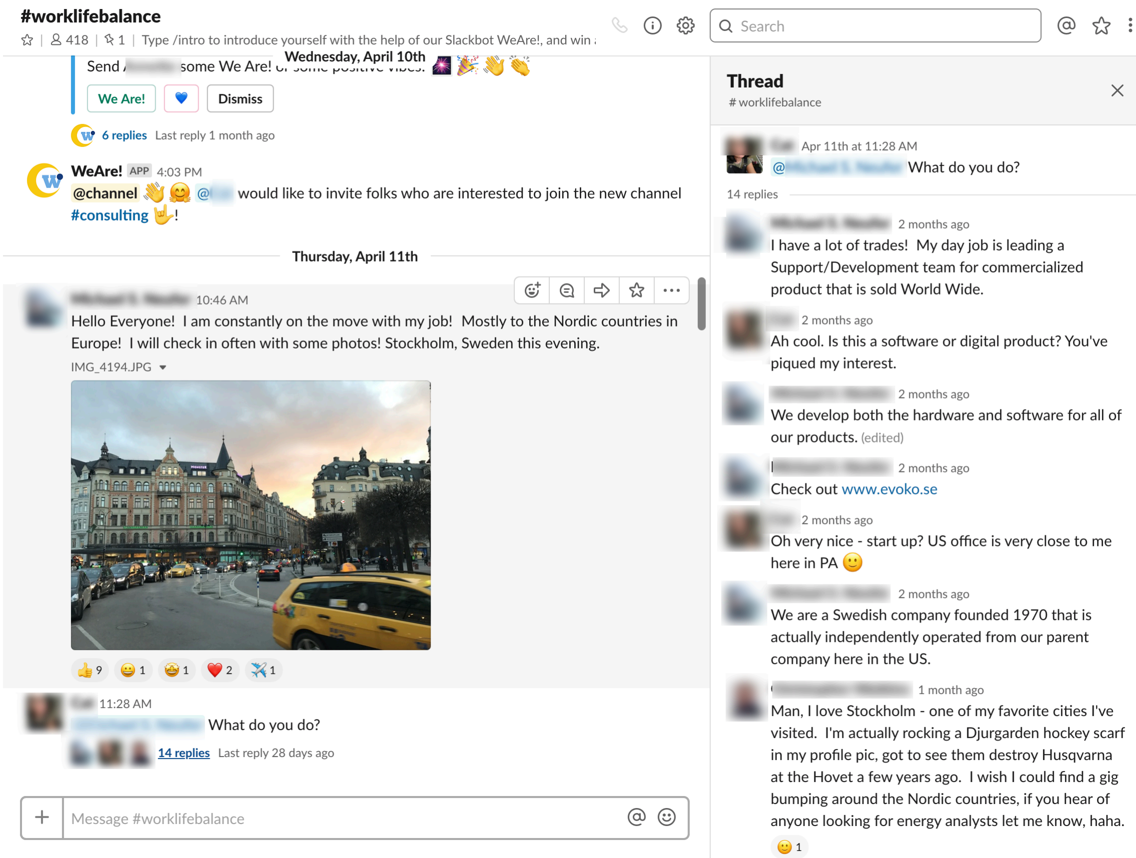



**Figure 7.6** Conversation around a picture sent from a nomadic worker

or *"Any one from Texas?"* (see Figure 7.7). As the conversations in the Figure 7.7 shows, P43 seemed to get interested in finding people in nearby cities of the same state (i.e. Texas), and was surprised by how much affinity he can relate with (e.g. *"no way, I'm originally from Palestine"*. Therefore, it is not merely about how geographically proximal these students are, but more about the place identity associated with the places they inhabit.

We also found that at times students share future plans about where they will be, suggesting a hope for possible meet-ups in person. As an example that illustrates the tendency of finding peers to actually meet in real life, [Juan] shared a picture of ferris wheel in Coachella on a sunny day in channel `#westcoast`[16], asking: *"Anybody*

[16] Coachella is a Art Festival located in California, taking place in April.

*else going to Coachella weekend 1?"* Sadly but not surprisingly, with a low number of people (only 8) participating in `#westcoast`, no one responded.

Some students shared their plans of visiting Penn State campus locations, and asked for company or local recommendations:

> *"Good afternoon! My name is [C90] and I currently live in Baltimore but relocating to Fort Wayne this summer. I'm graduating next week earning my A.S. in Business Administration. I'm re-enrolling to earn my BS in Organizational Leadership next fall. Will anyone else be attending the graduation dinner on May 3rd?"*

As a result of this question, C90 got a emoji from five of his peers, but did not get public responses indicating shared attendance. It is possible that he received private messages, but I cannot know about these given my study design. However, CP38 did get some enthusiastic replies when she shared her travel plan as part of her self-introduction inside `#general`:

> **CP38**: *"I'll be visiting campus for a summer intensive course during 6/16 - 6/21. Super excited to make my way out to Pennsylvania since it'll be my first time there."* [04.12.2019]
>
> **CP3**: *"Ah you will love the campus! I visit there often for different Univeristy Park events. It's so much fun. That's a very short course haha good luck!! Hope it goes awesome for you. If you need ideas on what to see while there i got a ton of them :)"* [04.14.2019]

As these episodes indicate, I found that questions, especially those in channels inhabited by relatively few members, often remained unanswered. Therefore, although people might gain more relevance in subscribing channels associated with their own interests, the potential lack of critical mass is a challenge for them to obtain benefits or satisfy more general information needs. I speculate that the *specific timing* of a post, with respect to availability of relevant peers, may be an important determinant in lack of responses. Lack of response might also be due to students' general time commitments to other duties in life. The distribution across 13 time zones creates yet another challenge with respect to the temporal constraints on interaction.

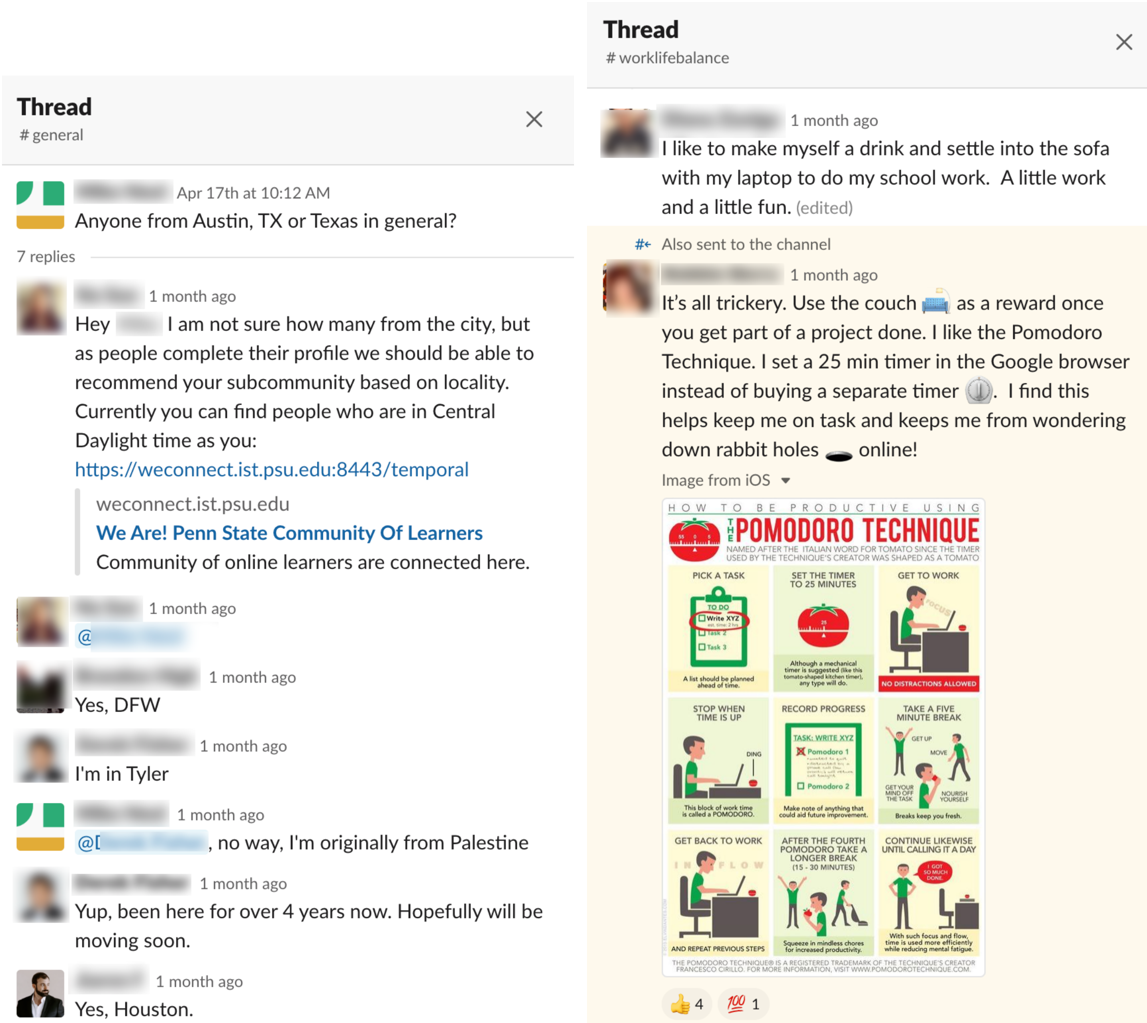



**Figure 7.7** Conversations about local neighbors (left) and Replies to questions about productivity (right)

#### 7.5.3.2 Information Seeking and Sharing for Professional Development

I found many messages related to information seeking, ranging from professional development consultation, to class-taking advice or coursework (e.g. *"@channel anyone enrolled in MKTG 811 for Fall 2019?"* or *"Anyone finish reading the Mueller report yet?"*. Specifically, a good number of students shared LinkedIn profiles, asking others to also share, and at times even explicitly asking for endorsement with promise of returning the favor:

**CP27**: *"Hey everyone! My name is [CP27] and I'm a freshman at PSU WC. Feel free to add and endorse me on linked in* ***`http://www.linkedin.com/in/[anonymous]`*** *. If you endorse or recommend me I will return the favor 100%!! Also is anyone here in the finance degree program???"* [04.11.2019]

**CP20**: *"Hi [CP27]! I added you on LinkedIn! Congratulations on beginning your education!"* [04.11.2019]

**CP27**: *"Thank youuuuu!!! ☺️☺️🤗🤗"*

As this conversation shows, some students used the *WeAre!* to expand their professional network, while planning also to reciprocate with mutual endorsement. They introduced themselves while also sending a LinkedIn link for potential connections. Beyond sharing LinkedIn information, P41 also made efforts to advertise his Youtube channel in two channels (i.e. `#announcement` and `#general`: *"Here is my linked in if ya want to connect!* ***`https://www.linkedin.com/in/thejonathancelestin/`*** *or my YouTube channel whenever ya need some motivational vitamin!!!!* ***`https://[anonymized]`*** *"*.

Many members of *WeAre!* seemed to be seeking tips for professional novices, or specifically in a channel associated with a major or a profession. For example, when CP20 introduced herself in the `#general` channel, she not only introduced her progress (*"heading towards the end of my journey to gain my BA"*), location and current employment (*"worked at a Hyundai dealership."*), but also sought advice on starting a new career as she started to look for a new job:

**CP20**: *" I'm starting to look for new jobs, specifically in Human Resources or academic advising. Does anyone have advice on starting a new career? I've sent out about 15 resumes, and I'm feeling kind of lost at this point."*

**CP38**: *"Network, network, network! Go to networking events (ex: Eventbrite, MeetUp, SHRM chapter meetings), reach out to people on LinkedIn, join some mentorship programs via local NPOs. Most jobs are gained through the internal market (i.e. internal job referrals, internal postings). Submitting resumes online is tough since many companies use keyword matching* *bots to help shortlist candidates. Depending on the company, your resume might never see human hands. It's best to network because someone is sure to have seen it (the person you're talking to, or if they give an internal referral, their internal recruiter)."*

Two days after she posted the question, CP38 replied (as above) with genuine suggestions about what she could do in the job-hunting process as a newcomer to a field. CP20 then reacted with a heavy check mark emoji ✔, indicating indicator acknowledgement and agreement. Although I did not observe a public conversation following up this exchange, it may be that they shifted the conversation into a more private space. As a Human Resource specialist in a tech company, CP38 offered to give internal referrals to other fellow students in her introduction in multiple channels.

Depending on the availability of peers who can answer a question, and the personality of the individual who is available, some requesters are not as lucky as CP20. CP83 asked members in the `#acct-fin` channel (accounting and finance) for internship or employment opportunities, but no one responded:

> *"Fellow Finance major here. I will be finishing up this summer and fall and graduating this coming December. Looking for internship and employment opportunities in Chicago, IL. The Field I am most interested in is Financial Analysis. Anyone interested in the same field/already working in Analytics? Let me know!"*

#### 7.5.3.3 Sharing of Practical Tips, Tricks and Deals

One representative case *WeAre!* conversations that has a utilitarian value is in the `#textbook-swap` channel. Among the 17 messages, some students offered to send out old books for free as long as other students will pay for the shipment; other students reply with interests and inquiries about the availability of the listed books. Sometimes, the offer and replies take place over nearly a month, as shown in the snippets below.

> **Student who offered old books**: *"I have the following textbooks to give away, you just pay for shipping: ANTH 045 Cultural Anthropol-*

*ogy, EGEE 120 Oil: International Evolution, PLSC 1 Intro to American National Government, PLSC 14 International Relations, and CAS100B Effective Communication."* [04.21.2019]

**Student who showed interest**: *"Hi [John Doe]! Has anyone claimed your ANTH 045 textbook?"* [05.17.2019]

World Campus students also shared coping strategies of a common challenge, such as productivity and the tricks of dealing with time zone differences with the main campus (i.e. for students living in a different time zone with Eastern Daylight Time, which is the time zone of main campus University Park of Penn State). For example, CP54 asked for suggestions about his stay-at-home working style in `#worklifebalance` channel, after disclosing that he is a single stay-at-home Dad with (in his words) *"the cutest Twin Girls :)"*:

**CP54**: *"So work love balance is something I really struggle with. I work from home 3D printing and making models and props but I'm also a stay at home dad and then we have World Campus. Let's just say the couch looks overly welcoming sometimes. Anyone have any suggestions?"* [04.19.2019]

**CP2**: *"I like to make myself a drink and settle into the sofa with my laptop too do my school work. A little work and a little fun."* [04.19.2019]

**CP63**: *"It's all trickery. Use the couch :couch_and_lamp: as a reward once you get part of a project done. I like the Pomodoro Technique. I set a 25 min timer in the Google browser instead of buying a separate timer :timer_clock:. I find this helps keep me on task and keeps me from wondering down rabbit holes :hole: online!"*. [04.20.2019]

Both CP2 and CP63 shared thought on balancing work and life, while *"getting the work done"*. CP63's strategies (along with her picture of the technique, see Figure 7.7) also evoked many emoji reactions, such as 👍 and 💯. These conversations showcased the struggles with shared challenges and demonstrated mutual support among *WeAre!* members. Meanwhile, emoji reactions symbolized acknowledgement and also seemed to indicate a sense of participation in the conversation. Similarly, CP57 took the initiative to share his strategy of purchasing textbook in `#general`, and also got awarded with appreciative emojis (e.g. 👍, ❤️) and comments that showed agreement and alternatives of saving money:

> **CP57**: *"MBS*[17] *is such a racket; $175 for a book rental. Make sure you are checking Amazon. They rent books on the cheap (around $20-$50 for most books)."* [04.25.2019]
>
> **CP90**: *"I agree 100%. Even if the ISBN doesn't match the books are fine if the edition and year match. I always try Amazon first."* [04.26.2019]
>
> **CP91**: *"Official online bookstore of Pennsylvania State University World Campus."* [04.26.2019]
>
> **CP93**: *"Don't forget you can get a discount on Amazon Prime for being a student. First 6 months are free!"*

#### 7.5.3.4 Friendliness among Shared Majors

One of the *WeAre!* channels, `#plsc`, was clearly the most 'talkative channel,' comprised of a member group size of 8, and with considerable messaging (n = 511). A closer look at the conversations revealed that they took place primarily among three participants: CP31, CP37, and CP105 (98.43% of the messages in this channel). The content of their conversations covered a wide range of topics, including *exchanging thoughts about assignment*, *talking about post-graduation plan*, *watching shows*, *exchanging thoughts about election*, *textbook-swap*, *scheduling for study conference call for a course*, *discussion about study materials, exams and final grades*, *sharing of strategies and mindset* and even *games*.

Starting from exchanging thoughts about assignment, these students seemed to quickly form a solid friendship with one another, exchanging thoughts about courses, study tips and events unrelated to schoolwork (e.g. election, families). This is promising with respect to the use of *WeAre!* (or a variant) as a team coordination tool for distance learning. In contrast to most of the other conversations, these students not only shared the same major, but also have a great amount of overlap in the

[17] MPS is an official online bookstore of Pennsylvania State University World Campus.

courses they are taking and in the professional interests and knowledge. The activities in this channel shows the promise of tools like *WeAre!* for fostering interpersonal relationships, by allowing students to find one another, develop common ground, and build ties based on shared interests. However, I recognize that it is difficult to generalize from the three active people who engaged in these broad discussions, It remains an open question as to what can provoke individual community members to find and bond with similar others.

In summary, members' conversations demonstrated prosocial behavior and social interactions among this emergent WC student community. First, the many user-created channels showed students' interests in developing more self-organized and close-knitted subcommunities. Second, most of the subcommunities failed to prosper, suggesting that further research needs to unravel the factors that affect the process of creating effective smaller groups (though it may also be that the duration of the deployment study was too short).

Looking at conversations across the channels, I identified three primary patterns of community involvement based on open-coding of the conversation content: introduction conversations that often relate to majors, locations and disclosures about life invite lots of peer engagement of various sorts; students in the community have strong interests in professional developments within *WeAre!*; students are willing to share practical strategies, tips and tricks about being an online learner. These findings extend my prior understanding of the importance of peer information, especially pointing out the relevance of critical mass (as opposed to temporal and social availability), educational and professional interests, as well as the practices-driven information needs. Overall, my observations of *WeAre!* usage patterns yielded richer insights about building community with messaging tools; these data also point to the tension between users' needs to foster subcommunities and the challenges of doing so (e.g. lack of response or prosocial peers).

## 7.6 Comparing Community Perceptions Before and After Deployment

In this section, I address a central research question underlying my dissertation, namely is it possible to design tools that promote feelings of community among distance education students? To assess this, I first report the results of a series of Wilcoxon signed-rank tests used to measure the reliability of changes in community perceptions over the course of the deployment of the *WeAre!* and *WeConnect* tools, looking separately at groups containing users and non-users (i.e., the control group). Following this, I present the results of five related ANOVA analyses I carried out to explore predictors of the variance of these differences in community perceptions.

### 7.6.1 Increased Community Perceptions with System Intervention

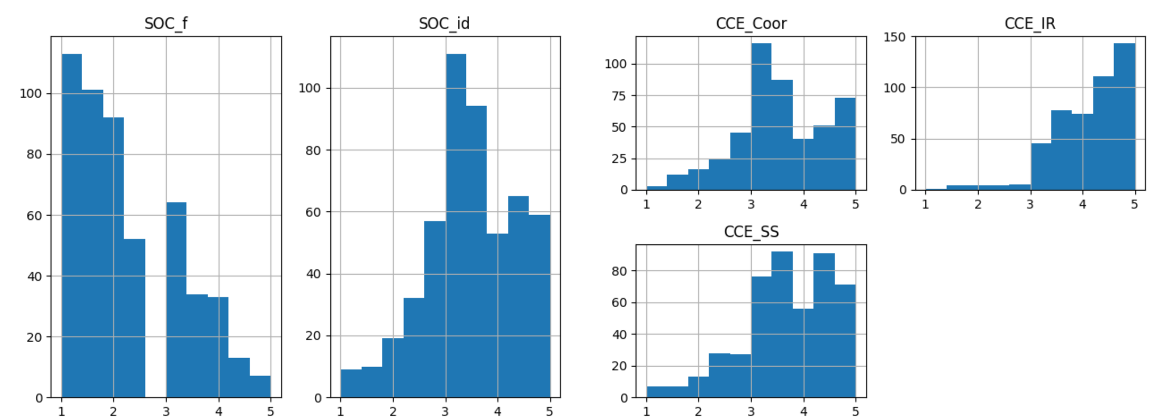



**Figure 7.8** Frequency of community perceptions prior to the Slack deployment

To tease out the effect of user versus non-user, while controlling the random errors introduced by the time duration in between, I conducted instances of the Wilcoxon signed-rank test, a widely used nonparametric test for one within-subjects factor with two levels (in our case, the factor is time span before and after the design intervention period, see [Wilcoxon, 1945]), five repeated measures nonparametric tests to compare the community perceptions before and after the Slack deployment period. As the five community scales for Friendship, Common Identity, Identity Regulation, Coordination, Social Support are extremely negatively skewed (as shown in Figure 7.8), I used This test is very appropriate as the collected data met the following assumptions:

1. The dependent variable (DV) must be continuous which is measured on an ordinal or continuous scale;

2. The paired observations are randomly and independently drawn;

3. The paired observations come from the same population.

Because a pre- and post- comparison requires repeated measures for the same participants, I had to exclude any participant who was missing either in the pre- or post-usage survey. Therefore, the participant numbers in Table 7.6 (n = 121) differ from those in Table 7.1 (n = 509), not only in categorization, but also in the reported participants; the former is a subset of the latter.

121 participants provided community perceptions of SOC in both surveys; a slightly smaller number (n = 115) completed both sets of CCE scales. As seen in Table 7.6, in general the community perceptions tended to increase over the duration of the deployment, consistent with the effect of time on interpersonal relationships [Walther, 2002].

However a closer examination of the data (contrasting values for WeAre! and Control), suggests that sizable increases appear only for participants in the WeAre! condition. The Wilcoxon test confirmed that these increases were reliable for the constructs of Coordination (Wilcoxon Z = 233.500, $p < .01$) and Social Support (Wilcoxon Z = 199.500, $p < .01$). No other pairings were significant for either the WeAre! or Control condition. In other words, the use of *WeAre!* had a significant positive impact on some aspects of students' CCE, but not on the factors underlying SOC.

**Table 7.6** Community Constructs Means, Before and After Tool Deployment

| Perception | | Common Identity | | Friendship | | Identity Regulation | | Coordination | | Social Support | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| # | | 121 | | 121 | | 115 | | 115 | | 115 | |
| **Conditions** | | Pre | Post | Pre | Post | Pre | Post | Pre | Post | Pre | Post |
| **All (121)** | *Mean* | 3.38 | 3.49 | 2.24 | 2.28 | 4.05 | 4.10 | 3.48 | 3.71 | 3.67 | 3.82 |
| | *SD* | 0.93 | 0.98 | 1.07 | 1.02 | 0.73 | 0.82 | 0.93 | 0.96 | 0.95 | 0.96 |
| **WeAre! (49)**[18] | *Mean* | 3.45 | 3.55 | 2.17 | 2.36 | 4.13 | 4.18 | 3.45 | 3.77 | 3.60 | 3.96 |
| | *SD* | 1.05 | 0.93 | 0.92 | 1.04 | 0.69 | 0.84 | 1.01 | 0.96 | 0.97 | 0.96 |
| **Control (72)** | *Mean* | 3.33 | 3.45 | 2.28 | 2.23 | 4.00 | 4.04 | 3.50 | 3.66 | 3.71 | 3.71 |
| | *SD* | 1.08 | 1.08 | 0.95 | 0.94 | 0.81 | 0.76 | 0.96 | 0.88 | 0.95 | 0.95 |

### 7.6.2 Factors Predicting Change in Community Perceptions

Recall that the design intention of *WeAre!* and *WeConnect* was to promote community. In this section, I consider what other variables might influence the observed changes in community perceptions. As a step toward doing this, I first calculated a delta value for each respondent who provided SOC or CCE data in the two surveys. The resulting variables for each of the community constructs are visualized in a histogram in Figure 7.9.

After confirming the normal distribution of these delta values (the analyses met the normality and equal variance assumptions based on Shapiro test and Levene test), I conducted five separate ANOVAs to examine factors that might predict the differences from pre- and post-deployment. Note that these analyses were conducted using only data from the 49 students in the intervention group (i.e. WeAre! users) and incorporated their self-reported estimates of usage and competency; as a result the number of observations and corresponding power to detect differences is relatively modest. Two of the analyses yielded significant models:

1. For the analysis of Identity Regulation changes, I found main effects of self-reported WeAre! usage frequency and WeAre! membership; however these main effects must be interpreted in light of a significant interaction between WeAre! usage frequency and users' self-reported competency with Slack in general (*Identity Regulation* [$R^2$ = 45.7%, F(8,40) = 4.810, $p < .001$].

2. The interaction of WeAre! usage frequency and Slack competency was also a significant predictor of increases in *Coordination* [F(3,40) = 3.569, $p < .05$],

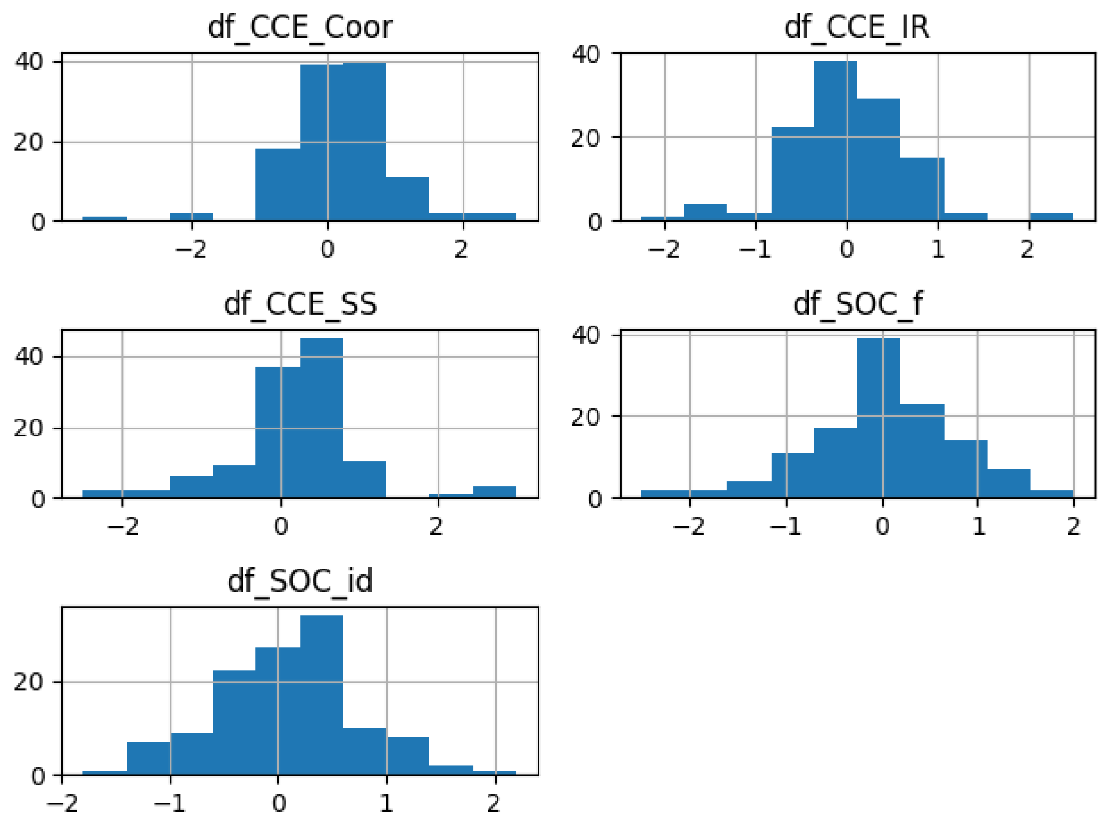
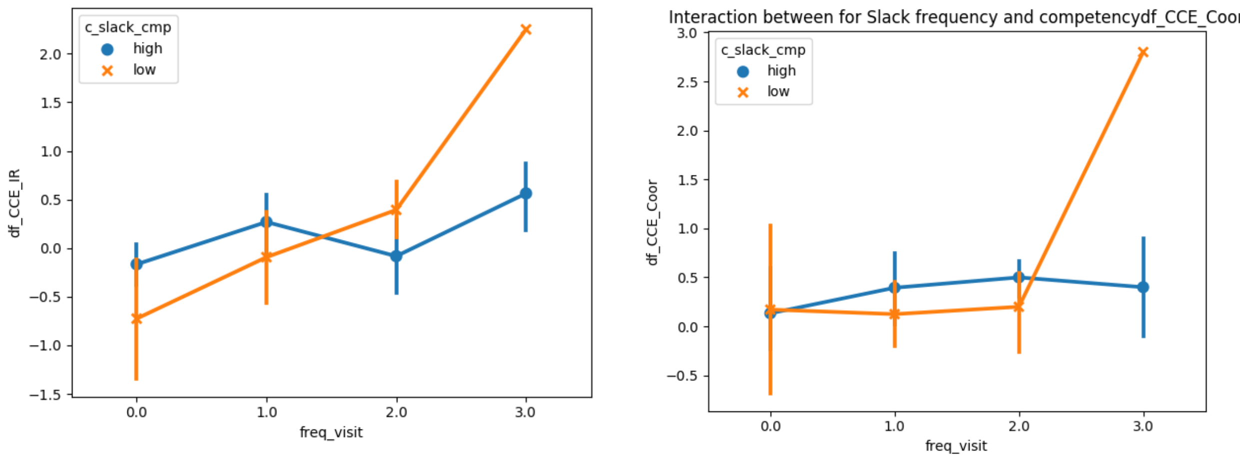



**Figure 7.9** Frequency Count for Five Community Perception Differences

but the main effect of the two predictors just approached significance [$R^2$ = 27.0%, F(8,40) = 2.110, $p$ = 0.06].

For students who chose to join *WeAre!*, the results of these secondary analyses (see Figure 7.10) reveal an interesting pattern associated with the increase in scores for Identity Regulation and Coordination. While those reporting more frequent use of *WeAre!* tended to have higher scores, this relationship seems to be limited almost entirely to those who rated themselves as *low* in general Slack competency. Although it is difficult to know exactly what led to this pattern, it may be that the choice to join *WeAre!* led to exploration of a new type of online communication technology (Slack), and that their growing ‘success’ in using this environment to interact with peers helped them to feel more efficacious as a community.




(a) ANOVA for Identity Regulation

(b) ANOVA for Coordination

**Figure 7.10** Interaction between Slack self-reported use frequency and competency

## 7.7 Summary

In this section, I refined my findings of perceived community for distinct community definitions in Section 4, and concluded that perceived community via Common Bonds (i.e. ABC) are significantly higher than those via Common Identity in the context of online education program, regardless of Community Dimensions. Leveraging pre- and post-survey responses and the logging data of the system, I herein present how *WeAre!* enabled World Campus students to actively seek and foster relationships with their peers and also increased their community feelings about their collective efficacy in Coordination and Social Support. In addition, I found significant interaction effect of (self-reported) Slack use and Slack competency prior to the experiment in predicting the variance of perception differences (for Identity Regulation and Coordination).

Based on observation and qualitative analysis of both survey responses and conversations, I found evidence of increased awareness about one another's identity-based background and activities, and identified patterns of community involvements. In particular, World Campus students' activities and reported experiences of *WeAre!* demonstrated the ability of Slackbot in breaking the ice and facilitate peer interaction (e.g. facilitated introduction). However, carefully design bot-facilitated dialogues also raise questions about authenticity of social interaction, compared with natural conversational interaction supported in basic Slack functions. Frequent Slack notifications (sent in the form of App notification or Email Daily Digest) help inform community members of important events and engage peer interactions, but also risk distracting or interrupting users from their own work.

I cannot conclude with the effect of visualizations for peer information in *WeConnect* because of the low use and sparse self-report from the participants. Reflecting on the design and evaluative process, I argue that the relatively low use of *WeConnect* may arise from short period of deployment (several weeks), general low competency of using Slack, and the hurdle of multi-site navigation from Slack to an external web app.

# Chapter 8
# Discussion and Implications

Although I have embedded preliminary discussions of many findings within the separate chapters reporting my empirical studies and design work, I will use this chapter to reflect more generally on my findings. In the following I discuss the implications for the multiple research areas I used as a foundation for my work, ideas about design directions that seem particularly promising, and opportunities for future work.

## 8.1 Supporting Instructors' Orchestration of Online Community

The study of online instructors' perceptions on building and sustaining community fills a research gap between the expected benefits of enhancing learners' social experience [Kreijns et al., 2003], and the tendency for it to be neglected in distance education practice [Stephens-Martinez et al., 2014].

### 8.1.1 Toward More Sociable CSCL Environments

Instructors demonstrated their capability to perceive and infer the presence of community based on their direct experiences and observation. In particular, for relationships between instructors and learners, they affirmed students' appreciation for interpersonal outreach though they also conceded that such outreach often comes with a substantial amount of extra time and coordinating efforts. For student interconnections, their one-to-one connections may grow during a class together, as evidenced by the emotional supports they offer or simply calling one another by name. However, even if strong ties are built there are not obvious channels for continuing these relationships, short of starting an extra curricular organization or trying to schedule courses together. At the institution level of community, ties are difficult to discern or celebrate due to the absence of institution-wide activities and interactions, which also partly accounts for why social connections are hard to maintain once a class session is over. Looking across these findings, it is important to support the desires for interpersonal communication initiated and managed by students on their own, instead of depending on what an instructor can organize or lead.

### 8.1.2 Helping Instructors Nurture Student Engagement

I found that instructors may be promoting feelings of community by publicizing bits of students' work, by injecting more of their "real" selves into their communications and by personalizing the communications they use. Class-level connections seemed to be most strongly created and reinforced through real-time activities that attracted good attendance and high engagement. In particular, instructors can enhance the visibility of implicit social content by publicizing cognitive and socio-emotional knowledge in a larger context. For example, organizing a discussion about someone's submitted project not only brings social attention to that student at that moment, but also invites peer-based evaluation, reflection, and learning. In this sense, instructors may shift interactional resources across different levels of connected learning group units, assembling isolated individual learning processes towards collective knowledge construction at a community level [Stahl & Öner, 2013]. Whether it is students' knowledge artifacts (e.g. assignments) or personal background, instructors' public sharing action serves as a social overture that may provoke synthetic discussions or other collaborative learning opportunities.

With respect to CSCL design implications, my interviewees' practices suggest specific enhancements of technology support to attract students socially into the learning subject and with one another. For instance, a synchronous meeting room might include channels to convey contextual cues on what is happening on the other end to make it feel real, such as status updates on specific activities. Text-mining techniques can be applied to self-introductions or other coursework for instructors or students to customize communications with socio-emotional background and predictions. Another design option is to increase the visibility of individual contribution based on instructors' endorsement or teaching assistants' grading outcomes.

### 8.1.3 Shooting In The Dark?

Taking another perspective, I heard concerns from some instructors about students who are *"silent"* online or fail to even show up for online activities. As a result, these teachers questioned the benefits of organizing class-wide activities in the sense *"if we built it, will they come?"* Researchers know that heavy reliance on remote communication may encourage students to filter out the cues that indicate presence or attention [Walther & Parks, 2002]. Just as there will always be students who "take a back seat" in residential courses, similar students will enroll in online courses. At the same time, computer-mediated communications often leave a historical record so that students who do not attend a live session can engage later on. Nonetheless, lurking behaviors in online courses are even less visible than sitting in the back of a resident course; they require extra efforts by other students and instructors to assess and respond [Tynan et al., 2015].

Interestingly, the time over which computer-mediated interactions take place, and the visibility of personal cues within those records may contribute to the growth of social schemas about others [Liebman & Gergle, 2016]; these then can be used to draw inferences about affinity [Walther, 1996]. Without records of students' time investment and social presence, instructors and peer learners are less able to respond to individual needs. An individual's silence (e.g. not attending a real-time session) may be due to time constraints or other reasons, but attendees at the session are left in the dark about these factors. I suggest that CSCL design enhancements are needed to convey online students' real-time constraints and subsequent activities (e.g., viewing or commenting a recorded session) in ways that make their presence visible and accountable [Erickson & Kellogg, 2000]. As an example of a specific design direction, the sharing of explicit declarations about time availability may help both instructors and peers adopt a more nuanced set of participation expectations.

### 8.1.4 Limitations and Future Work

Because college of IST is a part of an interdisciplinary college, the instructor-participants reported here represent a range of course topics, teaching experiences and approaches; even so, this exploratory study was conducted within one college at Pennsylvania State University, and I call for future research to generalize and extend my exploratory analysis. I also recognize that this piece was a descriptive study that relied extensively on instructors' opinions and my own interpretation of the same; I cannot draw firm conclusions about causal factors or mechanisms in building community in online courses.

## 8.2 Fostering Peer Relationships in Remote Team, Class and Campus

My analysis of the student interviews revealed that students come to feel connected as a result of goal-directed interactions with one another, especially when their social actions yield immediate responses (e.g. in a live session, video-meeting or chat). At the same time we found that feelings of connection can arise from coincidental encounters (e.g., recognizing the name of a former classmate, showing up early for a live session). With respect to variations in scope, I found that a shared choice of university creates a sort of general brand commitment, but that the feelings of connection at this level are diffuse.

Some of the ties described to us seemed to be evoked by discovery of shared identity, such as introductory posts of being veterans. Other ties are deliberately established and maintained, especially if a peer is viewed as a useful contact for future learning or professional activities. Students certainly seek "good" partners for shared work, but also help one another at times of difficulty. In the following sections, I discuss these findings in more detail, and consider design directions that might promote a collaborative and connected distance learning environment.

### 8.2.1 Promoting Discovery of Shared Identity

Distributed teams who are given a basis for shared identity will prioritize interaction reciprocity and resource sharing relative to teams without a shared identity [Bos et al., 2010]. My findings contribute to this body of work, documenting how distance learners seek out – or simply discover along the way – elements of shared identity that contribute to feelings of connection with one another. Construal level theory [Fujita et al., 2006] suggests that prior to interpersonal interaction, gleaning any information that reveals similarities with an unknown individual can engender a more vivid and accurate perception of a remote person [Marlow & Dabbish, 2012]. This suggests a direction for CSCL designers to promote a sense of community: despite the potential learning gains from heterogeneous group composition [Manski, 2000], designing ways to convey shared identity, especially in the early stage when learners have little ior interaction with one another is critical to foster connections in distance education.

Even before encountering peers in online classes, distance learners share the identity of being a student at their university, and an additional identity of choosing an online program. Organization commitment theory posits that members' affinity to the "brand" of the organization predicts their contribution to that organization [Allen & Meyer, 1990], especially with regard to people's willingness to provide information [Lampe et al., 2010]. Perhaps distance learning programs could offer students ways to reinforce this rather diffuse connection, for example sponsoring online meetups to "attend" a university-sponsored event together, or creating clothing or other collectibles to celebrate the distance learning mission of the university. The online student organizations described by some of the students is an example of how this is already happening, but designers could enhance those opportunities.

Interviewees also found affinities with peers by reviewing self-introductions. But it can be tedious to sift through one introductory post after another. Some students also object to being asked to introduce themselves time and time again. I see this as a clear design opportunity for a program-wide profile module, perhaps one that can be searched or filtered to highlight particular aspects of shared identity (e.g., single parent, job switching). Meanwhile, using a universal profile may introduce privacy concerns and maintenance efforts for different course subject, which may call for designers' further consideration when promoting the sense of community.

Feelings of shared identity can also emerge from tacit knowledge of a shared situation (e.g. a tight schedule, comparable personal challenges). Although this is not unique to distance learning, I was struck by the *"we are in this together"* bond from observing what their peers are able to do and accomplish even when stressed. They hold the "team spirit" in their efforts to succeed in a similar challenging situation even when working individually on a coursework. As the theory of *expectation interaction* [Manski, 2000] suggests, people set up similar expectations when they observe others' outcomes. Finding ways to celebrate these feelings of accomplishment in a virtual context could also enhance distance learners' shared identity (e.g. annotating chat messages with emojis to show their pride or joy [Zhang et al., 2017]).

It is interesting to compare the facets of identity that were discussed by the interviewees with studies of other online platforms, where researchers have noted that some members prefer to keep different facets of their lives distinct from one another [Farnham & Churchill, 2011]. My participants seem to be routinely exposed to the blended identities of their peers. These facets of identity may be deliberately disclosed (e.g., in an introductory post), may be shared incidentally as part of group activities (e.g., special expertise or important life circumstances), or may even be unintentionally shared as unrelated context (e.g., taking care of a sick baby).

I suggest that the blended identities shared by the distance learners may be an acceptable version of *context collapse* [DeVito et al., 2017; Marwick & Boyd, 2011]; this concept refers to the conflict that may be experienced by members of social network sites (e.g., Facebook, Twitter) who post content intended for a specific audience to a more general audience. In contrast, many of the interviewees seem to interweave their multi-faceted real world roles (e.g. being an employee, a parent, a community committee member [Carroll & Rosson, 2013]), making context collapse both more appropriate and at times even intentional (e.g., when managing expectations a shared timeline). This difference may be explained at least partly by the distance learners' instrumental goals: they choose online education for convenience of time and space. At times their learning goals must have lower priority than other matters (job, family). To manage their relationships and commitments with their learning partners, it is often critical to share this "outside" information.

### 8.2.2 Leveraging the Connectivity of Small Groups

Not surprisingly, connections situated in small groups appear the focal point for feeling connected to peers in distance learning. These groups are the primary source of social relationships, at times leading to an "almost friends" degree of interpersonal bonds. With respect to these feelings, I found an important role of real-time or close-to-real-time interaction (e.g. live sessions, messaging), suggesting a need to emphasize and enhance tools for real-time communication among online learners (see also the study of video meetings [Cao et al., 2010] and video-based discussion activity in MOOCs [Kulkarni et al., 2015]). The immediacy of interaction is important for online students to initiate conversations [Lampe et al., 2010], and to tolerate the occasional overdue contribution [Strijbos & De Laat, 2010]. When a group meeting is scheduled, members may also arrive early or stay late, adding the opportunity for more casual interaction about their lives. However, "being there" for interaction with peers competes distance learners' time resources against their external life commitment [Benda et al., 2012]. Distance learners in a busy life may need to manage their own time while being aware of their peer learners for the benefits of real-time small group interaction.

An interesting compromise was the regular use of persistent work spaces, where team members could assume that their queries or offerings would be seen and addressed soon, even if not in real-time. This creates an interesting extension of the familiar goal of "being there" to something that is better described as "being there when you need me". By setting up a space such as this and reliably using it for project-related work, the team members can trust that their concerns and contributions will be seen and integrated in a timely fashion. If the shared space also has a mechanism for real-time communication history (e.g., recorded video/audio sessions, or even chat), members who miss a live session can use the recorded information to catch up.

### 8.2.3 Evolution of Distance Learning Communities

Another question raised from my findings concerns the possible future of group-based peer connections that are nurtured by small group interactions. In some cases these ties are continued as general social media connections at the end of a class; however, there is no guarantee that this will happen. Indeed a downside of forming close bonds with project members is that class-based groups are by their nature transient groups that form and die off each semester [Ren et al., 2011]. Some interviewees shared feelings of regret at having to leave these peers behind. However, note that many online systems now offer new members the option to register with "regular" accounts (e.g., from Google or Facebook); if CSCL environments provided similar options, one side effect could be that group members are more likely to continue their association beyond the rhythm enforced by the semester structure of higher education.

More broadly, the varied types of connections described by the participants suggest an evolutionary view of connections between distance learners, where initial vague affinities grow into stronger team-based ties, and in some occasions are carried forward into longer-lasting connections. Initially, my participants expect more closeness with classmates or team members with whom they share common identity (e.g. being a parent or coming from military background). Building from these early affinities, the learners strengthen their ties through focused group work. After initial class-based interaction is over, the learners may continue to recognize and reinforce connections that are now grounded in their shared educational goals and experiences. Eventually, the bonds originally established during tightly coupled group work may fade away or become a low-stakes resource for ongoing professional development (e.g., job networking). My work thus extends the temporal pattern discovered by Haythornthwaite [Haythornthwaite et al., 2000], who found that the underlying connections in an online learning community emerge but fade away in a temporal dimension with a refined evolutionary view.

I propose that CSCL learning platforms could take into account this evolutionary view of building and maintaining peer connections of varying types and strengths. I suggest that more "sociable" CSCL environments for distance education might store a history of students' social interactions or online encounters (e.g. attending the same online activity in the same class), reinforcing students' inclination to strengthen and maintain well-developed interpersonal connections that grow out of their interactions as classmates (I recognize that such histories and resulting access would raise many issues of privacy, but the general concept is valuable nonetheless). More persistent social venues, such as cross-course or cross-program forums or chat rooms, could be used to reinforce and sustain the relationships developed through the camaraderie of shared coursework, which otherwise would dissipate at the end of a course. For example, the platform could enact an explicit lifelong learning paradigm, wherein learners begin by taking classes and connecting with each other through shared identity and small group activities; move on to an intermediate phase where they can find and enhance ties with former classmates while building their network through new courses and group activities; and finally graduate into an alumni role as a provider or recipient of more broad-based career advice. The continuation of relationships that originate from (virtual) classroom encounters has been proved to be possible over chat even in MOOC settings, in which alumni in programming classes linger as part of an extended learning community to both contribute and learn from others [Nelimarkka & Vihavainen, 2015]. I recognize that not all distance learners will want to participate in such an extended learning community, but the small sample did contain some who expressed this sort of interest; even a modest presence of "old-timers" could make the environment more welcoming to new and intermediate learners.

## 8.3 Injecting Social Context to Encourage Peer Interaction

I synthesized multiple phases of the user-centered design process and now discuss the implications for the design towards providing socially integrated learning experiences.

### 8.3.1 Support Peer Information-Seeking

A major theme in my research has been that knowledge of one's peers is a critical enabler of social connection and collaborative learning. Berger and Calabrese's developmental theory of communication about interpersonal knowledge [Berger & Calabrese, 1975] is useful in elaborating this theme. These theorists argue that when communication partners have greater certainty about each other, they are more comfortable reaching out to and interacting with each other. This increased comfort is evidenced by a strengthened ability to appropriately tailor their communication content, level of intimacy, and information-seeking. A similar phenomena is seen in academic communities of globally-distributed scholars, where socio-emotional knowledge is a strong predictor for sense of community and knowledge sharing acceptance [Nistor et al., 2014]. Researchers have also reported that in virtual seminars, students with more interpersonal knowledge about other participants take more initiatives in shared discourse, for example, critically evaluating contributions of others, asking questions and making new proposals [Diekamp et al., 2008]. With respect to distance learning, an "interpersonal knowledge gap" might account for why online students are less likely to interact with peers and faculty than traditional students, even though they are positive about the idea [Rabourn et al., 2015].

Both my student interviews and comments about the visualization prototypes revealed that simply knowing that a group of peer learners is "there" is not enough to assess social needs and orchestrate effective social behaviors. Without sufficient interpersonal knowledge, a shared online space may never invite meaningful social exchange. I suggest that CSCL designers should support the creation of online peer information structures that contain the kinds of information the learners found to be helpful in feeling more connected (e.g., age, course background, gender, geographic region). Such structures can be recruited for a variety of social behaviors, both within and across small groups and classes.

In addition, my interview results revealed that social connections can be built through the bits and pieces of shared learning experiences, either in the past (e.g. common class, major or instructors), the present (e.g. same group members across different classes) or when planning towards the future (e.g. electives recommendation). These findings suggest that a starting point for connecting online learners at a large scale might be to gather and externalize information about their histories and future goals; this could promote discovery of shared identities and subsequent social interactions. This implication overlaps one drawn from a longitudinal study of MOOCs: if designers of online learning environments want to encourage social interaction, they should find ways to promote development of interpersonal knowledge [Celina et al., 2016].

Of course, gathering this information about learners is more easily said than done. Currently, distance learners weave together information that they discover about their peers in bits and pieces - from introductory posts, chit chat at a real-time meeting, or other side comments. Even if I can establish what is most useful to know about peer learners, as designers I will still have the challenge of evoking and storing this content in a relatively simple and low-cost way. While some of the initial content might be available through the educational program itself (e.g., learner demographics) or by logging of learners' interaction habits, important issues such as privacy will inevitably arise. Supporting the secondary task of personal information privacy management may become complex and will likely require its own specialized design attention.

### 8.3.2 Learning Goals as a Mediator of Social Interaction

My interview study and exploratory design evaluation pointed to the importance of knowing fellow students' online education history when connecting with others in an online learning environment. The learners enrolled in online degree programs are primarily driven to earn their degrees [Moore, 1993]; it is the social activities that serve these instrumental goals that are likely to be most rewarding. In this sense, these students' community goals may be quite different from members of other online communities that rely extensively on common bonds, emotional support and friendship [Brzozowski et al., 2015a; Celina et al., 2016]. More specifically, distance learners' community perceptions may depend largely on the benefits inherent in their peers' special bodies of knowledge or professional connections. Similar to strategies that use opportunities for information exchange to attract participation in online communities [Lampe et al., 2010; Ridings & Gefen, 2004], I propose that distance learners should be encouraged to share their online learning experiences and professional backgrounds (as they did in response to the surveys). By making these histories and goals more visible, the system might invite social interaction aimed at information exchange; this in turn could help to attract and maintain a large and diverse group of students in the education program.

As weak ties tend to serve instrumental goals more than strong ties [Granovetter, 1983], my design investigation shows CSCL designers one way of helping learners create a large number of weak ties (to support basic information exchange) without the necessity for small group interaction. In this sense, conveying to a learner the diversity and richness of peer information available on a large scale might evoke social interactions aimed at specific knowledge gains (e.g., what it is like to work for company X) [Celina et al., 2016; Kulkarni et al., 2015].

At the same time, students' feedback about the visualization tools pointed to a desire to refine the larger community into peer group of interest (e.g., by age, residential locale). This suggests that students' interest in reaching out to peers might be associated with multiple criteria at the same time, including personal characteristics but also physical or temporal proximity. These refined social groupings also suggest the design requirement to support a "drill down" from a large amorphous community to one that is at a smaller scale, similar to the design implication proposed in a MOOC study in providing navigation across different layers of community engagement [Celina et al., 2016].

### 8.3.3 Learner-centric Information Visualization

Students found the network visualization to be helpful in revealing connections that they would not otherwise know about; this is a promising outcome with respect to my design goal of externalizing implicit social connections. Distance learners seem to lean toward sub-groups that have clear benefits for personal and professional development. To support these goals, designers need to provide a clear sense of personal relevance in the visualized structures, such as exploring the use of an ego-centric view using a radial layout instead of the third-person viewpoint offered by my first prototype [Heer & Boyd, 2005]. Another option is using color-coding to identify the most "promising" peers in a network (e.g., with respect to a given instrumental goal such as getting homework help). The use of ego-centric representations may suggest the behavior of activating one or more implicit ties, which might in turn guide designers to build tools beyond social presence [Garrison et al., 2001].

Although the map visualization was used to scale down a geographically distributed community to regional sub-communities, that feature seemed most desirable as a piece of a composite filter that could generate sub-groups for a specific goal (e.g., an offline meet-up). To address this, one option might be to design dialogs that allow learners to specify combinations of criteria that create juxtaposed views of different sub-communities. Thus I propose giving learners more dynamic control over the information display as an active invitation to "get to know your community." The capability of drilling down to learn more about details, after starting from an overview is also necessary as visualizing community beyond class units encounters the issue of scalability. For example, additional interaction options for network views include collapsible nodes [Heer & Boyd, 2005], and nodes with different transparency values that depend on their network structure [Schreurs et al., 2013].

## 8.4 Availability Matters

Focusing on the important factor of availability, I conducted follow-up interviews to uncover the underlying reasons and strategies for distance learners to predict others' availability and to manage their collaborative work. Interestingly, my qualitative work not only deepens my understanding of how peer availability plays an important role in feelings of community for distance learners, but also points back to the other factors that emerged in my earlier exploratory factor analysis. For example, availability entails richer meanings than literal temporal co-presence. It includes things like the availability of other peers' expertise for them to have "intellectual match" to collectively work on issues or problems. Distance learners, who are for the most part adult learners with full-time jobs, have expertise that relates to both their Learning Background and Profession.

I found that distance learners' unavailability to distance education peers is a function of balancing their education workload with other professional or personal duties; the experience of having a peer who is unavailable can contribute to unpleasant collaborative learning experiences (e.g. failed contribution in the group work, overdue submission) and ultimately jeopardize the community atmosphere. My triangulated results indicate that a peers' area of expertise (including learning background and profession) and current educational goals influence the use of peers' availability information, especially when the learner is pursuing instrumental goals (e.g., organizing a group project). This suggests that the overarching process of building connections is more nuanced. Finding people who are available at certain point is essential, but in the meantime they should preferably be relevant with regards to the other factors I uncovered, such as Learning Background and Profession. I speculate that perceived overlap in all of these ways will contribute to the richness of learners' social interactions and reinforce positive community feelings.

### 8.4.1 Beyond Distance, Time Matters

In spite of the oft-cited caution that "distance matters" [Olson & Olson, 2000], My interviews underscore the issue of *time* for online learners. Because they have by choice sought out distance education from arbitrarily distinct locations, this distance factor may not be something they expect to overcome. However, collaborating within the same temporal rhythm may help to compensate for the fact that they are not "together" in the real world.

The evidence for the importance of a shared time framework comes from these learners' efforts to form teams and groups that are compatible with respect to time available for meetings or for organizing to achieve project deadlines. Of course, dealing with tight schedules is a problem for both residential and distance learners [Nathan, 2006]. In the residential case, mobile access to social media and shared geography allows groups to initiate ad-hoc social gathering grounded in the shared physical setting [Barkhuus & Tashiro, 2010]. I propose that a shared temporal rhythm (i.e. matching days and times for study or relaxation) may enable similar ad hoc social gatherings that are grounded in expectations of mutual availability. For example, interviewees told me that at times they may arrive at a live session a bit early, discover one or more partners, and enjoy a brief social chitchat. If the tool made it easy to know when a team member was available, this sort of ad hoc socializing might become even more common and natural.

Note that the notion of temporal rhythm is a refinement of the time zone mismatches that complicate globally distributed cooperative work [Herbsleb et al., 2000]. Sharing a time zone sets the scene for overlaps in availability, increasing the chances for real-time or close-to-real-time interactions and affecting the believes in getting in-time help [Herbsleb et al., 2000]. However, simply living in the same time zone does not guarantee that your real world schedule will match that of nearby peers. In this regard, my work extends other studies of collaborative work that caution designers to consider actual work schedules and not just geographic time zones [Espinosa & Pickering, 2006].

Furthermore, my interviewees did report that synchronous communication with team members or the entire class was important in feeling connected, reinforcing other findings that communication immediacy is a core aspect of social presence [Lowenthal, 2010; Short et al., 1976]. Distance learners' inclination to IM for quick turnarounds also seems to reflect a desire for being there "at the moment", perhaps contrasting to a preference by international workers for the explicit and clear channel of email communication [Tang et al., 2011]. Most social interaction in distance education is student-initiated in the service of shared learning goals or even just learning about each other, and may not need the more formal tone of asynchronous email "memos" that are common in industry.

In response to these observations, I propose that CSCL designers should investigate options for time-aware communication services that can support impromptu requests (e.g. synchronous social Q&A [White et al., 2011]) and immediate action (e.g. reminding of an upcoming event). As an example, a shared calendar through which students indicate their preferred study times and personal availability might increase collaboration awareness, thereby inviting (or discouraging) impromptu outreach for small questions. Future research is needed to address these possibilities and account for how students' availabilities are tied to with their social behaviors.

### 8.4.2 Temporal Proximity Matters

My research findings underscore the matter of *temporal distance* for online learners, complementing the oft-cited caution that spatial *"distance matters"* [Olson & Olson, 2000]. As distance learners by choice seek out online education options from arbitrarily distinct locations, it is possible that physical separation is already well anticipated, especially for a learning population that is distributed across the world. Consistent with this view, prior research suggests that the negative effect of physical distance is dampened once it exceeds city boundaries, while the role of temporal distance becomes essential [Huang et al., 2013]. In my study, distance learners seek to arrange collaborations by arranging for a shared temporal rhythm (even in different time zones), to compensate for the fact that they are not "together" in the real world.

In parallel to how residential college students leverage shared geographical locations to initiate ad-hoc social gathering on campus [Barkhuus & Tashiro, 2010], distance learners' social interaction and felt connectedness are primarily influenced by *temporal proximity* in coordinating *on-demand*, *regular* or *last-minute* collaborations. Temporal proximity is a term that refers to the overlap of waking hours, and is measured in terms of time zone difference [Huang et al., 2013]. I found it to be a useful reference for me to understand learners' preferences for interacting with peers sharing the same or near-by time zones - a pattern that emerged in both my quantitative and qualitative studies; this extends findings in the context of online gaming, where individuals with temporal proximity were more likely to interact with one another than those with significant time zone differences [Huang et al., 2013], presumably because these individuals tended to be online in the gaming environment at the same time. Obviously, sharing a time zone increases the chances for real-time or close-to-real-time interactions and affects expectations for receiving just-in-time help [Herbsleb et al., 2000].

I also noted that simply living in the same time zone does not guarantee a matching schedule. Prior work has indicated that online students are often confronted with social issues derived from other facets in their immediate life (e.g., a sick baby) [Sun et al., 2019], and suffer from insufficient time for learning [Benda et al., 2012]. In this regard, my work extends other studies of collaborative work that suggest designers to consider actual work schedules besides geographic time zones in online learning settings [Espinosa & Pickering, 2006]. However, I suggest that distance learners might feel these constraints more severely, because they choose to enroll in distance education to better juggle other real world distractions (e.g., busy Mom, primary caregiver, on-call service provider).

I recommend that online learning platforms promote awareness of learners' availability (e.g. available vs unavailable, flexible vs fixed, working days of a week and working hours of a day) while taking account of the time zone differences. Similar to the case of family communication across time zones [Cao et al., 2010], sharing daily routines as well as occasional exceptions in time is important to establish effective communication among learners. Nonetheless, updates from remote learning partners may not be as salient or prioritized as communications with intimate family members. Future work needs to consider effective ways of drawing attention, but not interfering with ongoing life activities. A possibility might be to add social features or transparency to the calendar tools that are already used in online learning platforms (e.g., as used to track class schedules and due dates). For example, a shared calendar can be used to indicate preferred log-on times or broadcast a call for participation of study sessions with the available options or unavailable markers. Those indications of workday settings are common in office environment [Tang et al., 2011], and could also be adapted for distributed learners.

### 8.4.3 Supporting Synchronous Sessions for Distance Learners

Similar to the social gatherings among residential college students [Barkhuus & Tashiro, 2010], I found online learners arrange for "meetups", while managing their availability within and outside of learning tasks, to interact with their peers synchronously. Different from how residential students join local gatherings to socialize and have fun, online learners must be more deliberate in planning social activities, which tend to be driven by learning needs. My findings about the role of peer availability in promoting peer connections align with previous literature in which researchers identified schedule compatibility as the most popular team selection criteria for both students and instructors in the residential courses [Jahanbakhsh et al., 2017]. Further, my research extends previous understanding of availability beyond temporal meanings (*when*), and further reveals how other facets of peer information (*who*) interact to influence social interaction and perceived connections. I now discuss design implications from these identified availability needs.

#### 8.4.3.1 Ad hoc immediate discussion

To receive timely help from peers, distance learners expect someone to be online at the moment when she or he needs help. It resonates with the positive effect of receiving rapid feedback for MOOCs learners on their course outcomes [nmay

Kulkarni et al., 2015]. More generally, fulfilling learners' information-seeking needs rapidly has been found to be important for online degree learners. Finding ways to encourage and support synchronous interaction in online learning environments remains a significant design challenge. For example, researchers found that always-available discussion sessions struggle to achieve critical mass: Even with a large-scale learner population, as is typical for MOOCs [Kulkarni et al., 2015], only a small proportion of students actually overlapped for an immediate discussion session (e.g. out of 2940 visitors only five students overlapped). Despite their smaller scale, online degree courses may attract people of less geographic dispersion and, thus, are more likely to yield study groups of overlapping sign-in time. However, current Course Management System (e.g. Canvas) do not support presence transparency for students to recognize their online peers, although the need for such presence indicators is sometimes fulfilled in Hangout sessions and Slack teams. This finding may partly account for why Canvas has not been found to help learners to connect with their peers and instructors [Wilcox et al., 2016].

As an alternative to on-demand discussion sessions, persistent unstructured chat modules have also been observed to have no effect on sense of community among online learners; they tend to result in minimal participation, perhaps indicating the difficulty with aligning a concurrent available time slot [Coetzee et al., 2014b]. To address the issues of critical mass, a possible direction might be to extend the scope of social learning experiences beyond a single class section by including past classmates and class alumni [Learning or lurking? Tracking the "invisible" online student, Sun; Gašević et al., 2013; Nelimarkka & Vihavainen, 2015]. Note that such on-demand availability needs may anticipate online learners' "*sociotechnical state of being constantly connected and accessible to others*" [Mazmanian & Erickson, 2014]. One way to ensure peer availability is to set up a rotating system whereby everyone agrees that someone will be available some of the time. This is similar to how service industry professionals maintain their client expectations of 24/7 service while ensuring time off for each individual [Perlow & Porter, 2009]. However, unlike social industry professionals, online students juggle multiple social worlds with different roles they need to play [Kazmer & Haythornthwaite, 2001]. In reality, it is a taxing and almost impossible task to be constantly available for their learning peers when other roles (e.g. parent, spouse, employee) may take precedence over a learning activity [Benda et al., 2012; Sun et al., 2019]. Therefore, keeping a balance between rapid response time in cooperative learning tasks and managing the boundaries across different individual identities is both a practical challenge for distance learners and a design challenge for the HCI community.

#### 8.4.3.2 Planned online social gatherings

As described by my participants, planned social sessions among online learners often rely on a leader's efforts to propose and team members' cooperation to reach consensus on a time slot. This consumes additional time and coordination efforts from all parties. To alleviate the pressure on an already busy online learners' time, future systems could leverage distributed learners' time zones (or temporal proximity), logged-on patterns, relevant social meta-data (e.g. common classes), shared enrolled courses and/or explicit preference input and recommend pairs or groups of learners with days and times for study or general social talk around learning topics using computational models [Jahanbakhsh et al., 2017].

#### 8.4.3.3 Just-in-time broadcast of unavailability and expertise availability

I found an association between felt disconnection and moments of dealing with unavailable peers. Despite the fact that distance learners might have stronger intrinsic motivations to learn [Wighting et al., 2008], My interviews depicted the detrimental social consequences of being distracted by multiple, offline responsibilities, ranging from job duties, care-giving for their dependents and other commitments. From another perspective, such personal, learning and professional background also provide foreground for distance learners to reach out to others more confidently to get their intended answers [Berger & Calabrese, 1975]. In other words, many personal and professional responsibilities contribute to both a peer's unavailability and his or her expertise availability for collaborative learning.

However, unlike residential work or learning environment, current online learning systems do not encourage casual information exchanges unrelated to learning tasks, such as one's past or current profession, or their particular conditions that might lead to unpredictable emergency (e.g. pregnancy or military service). For example, online learners are not able to infer their peer learners' personality, working condition or cultural values in an online class encounter without them explicit disclosing information about themselves. Another qualitative study about online learners' social strategies also found the importance of successfully establishing common ground and setting up appropriate expectations about one another within a team for online learners [Sun et al., 2019]. However, even when provided with such information-sharing tools (e.g. profile), online learners are not completing or using it often, perhaps due to the incapability of enacting meaningful social interaction from here [Learning or lurking? Tracking the "invisible" online student, Sun]. My findings validate the challenges and needs of facilitating such non-task communication or information exchanges for the design of sociable CSCL environment [Kreijns et al., 2003].

In sum, I propose that CSCL designers should investigate options for time-aware communication services that can support impromptu requests (e.g. synchronous social Q&A [White et al., 2011]), automated scheduling mechanisms, non-task related information exchange and unavailability notifications for online learners. Future design research is needed to address these possibilities and to test these design claims; more work is also needed to further explore and generalize how students' availability is tied to their social behaviors.

## 8.5 Quantifying Perceptions of Student Community in Online Learning

My efforts to operationalize and measure perceptions of community by distance learners have led to four general contributions: First, I found that common identity is important in fostering sense of community and identity regulation as part of collective efficacy. Second, I raised questions about the role of personal ties in online learning communities, calling for future research to probe the value of building friendships as part of online learning. Third, I have created assessment instruments that articulate an expanded analysis of community to guide future research and practical efforts. For example, researchers can leverage these instruments to examine learners' perceived community at the end of their degree programs, perhaps examining to what extent designs for promoting community have an effect. Fourth, my approach to these community constructs can be adapted for use in near-neighbor domains, such as MOOCs, where the presence or impact of community is still relatively unexplored [Coetzee et al., 2014c; Zheng et al., 2015]. In the remaining discussion, I highight opportunities for future research along with preliminary insights for enhancing the design of online learning environments.

### 8.5.1 Putting a Spotlight on Common Identity

Across my analyses, I saw that shared identity plays an important role in online learning communities. Ratings of the SOC construct Common Identity and of the CCE construct Identity Regulation tended to have higher means than other constructs. This may indicate that identity is a core driver of connections among online learners –above all, they share an understanding of what it means to be an online learner; and they are a part of a collective that is pursuing common endeavors and shared interests [40]. They can relate to one another with respect to how they experience self-directed learning, with all members expected to learn at the course scheduled pace using the same materials, while understandably juggling through everyday responsibilities that compete for their time and attention.

Moore and Kearsley [Moore & Kearsley, 2011, 213] argued that online learners have many idiosyncrasies due to their life situations, and that they are more likely to behave independently of one another; they argue that these characteristics should be accepted as a valuable resource rather than as a distraction. Indeed, heterogeneity in a group contributes to a richer information repository with which learners can develop and coordinate their own learning plans and manage social resources in a community environment. The implication in this case is relatively straightforward: learning technologies should offer features that help students to recognize and express a shared identity as online learners, celebrating commonality while respecting differences.

Of course, recognizing and expressing a community identity implies extra activities or efforts, which may be experienced as an additional burden. Researchers have documented identity-related content *"between the lines"* in online discourse forums, but it is very sparse: only 15.6% of discussion posts manifest identity presence, and distance learners may not think that content that contains cues to inner values is appropriate to post [Ke et al., 2011]. To address this, learning technologies could encourage a communication protocol that portrays common identities and acknowledges one another's idiosyncrasies. One example could be a shared testimony of why the online students have chosen to study online. Another possibility is to signal identity as a function of shared connections (affiliations) to other individuals in an organization, for example through a public display of a social network [Heer & Boyd, 2005].

### 8.5.2 Tenuous Friendships in Online Learning Communities

I found that the SOC construct of Friendship evoked relatively low ratings from survey respondents (overall mean of 2.01 on a 5-point scale). This may indicate that feelings of friendship are tenuous as a part of an online learning community, despite any collaborative learning activities that may have taken place, and despite repeated encounters with peers on the same learning trajectory (e.g., cohorts of a major). Does this imply that online learning communities should not be viewed as a context for genuine friendship and inter-personal ties? I think not, and in fact see this as a missed opportunity for organizations that host and support online learning. At the least, counter-evidence for social purposes of online learners can be found in the related online context of MOOCs [Zheng et al., 2015]. To speculate on this phenomenon, I contrast two possibilities: one is that the intimate social bonds needed for friendship simply do not form due lack of expectation and corresponding outreach efforts; this is quite different as an explanation to the conclusion that friendship would never emerge in this community context. Future work is needed to explore these alternative explanations.

#### 8.5.2.1 Friendship as Serendipity or as a Pursuit?

I observed relatively low use of the Profile tool for social purposes; this aligns with prior work about how friendship is often celebrated as a pleasant surprise, rather than nurtured through intentional outreach [Stodel et al., 2006]. Perhaps friendship is best seen as serendipity, rather than something online students seek with intention. With this view in mind, tools for online learners might try to leverage the *"upfront"* orienting construct of shared identity, using knowledge of shared community commitment as a basis for interpersonal connections. At the same time, the environment might at least consider the benefits of adding more support for friendship building, for instance addressing the relative lack of social cues that foster interpersonal affinity [Liebman & Gergle, 2016]. As an example, private messages (e.g. whispering functions [Haythornthwaite, 2000]) and backchannels [Du et al., 2012] appear to facilitate a sense of community, but such messaging services are rarely available outside of a group of classmates.

#### 8.5.2.2 Connecting Across Distance, But Staying at a Distance

It is not surprising that online learners come to an online education program for its instrumental merits (intellectual knowledge) and pragmatic outcomes (e.g. an eventual degree) [Shachar & Neumann, 2003]. Given this general mindset, why would online learners seek intimacy that would result from making friends? Whatever connectivity there is in one's online learning network might arise from weak ties rather than strong ties. Weak ties tend to exhibit more instrumental values than strong ties [Granovetter, 1983], allowing people to work together without first forming close bonds [Haythornthwaite & Wellman, 1998]. For a large and diverse community such as World Campus, a set of weakly tied connections may be a perfectly acceptable and even desirable state. Accordingly, with respect to getting to know each other, these students may simply stop at a comfortable distance of *"knowing"* just enough without too many expectations on continually becoming closer and closer.

In other words, a community of online learners in distance education programs may have rather different structural characteristics than other communities that emerge as inter-dependent structures (e.g. a Wikipedia community that builds shared artifacts, or Quora contributors who support those in need). Another example of community that seems to emphasize identity over interpersonal ties is the YouTube community [Rotman et al., 2009]: although YouTube is perceived strongly and unanimously as an established community by its users, its friendship and subscription networks are quite chaotic when compared with a typical social network structure.

In this setting, it seems that rapport can be conveyed even when a typical personal social network (e.g., with interlinking ties among many members) does not seem to be present. More broadly, reaching out and creating interpersonal ties among members might not be a relevant goal for every community, especially when members may be heterogeneous and extrinsically motivated (e.g., *"just get the degree and move on"*). Although sociality and bonds are often created as communities of this kind emerge and develop, distance may not be a critical issue because interpersonal attachments were never part of what brings the participants together.

In either case, it is an open question as to why friendship in a distance education program is missing or diffuse. I call for a broad set of studies of different kinds of online community (e.g., more or less intrinsically or extrinsically motivated, with more or less focus on information exchange versus helping or other interaction modes), so as to develop a more nuanced understanding of the design space for online communities in HCI.

### 8.5.3 Community Collective Efficacy

In the process of doing a capacity analysis of online learning communities, building scales and conducting exploratory and confirmatory factor analysis, I uncovered three constructs that underlie Community Collective Efficacy for online learners. Due to the domain specific nature of collective efficacy, it is possible to use the resulting CCE constructs as a self-reported surrogate of complex behavioral phenomena in online learning at a collective level [Prentice et al., 1994]. In this sense, the specific CCE scale I produced as part of my dissertation enriches the toolkits that other researchers can use or adapt for future studies of online learning community.

Empirically, identity regulation, coordination and social support account for what online learners as a community are aiming to achieve. In addition, the finding that only Identity Regulation was sensitive to the two contrasting community definitions (acquaintance-based and virtual campus) helps us see why a community of larger size might induce more continued engagement in MOOC discussions [Baek & Shore, 2016]; I speculate that evidence of identity regulation is more likely to be experienced in the larger community. Most studies that have examined socially shared regulation of learning have focused on group-level effects [Järvelä et al., 2016]. Thus I emphasize the need for research efforts that are directed at the community level, with specific attention to identity regulation.

Scrutiny of the three CCE factors also reveals overlap with an earlier CCE study of geographic communities [Carroll et al., 2005]. For example, one item of Identity Regulation is the capability for the online learners to construct socio-technical infrastructure; this is similar to building and maintaining an economic infrastructure as expected in a neighborhood community: *"the community can create and maintain an adequate physical and social infrastructure"* (p.4). Coordination is similar to both the factor of *"consensus and united action"* and the factor of *"managing tradeoffs and conflicts"* [Carroll et al., 2005]. It is not surprising that both online and offline communities need to coordinate to address conflicts to act on behalf of a shared goal using a united voice. What distinguishes the individuals who come together to pursue distance education is that they share a vision for learning achievement, and they expect their online student peers to be equally responsible and committed.

Community collective efficacy does not have an extensive tradition in community informatics research. However, I offer it as a useful complement to the *"feelings"* measures that have been prevalent [Hamilton et al., 2014; Rotman et al., 2009]. With its focus on what members expect of each other and how well they feel the collective can meet these expectations, such measures may help to shift the research conversation from what *"feels good"* as a community member to also include what is *"responsible behavior"* for a community member. Moving in this direction toward responsibility and participation are reminiscent of early design advice on how to build successful web-based communities [Kim, 2000; Kraut & Resnick, 2011].

With respect to the longer-term use of the instruments, I must emphasize the self-selection bias in any study of this sort: only learners who are interested enough in questions about community are likely to respond. This means that my community measures should be seen as representative of learners who already have some interest in community with their peers. Further, my contrasting definitions of community (VCC versus ABC) may have been too weak to distinguish the acquaintance-based community from a more general understanding that includes all World Campus students. Future research should examine more closely the nature of ego networks in online learning, for example to see whether and to what degree *"friendship"* plays a role among these long-term online learning peers. Finally, while the constructs I developed appear to be robust and useful, serving as an important step toward a comprehensive analysis of online learning communities, I cannot claim that the model is exhaustive.

My new constructs – and others that may emerge in related work – are in need of further empirical study and theoretical development, both in other degree-awarding programs and in related organizations that offer education certificates (e.g. MOOCs). For instance, the notion of cohort typically found in degree programs seems to also resonate in a MOOC context [Nelimarkka & Vihavainen, 2015]; even so researchers must be sensitive to expected differences in motivation for learners pursuing degree programs versus those engaged in interest-based learning [Zheng et al., 2015].

## 8.6 Limitations and Future Work

Although Penn State is typical of North American universities that provide online degree programs, I acknowledge that is is just one of a variety of online education providers, and caution readers not to overgeneralize my qualitative and quantitative findings that were gathered from this particular context for online learning. The WC programs are highly rated examples of online degree options, but Penn State as a university has its own unique characteristics. Indeed, even the ability to seize on a moniker like "We Are!" may have been an important factor, both in terms of how well the students were primed to identify with Penn State, and how strong an affordance the label and associated images were as part of the online tools. I relied throughout on self-selection, in all of the surveys and in the decisions to enroll in *WeAre!*. In the end, the length of the tools' deployment, and the number of participants who completed both surveys, was rather modest. I was only able to attract members through relatively simple recruiting and marketing efforts (mass emails) and observe behaviors that could emerge over a few weeks. Therefore, I note that future work will be needed to validate and generalize my behavioral data and design directions for promoting community among online students.

# Appendix A
# Interview Protocols for User Studies

## A.1 Interview protocol with online instructors

Before we begin, we would like to record this interview for later reference. Are you OK to be recorded?

In the interview session, we will go over a range of topics in the discussion, but keep the interview open-ended so that we can follow up on individual insights. 1. Thanks for your time today. Could you please introduce yourself and the online courses you have been teaching or taught in the past?
2. Thinking about all of your online teaching, what are your reflections on students' feelings of connection, to one another, to you or your assistants, or even more broadly to the college or university?
3. Have you used any sorts of tools or methods to increase these feelings of connection?
4. We are also interested in student engagement more broadly. What tools or methods do you use to track student engagement when teaching online? Can you give some specific experiences?
5. Thinking more about managing your awareness of how students are doing - what tools, methods or strategies do you use to capture students' level of progress throughout the semester? How about achievement in general, how do you track this for your

online students?
6. What student data do you have access to? How do you utilize these data in terms of frequency and importance to your teaching procedure?
7. What student data that would be useful to know about (and why) in addition to those you have mentioned? In other words, are there other data that you wish you had for supporting your teaching process? (Potential prompts for questions are demographic information, academic status, achievement/activity deliverable)

8. What online course has impressed you most with the students' sense of community, or active engagement with the course content?

9. How (often) do you make use of the statistic reports from Angel or Canvas if at all? What do you think of it?

10. How (often) do you make use of the visualization summary for students' activities? What do you think of it?

11. Thinking by comparison to your teaching in residence, what are your thoughts about students' feelings of community or engagement online versus in residence? Do the tools you rely on differ in the two settings?

12. Do you have anything that you can think of on how to improve students' engagement in online courses, or how to enhance their sense of community in an online learning environment?

## A.2 Interview protocol with online students enrolled in World Campus courses

### A.2.1 Interview Questions for World Campus students

**Introduction**

Hello, my name is [Researcher] from [name of college]. Thank you for taking the time to participate in our research project aimed at fostering a sense of community among Penn State's online education students.

We would like to audio-record this session to ensure we adequately capture your ideas. However, your comments will remain confidential and we will not attach your name to any comments you make. Is this OK? Do you have any questions before we begin?

**Question list**

1. Before we start, do you have any general reflections about your online education experience?

2. We know a bit about your online education experience from your sign-up survey, but for the record here, can you tell us in what degree program are you currently enrolled? For how long? When do you hope to finish?

3. Why did you choose to enroll in this online degree program?

a. Probe: what do you expect to get from here? (academic goals, social goals, career goals)

1. Social goals: meeting cohorts, peers with the same pursuit/goals, develop social ties for career and life, getting assistance from instructors, or teaching resources (is the student-faculty engagement adequate, is it necessary?)

2. Career goals: Are you looking to re-enter the workforce, change careers, increase your skills in your current profession, or something else?

b. Probe: what resources/credits does Penn State/World Campus have to make you pursue further education here? (financial and/or time cost, social capital)

4. Next, we'd like to hear about your experiences in the various online classes you have taken so far. (interaction) a. Can you describe your normal routine for

learning what you need to learn and getting the work done? b. In what ways were the courses helpful to you, for example given your learning goals? c. In what ways do you feel that the classes fell short in helping you reach these goals? d. Probe for specific examples if they have not already come up: class size, group activity, group vs individual assignment, live session, flexible deliveries, online technologies for group conferencing/introduction, QnA, video/audio call how do you proceed with such info?

5. Have you ever been part of a team project in an online course? (group formation)

1. If yes, can you reflect on your most recent experience, and describe how it worked?
    - Amount of workload, easier or harder? In what ways did the teamwork help you? Why?
    - Strategies that make this work or not work?
    - What tools did you use to mediate the teamwork? How did they work?
2. Thinking about other cases: were they similar, e.g. can you generalize or does it depend on the specific course? How?
3. How did you interact with your group members (if not mentioned before)? Did you talk about course-only? What about life, work, career, values, experiences and others?
4. To what extent have you felt connected with your team members in these projects? Can you elaborate with specific experiences?

6. Thinking about connections to others more broadly, to what extent do you feel connected with the students in ALL of your current online classes? Why or why not?

a. Can you think of online course experiences that have made you feel more connected to other students or to instructors? (examples: talk to people online, know people by names, know their life/background)

7. Now think about students across the entire World Campus of classes, that is whether or not they have been in a class with you? To what extent do you feel connected to this larger group?

- What about the group that includes all students in <your current program, e.g. IST>?
- What about Penn State University as a whole?

8. To what extent do you WANT to feel more connected with other online students? Why or why not?

<if says yes: Can you think of connection-building things you might want to do if it was possible? (probe, taking classes with the same students semester after semester; attending an online event together; meeting up FTF)

9. What do you think of tutoring help you received from other students or people specifically assigned to help (e.g. group members, tutor or TA or LA)? Have you offered or considered offering help to other students? What about using your own work-based expertise or professional network, could these be valuable to other online students? Follow up with

1. If yes, what motivate you to act or think in this way? (Why)

   How did you offer tutor help?

   Did you anticipate recognition for work? experience (policies)

2. If not, what prevents you from doing so? (Why question)

10. We'd also like to hear about how your online experience has helped you prepare for your career.

a. Did you take advantage of any Career advice or other services offered by World Campus or elsewhere? Can you describe?

b. If yes, probe with: In what ways were the services helpful to you? And, in what ways do you feel that the services fell short in helping you reach your goals?

c. What about other people or resources - did you ever interact with professors, other students, or other information with respect to career preparation? Can you describe what happened?

11. World Campus is a campus of Penn State; to what extent do you attend or follow Penn State activities, e.g. athletics, politics, art or music events, news, student affairs?

a. How do you think about Penn State social media activity, do you contribute or would you like to? Are you involved in any World Campus or Penn State activities that are not part of your courses? b. Background: Penn State Sports Events (46%), Professional Conferences, (65%) Webinars/videos (53%)

12. Now imagine that you are part of a committee of people designing web technologies to enhance feelings of community in a virtual campus like yours. a. What would you propose to include in such an enhanced web environment? b. How would it change what you do on a day to day basis, with respect to either taking courses, or doing other sorts of career-related things like networking or preparing for internships or jobs?

13. Is there anything else we haven't discussed yet that you think is important for us to know about as we consider tailoring programs to foster you as a community or for the meet you educational goals in general?

# Appendix B
# Scale Development

The first section shows the full survey content. The next two sections entail the detailed items originally used for measuring Sense of Community and Community Collective Efficacy, both of which were developed based on prior literature and my empirical findings (e.g. interviews reported in Section 3.3).

## B.1 Measuring SOC and CCE Survey Content

Click to visit the Google Document with full description of survey content designed to measure SOC and CCE.

## B.2 All Sense of Community Items with Factor Loading Index

**Table B.1** Varimax Rotated Component Matrix of All Items for SOC

| Item Description | Component | |
|---|---|---|
| | 1 | 2 |
| **2. The community helps me fulfill my needs.** | **.804** | |
| **4. I feel like a member of World Campus.** | **.800** | |
| **1. I can get what I need in the World Campus community.** | **.797** | |
| **5. I belong in the World Campus community.** | **.773** | |
| 7. I feel connected to World Campus community. | .737 | .486 |
| **3. I feel that I can rely on others in World Campus.** | .717 | .386 |
| 6. I feel that I matter to other World Campus students. | .622 | .547 |
| **9. I have friends at World Campus to whom I can tell anything.** | | **.889** |
| **10. I feel close to others at World Campus.** | .717 | **.884** |
| 8. I have a good bond with others in this community. | .528 | .706 |

## B.3 All Community Collective Efficacy Items with Factor Loading Index

**Table B.2** Varimax Rotated Component Matrix of All Items for CCE

| Item Description | Component | | |
|---|---|---|---|
| | 1 | 2 | 3 |
| **13. We can follow social norms regarding appropriate online behaviors, even when there is no direct supervision by instructors, advisors or administrative staff.** | **.797** | | |

| | | | |
|---|---|---|---|
| **10. Our community can facilitate self-directed learning even when we are not fully aware of other learners' activities.** | **.655** | | .394 |
| 19. Our community can take full advantage of Canvas, despite occasional complexity or surprises in the user interface. | **.633** | .325 | |
| **9. Our community encourages each of us to pursue a degree, even though individuals may face unique challenges in doing so.** | **.596** | .315 | .383 |
| 18. Our community can adopt new tools to support our learning, even if most members are not experts in online technologies. | .594 | .737 | .486 |
| 15. We can work toward shared community goals, although each of us brings varied motivations, expectations, commitment, and needs. | .592 | .491 | |
| 14. When a member behaves unethically, our community can respond despite the distance that separates us. | .585 | .448 | |
| 8. Our community can broaden individual members' views, even though each of us has our own value systems. | .550 | | .531 |
| **24. Our community can sustain long term associations despite our virtual forms of interaction.** | | **.738** | .312 |
| **20. When a problem arises, we can coordinate amongst ourselves to gain help from outside entities, even when the problem is complex or multi-faceted.** | .398 | **.665** | |
| **22. The community gives members the opportunity to have individual influence, despite the general goal of speaking with a united voice.** | .375 | **.661** | |

| | | | |
|---|---|---|---|
| **21. Our community can convey its mission to outsiders, even when we have no obligations to do so.** | .423 | **.653** | |
| 17. We can self-organize into subgroups with more refined interests, even though it will take effort to manage a variety of groups. | | **.646** | .348 |
| 12. Our members try to help others with work-life balance, even if it may lead to extra workload for ourselves. | | .601 | .415 |
| 16. Our community can develop leaders even though individuals' potential may not be visible in the first place. | .491 | .560 | |
| 23. Even though the population of online students has been growing rapidly, our community is able to maintain its quality standards. | .490 | .497 | |
| 11. Our community can carry out collaborative activities, even though we have limited time available for interaction. | .376 | .443 | .408 |

# Appendix C
# Design Probes Screenshots

## C.1 Prototype Screenshots for Supporting Availability Information

I sketched using pencil and paper as well as commercial software to flesh out my design ideas. For example, towards the end, I designed multiple prototypes in the process of developing the concepts of Availability and other associated information (e.g. professional background).

### C.1.1 Design of User Login Screen

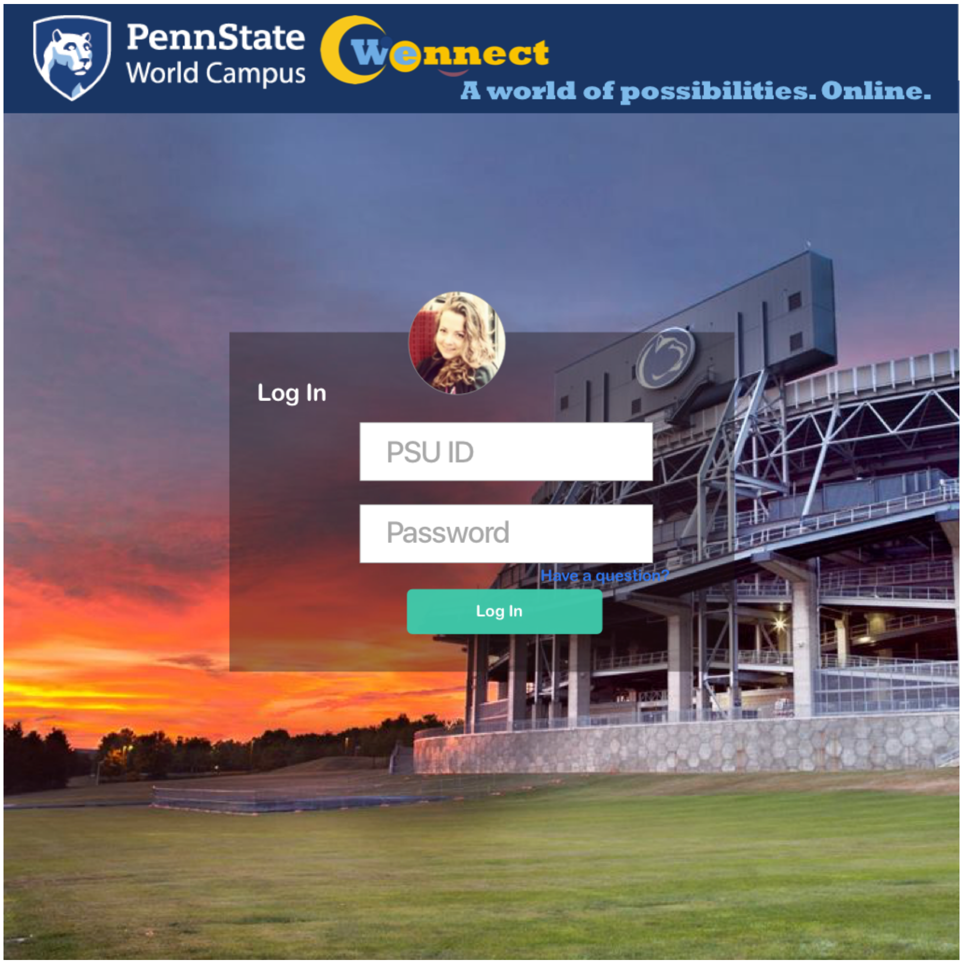



**Figure C.1** Iterative Design Dashboard for users' log-in activity

### C.1.2 Design of Entering Availability Information

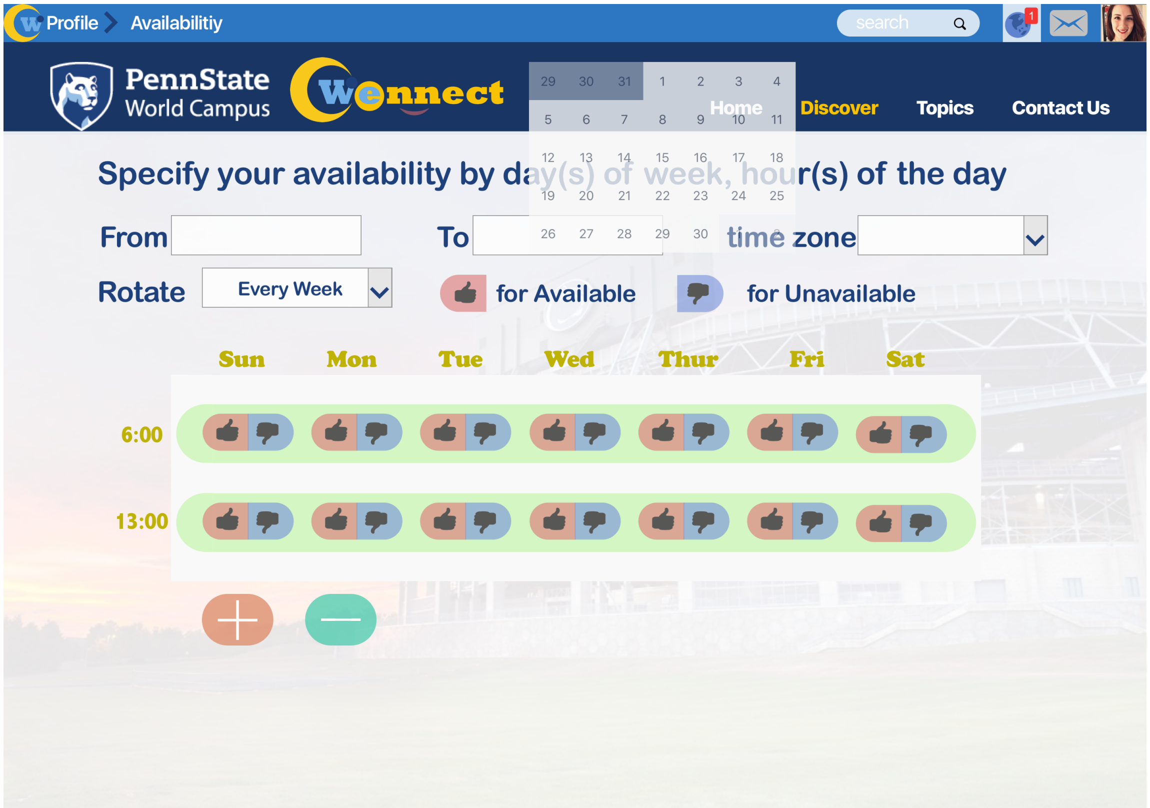



**Figure C.2** Choosing Availability status with Thumb Up or Thumb Down

### C.1.3 Design of Presenting Availability Information

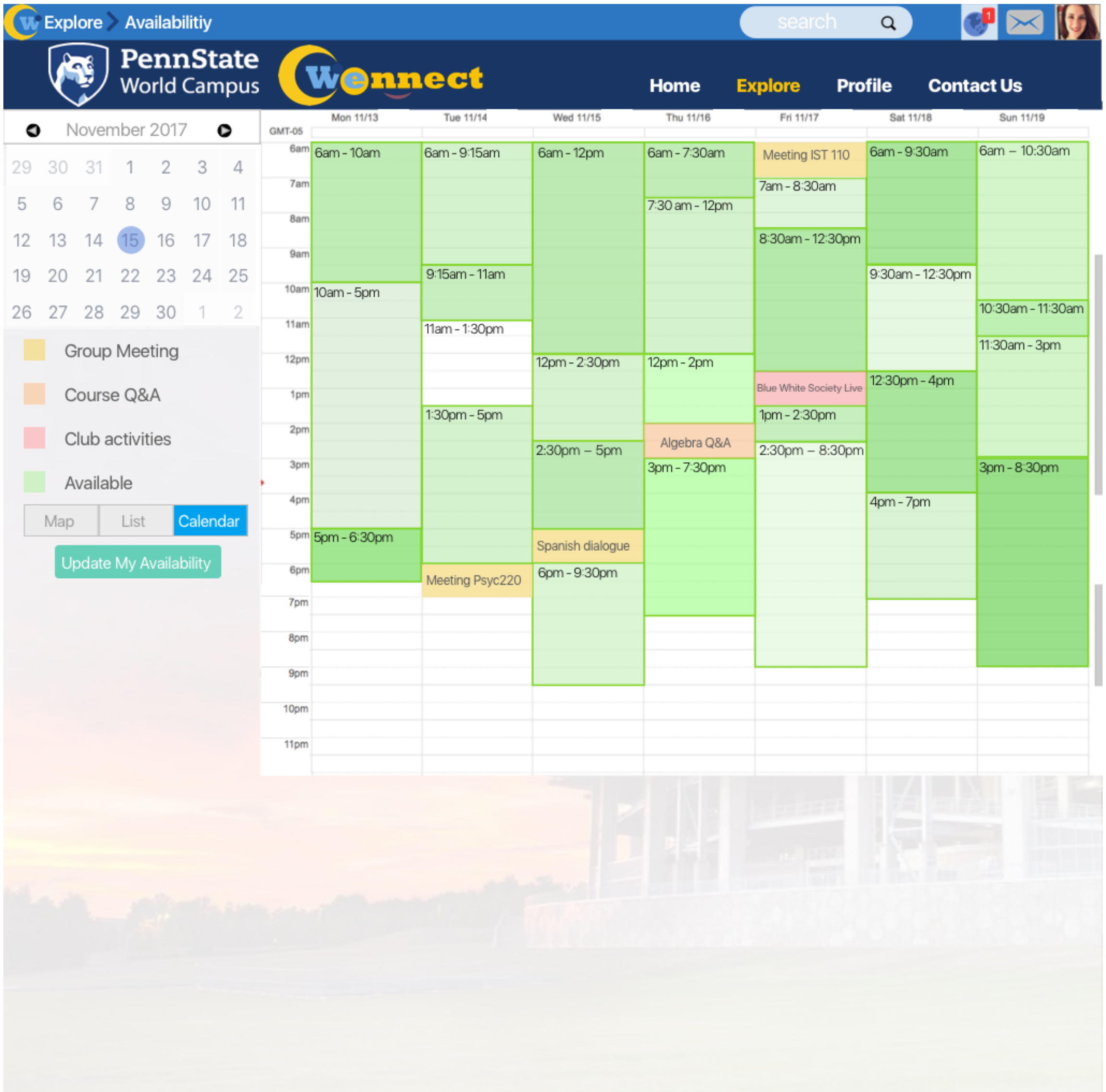



**Figure C.3** Present Availability through Familiar Calendar Interface

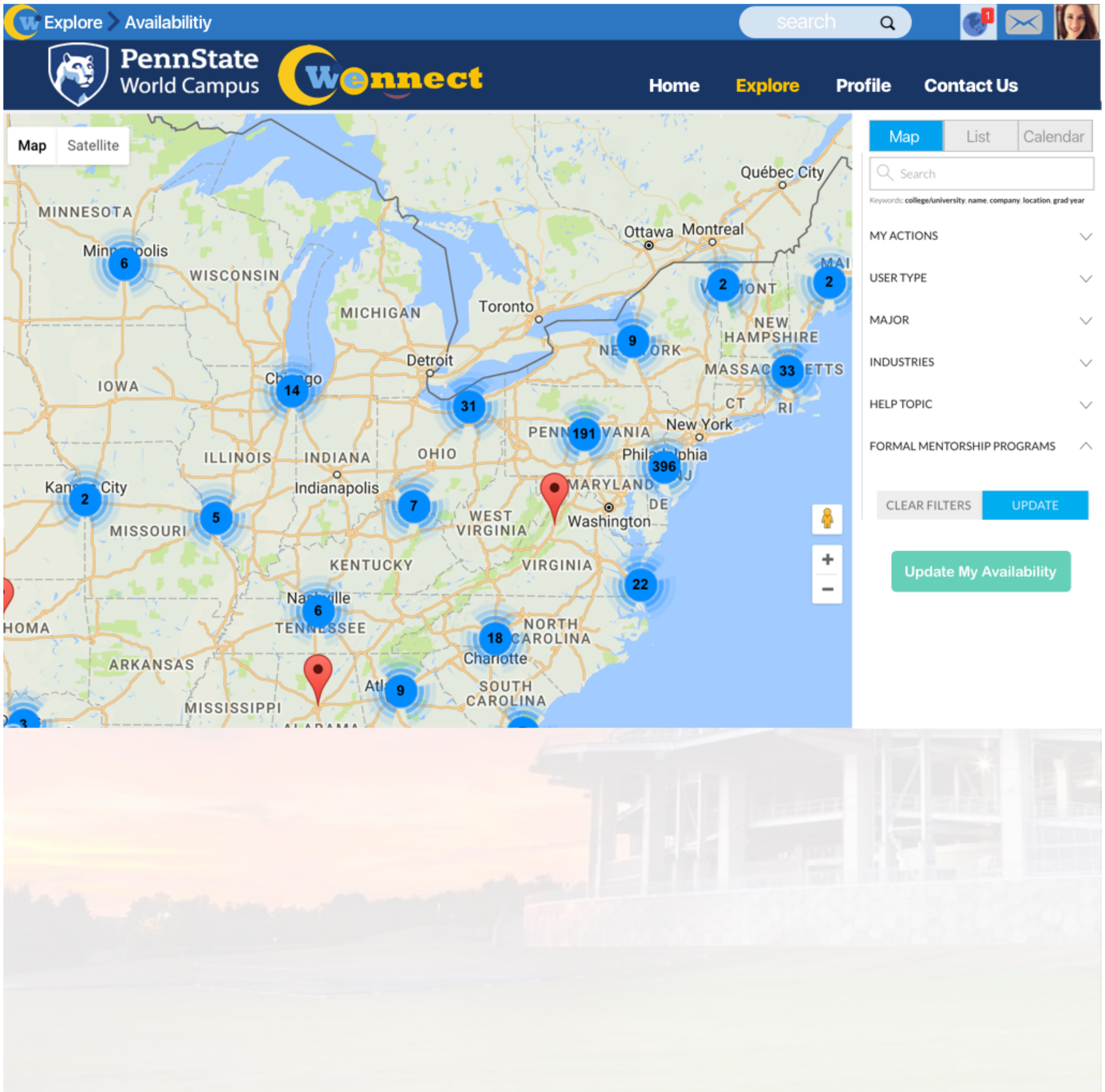



**Figure C.4** Present Availability through Map Interface

### C.1.4 Design of Sharing Availability Information

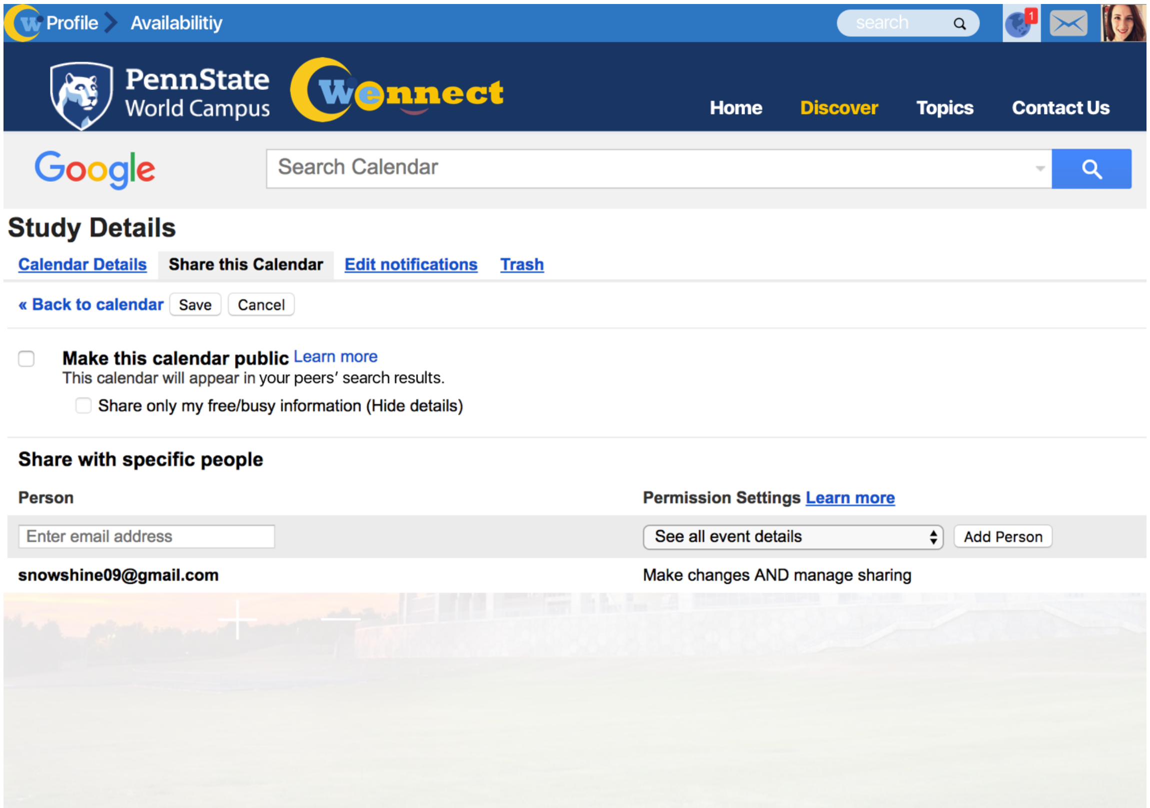



**Figure C.5** Share Availability through Google Account

# Appendix D
# WeAre! Deployment Study Materials

## D.1 Slack study Pre-usage Content

Click to visit the Google Document with full description of pre-survey content.

## D.2 Deployment study Post-usage Survey Content

Click to visit the Google Document with full description of post-survey content

The display logic of participants in different types are illustrated in Figure D.1(e.g. students who took pre-survey participants, students who registered in Slack Workspace *WeAre!* only, students who took pre-survey and registered in Slack Workspace *WeAre!*)

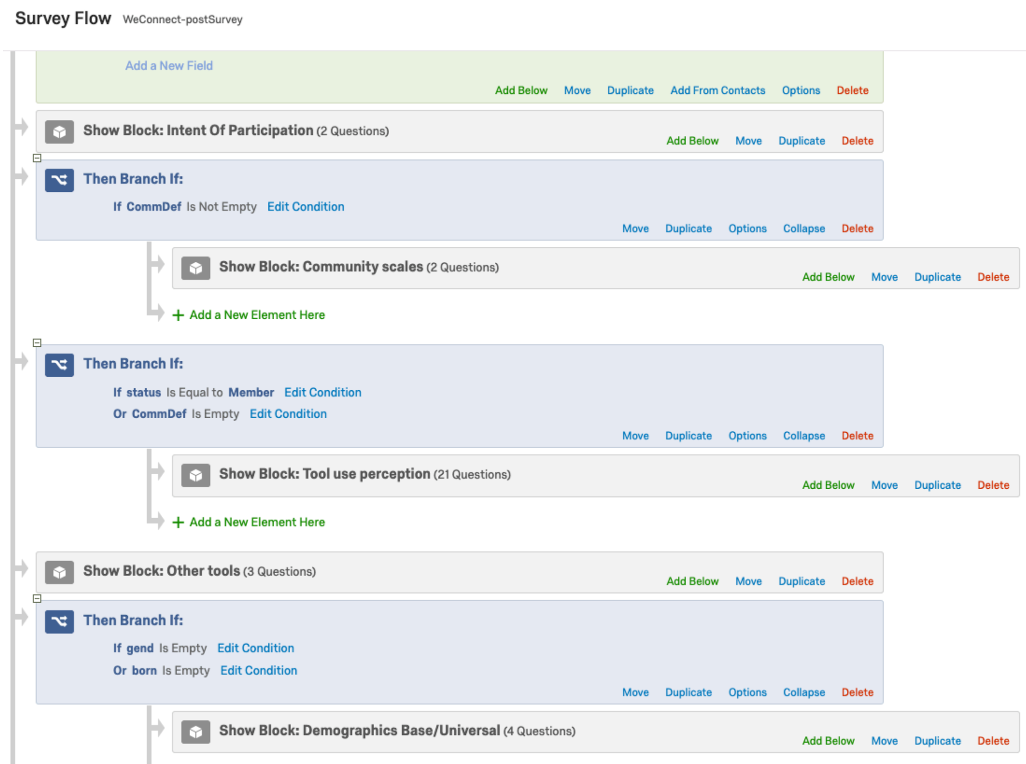



**Figure D.1** Display logic for Post-Survey Participants of different conditions

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Zinn, C. & Scheuer, O. (2006). Getting to know your student in distance learning contexts. *European Conference on Technology Enhanced Learning: innovative Approaches for Learning and Knowledge Sharing*, 437—-451.

# Na Sun

## EDUCATION

**Pennsylvania State University** — **State College, PA**
*Ph.D. in Human-Computer Interaction* — *June 2019*

**East China Normal University** — **Shanghai, China**
*B.S. in Computer Science* — *May 2013*

## KEY SKILLS

**Qualitative:** Interviews, Contextual Inquiry, Observation, Focus Group, Virtual Ethnography, Field Studies, Technology Probe, Persona, Affinity Diagram, Heuristic Evaluation, Remote Moderated Research, Think-aloud Protocol, Usability Test
**Quantitative:** Survey, Exploratory Factor Analysis, Structural Equation Modeling, Experiment Design, Log Analysis, Content Analysis, Regression Analysis, Cluster Analysis

## RESEARCH EXPERIENCE

**Pennsylvania State University** — **State College, PA**
*UX Research Assistant* — *August 2013 - Present*

### WeAre! Promoting Community Experiences for Distance Learners

- Collected users' needs of social connections with mixed methods and triangulation.
- Iteratively generated persona and design prototypes for facilitating students' connections.
- Developed and validated scales to define and assess students' community perceptions.
- Conducted field-experiment with Slackbot and web app using repeated measures and users' logs.

### Critical Thinker, Collaborative Sensemaking and Civic Engagement

- Conducted evaluation of Critical Thinker in online sessions and face-to-face classrooms.
- Co-developed a note-taking tool for sustainable intelligence analysis with multiple visualizations.
- Co-developed a claim-making based system to support community issues review for local citizens.
- Evaluated the system through field observation and interviews.

**LIRIS Lyon** — **Lyon, France**
*Visiting HCI Scholar* — *May 2017 - August 2017*

- Conducted interview and document analysis for online students' peer feedback activities on Emoviz, an annotation tool for students to comment on document areas with text & emojis
- Designed a visualization dashboard to present group members' feedback contribution.

## SELECTED PUBLICATIONS


[1] **Sun, N.**, Wang, X. and Rosson, M. B. How do distance learners connect?, in *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, CHI '19, 432:1–432:12, Glasgow, Scotland Uk, ACM, 2019, ISBN: 978-1-4503-5970-2, DOI: `10.1145/3290605.3300662`, URL: `http://doi.acm.org/10.1145/3290605.3300662`.

[2] **Sun, N.**, Rosson, M. B. and Carroll, J. M. Where is Community Among Online Learners?: Identity, Efficacy and Personal Ties, in *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, page 292, ACM, 2018, ISBN: 1450356206.